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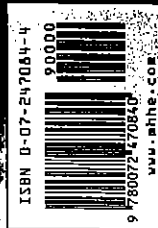
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Chapter 2

Descriptive Statistics



- Chapter Outline**
- 2.1 Describing the Shape of a Distribution
 - 2.2 Describing Central Tendency
 - 2.3 Measures of Variation
 - 2.4 Percentiles, Quartiles, and Box-and-Whiskers Displays
 - 2.5 Describing Qualitative Data

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- *2.6 Using Scatter Plots to Study Relationships between Variables
- *2.7 Misleading Graphs and Charts
- *2.8 Weighted Means and Grouped Data
- *2.9 The Geometric Mean

2.1 Describing the Shape of a Distribution

We begin looking at the characteristics of a population by describing the population's overall pattern of variation. That is, we describe the shape of the distribution of population measurements. We often employ a sample of measurements taken from a population in order to infer what the population looks like.

Several graphical devices—the stem-and-leaf display, the histogram, and the dot plot—are often used to portray shapes of distributions.

Stem-and-leaf displays We illustrate how to construct stem-and-leaf displays in the following examples.

EXAMPLE 2.1 The Gas Mileage Data

Table 2.1 presents the sample of 49 gas mileages that have been obtained by the new midsize model discussed in Example 1.4 (page 15). To graphically portray the pattern of variation in these mileages, we can construct a stem-and-leaf display. In order to do this, we first notice that the sample mileages range from 29.8 to 33.3. For this data we will (somewhat arbitrarily) construct a display having the first two digits of the mileages—the whole numbers 29, 30, 31, 32, and 33—as the stems. These are placed in a column on the left side of the display as follows:

29	
30	
31	
32	
33	

The following table shows the gas mileage data for the 49 cars. The first two digits of the mileage are the stems, and the digits after the decimal point are the leaves. For example, 29.8 has a stem of 29 and a leaf of 8.

VS
CHAPTER 1

TABLE 2.1 A Sample of 49 Mileages

	30.8	30.9	32.0	32.3	32.6
31.7	31.4	31.4	32.7	31.4	31.4
30.1	31.2	30.8	31.2	31.8	31.8
31.6	30.3	32.8	30.6	31.9	31.9
32.1	31.3	32.0	31.7	32.8	32.8
33.3	32.1	31.5	31.4	31.5	31.5
31.3	32.5	32.4	32.2	31.6	31.6
31.0	31.8	31.0	31.5	30.6	30.6
32.0	30.4	29.8	31.7	32.2	32.2
32.4	30.5	31.1	30.6		

The third (tennis place) digits of the mileages are the leaves of the display. These are placed in rows corresponding to the appropriate stems. Therefore, the first three mileages—30.8, 31.7, and 30.1—would be represented as

```

29 | 8
30 | 1 4 4 5 6 6 6 8 8 9
31 | 0 0 1 2 3 3 4 4 5 5 6 6 7 7 7 8 8 9
32 | 0 0 0 1 1 2 2 3 4 4 5 5 6 7 8 8
33 | 3

```

We continue this procedure, and we order the data within each row (although this is not required). We obtain the following stem-and-leaf display:

```

29 | 8
30 | 1 3 4 4 5 6 6 6 8 8 9
31 | 0 0 1 2 3 3 4 4 5 5 6 6 7 7 7 8 8 9
32 | 0 0 0 1 1 2 2 3 4 4 5 5 6 7 8 8
33 | 3

```

This display portrays the overall pattern of the sample mileages. It groups the mileages into classes, and it graphically illustrates how many mileages are in each class, as well as how the mileages are distributed within each class. The first class corresponds to the stem 29 and consists of the mileages from 29.0 to 29.9. There is one mileage—29.8—in this class. The second class corresponds to the stem 30 and consists of the mileages from 30.0 to 30.9. There are 11 mileages in this class. Similarly, the third, fourth, and fifth classes correspond to the stems 31, 32, and 33 and contain, respectively, 20 mileages, 16 mileages, and 1 mileage.

If we want a more detailed display, we can create more classes by using stems labeled as 29*, 30*, 30.1*, 31*, 32*, 32, and 33*. Here, for instance, in the row labeled 30* we place the mileages from 30.0 to 30.4, and in the row labeled 30 we place the mileages from 30.5 to 30.9. Doing this, we obtain

```

29 | 8
30* | 1 3 4 4
30 | 5 6 6 6 8 8 9
31* | 0 0 1 2 3 4 4 4
31 | 5 5 5 6 6 7 7 8 8 9
32* | 0 0 0 1 1 2 2 3 4 4
32 | 5 6 7 8 8
33* | 3

```

Looking at the stem-and-leaf display, the distribution of sample mileages appears to be quite symmetrical. That is, the upper portion of the display (the portion corresponding to the stems 29*, 30*, 30, and 31*) and the lower portion of the display (the portion corresponding to the stems 31, 32*, 32, and 33*) are nearly mirror images of each other. We will learn in later sections that

FIGURE 2.1 MINITAB and MegaStat Outputs of a Stem-and-Leaf Display of the 49 Mileages

Character	Stem-and-Leaf Display	Stem and Leaf plot for MPG
		stem unit = 1 leaf unit = 0.1
1	29 8	
5	30 1 3 4 4	
12	31 5 5 5 6 6 8 9	
21	32 0 0 1 2 3 3 4 4 4	Frequency
(11)	33 1 5 5 5 6 6 7 7 8 8 9	1 29 8
17	32 0 0 0 1 1 2 2 3 4 4	4 30 1344
7	32 5 5 6 7 8 8	7 30 566889
3	33 3	9 31 00123444
1		11 31 5556677889
		10 32 0001122344
		6 32 556788
		1 33 3
		49

(a) The MINITAB Output

(b) The MegaStat Output

symmetry or, more generally, the shape of the distribution of data influences how we measure both the central tendency and the variability of the data.

Finally, Figure 2.1 gives MINITAB and MegaStat outputs of a stem-and-leaf display of the 49 sample mileages. These displays are identical to the latter of the two displays we set up by hand. In the MINITAB output, the leftmost column of numbers summarizes how many measurements (mileages) are in the various stems. For example, the 11 (in parentheses) tells us that the stem for 31.5 to 31.9 mpg includes 11 mileages; the 12 (no parentheses) tells us that a total of 12 mileages are at or below 30.9 mpg; the 7 tells us that a total of 7 mileages are at or above 32.5 mpg. In the MegaStat output, the leftmost column of numbers tells us how many measurements are in each of the stems. For instance, 7 mileages are in the stem for 30.5 to 30.9 mpg.

We summarize how to set up a stem-and-leaf display in the following box:

1. Determine a unit will be used for the stems. The leaves should be single-digit numbers. If the data are not integers, the unit for the stems should be a power of 10. For example, if the data are 30.8, 31.4, 31.4, 32.7, 31.4, 30.1, 31.6, 30.3, 32.8, 30.6, 31.9, 32.1, 31.3, 32.0, 31.7, 32.8, 33.3, 32.1, 31.5, 31.4, 31.5, 32.2, 31.6, 31.6, 31.0, 31.8, 31.0, 31.5, 30.6, 32.2, 32.0, 30.4, 29.8, 31.7, 30.6, 32.4, 30.5, 31.1, the unit for the stems should be 0.1.
2. Place the stems in a column on the left. The stems should be in increasing order from bottom to top. For example, if the stems are 29, 30, 31, 32, 33, the stems should be listed in that order from bottom to top.
3. Enter the leaf values in the row corresponding to the proper stem. The leaf values should be in increasing order from left to right. For example, if the leaf values for stem 30 are 1, 3, 4, 4, 5, 5, 6, 6, 7, 7, 7, 8, 8, 9, the leaf values should be listed in that order from left to right.

Some stem-and-leaf displays do not appear symmetrical. This is illustrated in the next two examples.

Consider the sample of 65 payment times in Table 1.2 (page 6), and note that these payment times range from 10 days to 29 days and are expressed as whole numbers (of days). To construct a stem-and-leaf display of the payment times, we will use these whole numbers as the stems. Noting that a whole number, say 18, can be written as 18.0, we will use the

(implied) ten's-place digit, 0, as the leaf value. Therefore, the stem-and-leaf display is as follows:

10	0	
11	0 0	
12	0 0 0	
13	0 0 0 0	
14	0 0 0 0 0	
15	0 0 0 0 0 0	
16	0 0 0 0 0 0 0	
17	0 0 0 0 0 0 0 0	
18	0 0 0 0 0 0	
19	0 0 0 0 0	
20	0 0 0	
21	0 0 0 0	
22	0 0 0	
23	0 0	
24	0 0 0	
25	0 0	
26	0	
27	0	
28		
29	0	

Looking at this display, we see that the "tail" of the distribution consisting of the higher payment times is longer than the "tail" of the distribution consisting of the smaller payment times. This shows that a few of the payment times are somewhat larger than the rest of the payment times. Here we say that the distribution of payment times seems to be skewed toward the larger payment times. Alternatively, we say that the distribution is skewed with a tail to the right because, if we turn the display sideways so that we are looking at it with the stems forming a number line, the longer tail would extend to the right.

Consider the sample of 60 package design ratings in Table 1.3 (page 7). These package design ratings range from 20 to 35, so a stem-and-leaf display of the ratings is as follows:

20	0	
21	0	
22	0	
23	0	
24	0	
25	0 0	
26	0 0	
27	0 0 0	
28	0 0 0 0	
29	0 0 0 0 0	
30	0 0 0 0 0 0	
31	0 0 0 0 0 0 0	
32	0 0 0 0 0 0 0 0	
33	0 0 0 0 0 0 0 0	
34	0 0 0 0 0	
35	0	

Looking at this display, we see that the distribution of package design ratings seems to be skewed toward the smaller ratings. That is, we say the distribution is skewed with a tail to the left. This says that a few of the ratings are somewhat smaller than the rest of the ratings.

Frequency distributions and histograms The count of the number of measurements in a class defined by a stem is called the frequency of the class. One advantage of a stem-and-leaf display is that it gives the frequencies of the different classes and also lists the specific measurements in each class. However, such listings for the different classes can be unwieldy if we are portraying a large number of measurements. For example, while it is convenient to display the 65 payment times using the stem-and-leaf display of Example 2.2, summarizing 500 payment times with the same type of format would be difficult.

When we have many measurements, it is best to group them into the classes of a frequency distribution and to display the data by using what is called a histogram. These are also useful for summarizing more moderately sized data sets (such as the 65 payment times). We illustrate this in the following example.

Here is a frequency distribution for the 65 payment times:

Class	Frequency
10-12	3
13-15	14
16-18	23
19-21	12
22-24	8
25-27	4
28-30	1

The frequency distribution divides the payment times into classes, and the frequency for each class tells us how many of the payment times are in each class.¹ In order to group the payment times into the classes of a frequency distribution, we must first decide how many classes to use. A general rule says that the number of classes should be the smallest whole number K that makes the quantity 2^k greater than the total number of measurements n . Since $n = 65$, and since $2^6 = 64$ is less than 65 and $2^7 = 128$ is greater than 65, we should use $K = 7$ classes. We next find the length of each class by computing

$$\text{Class length} = \frac{\text{Largest measurement} - \text{Smallest measurement}}{K}$$

Since the largest and smallest payment times in Table 1.2 (page 6) are 29 days and 10 days, the class length is $(29 - 10)/7 = 2.7143$. This says that, in order to include the smallest and largest payment times in the 7 classes, each class must have a length of at least 2.7143. To obtain a more convenient class length, we arbitrarily round this value up to 3. To form the first class, we start with the smallest payment time—10—and add 2 (one less than the class length 3) to obtain the class 10-12. To form the second class, we start with the payment time 13 and add 2 to obtain the class 13-15. Continuing this process, we find that the remaining 5 classes are 16-18, 19-21, 22-24, 25-27, and 28-30. Note that each class contains 3 payment times, where 3 is the class length. For example, the first class of 10-12 contains the 3 payment times 10, 11, and 12.

In general, the smallest and largest data values that can belong to a given class are called the lower and upper class limits. For example, the seventh class of 28-30 has a lower class limit of 28 and an upper class limit of 30. The largest observed payment time—29 days—is contained in this last class. In cases where the largest measurement is not contained in the last class, we simply add another class. Generally speaking, the guidelines we have given for forming classes are not inflexible rules. Rather, they are intended to help us find reasonable classes.

Once we have formed the classes, we record the number of measurements that fall into each class. To do this, we look at each measurement and place a check, or tally, to the right of the class into which the measurement falls. For example, since the first four payment

¹Note that the classes must be defined so that any particular payment time will fall into one and only one class.

TABLE 2.2 A Frequency Distribution and a Relative Frequency Distribution of the 65 Payment Times

Class	Frequency	Relative Frequency	Boundaries	Midpoint
10-12	3	$3/65 = .0462$	9.5, 12.5	11
13-15	14	$14/65 = .2154$	12.5, 15.5	14
16-18	23	$23/65 = .3538$	15.5, 18.5	17
19-21	12	$12/65 = .1846$	18.5, 21.5	20
22-24	8	$8/65 = .1231$	21.5, 24.5	23
25-27	4	$4/65 = .0615$	24.5, 27.5	26
28-30	1	$1/65 = .0154$	27.5, 30.5	29

times in Table 1.2 are 22, 19, 16, and 18, we obtain the first four checks shown in the following table:

Class	First 4 Checks	65 Checks	Frequency
10-12		III	3
13-15		IIII IIII	14
16-18	II	IIII IIII IIII	23
19-21	I	IIII IIII	12
22-24	I	IIII III	8
25-27		IIII	4
28-30		I	1

If we examine all 65 payment times in Table 1.2, we obtain the 65 checks shown above. To determine the frequency for a class, we count the number of checks for the class. Table 2.2 lists both the frequencies and the relative frequencies for the seven classes. The relative frequency of a class is the proportion (fraction) of the total number of measurements that are in the class. For example, there are 14 payment times in the second class, so its relative frequency is $14/65 = .2154$. This says that the proportion of the 65 payment times that are in the second class is .2154, or, equivalently, that $100(.2154) = 21.54\%$ of the payment times are in the second class. A list of all of the classes—along with each class frequency—is the previously given frequency distribution. A list of all of the classes—along with each class relative frequency—is called a relative frequency distribution.

To graphically portray the distribution of the payment times, we construct a histogram. To set up a frequency histogram, we draw rectangles representing the classes. Here the base of a rectangle describing a class represents the payment times in the class, and the height of the rectangle represents the class frequency. A relative frequency histogram is drawn in the same way, except that the height of the rectangle represents the relative frequency of the class. To construct these histograms, we employ what we call class boundaries. Consider, for example, the first and second classes. The upper class limit of the first class is 12, and the lower class limit of the second class is 13. The class boundary between these classes is halfway between these class limits—12.5. Likewise, the class boundary between the second and third classes is 15.5, which is halfway between the upper class limit of the second class, 15, and the lower class limit of the third class, 16. Table 2.2 gives all of the class boundaries for the 65 payment times. As shown in the MINITAB output in Figures 2.2 and 2.3, these class boundaries are used to draw the rectangles that make up the frequency histogram (in Figure 2.2) and the relative frequency histogram (in Figure 2.3). Note that the difference between the upper and lower class boundaries of a given class is the class length 3. Also, note that MINITAB expresses the relative frequencies as percentages. For example, since the frequency and the relative frequency for the second class are 14 and $14/65 = .2154$, the heights of the rectangles over the second class in Figures 2.2 and 2.3 are 14 and 21.54%.

FIGURE 2.2 MINITAB Output of a Frequency Histogram of the 65 Payment Times

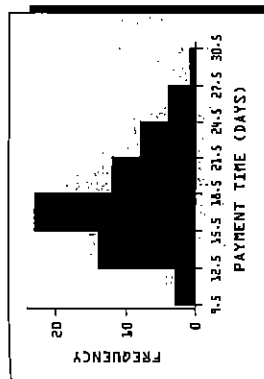


FIGURE 2.3 MINITAB Output of a Relative Frequency Histogram of the 65 Payment Times

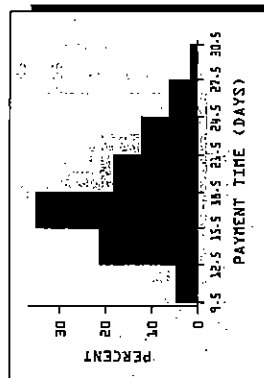


TABLE 2.3 A Frequency Distribution and a Relative Frequency Distribution of the 49 Mileages

Class	Frequency	Relative Frequency	Boundaries	Midpoint
28.8-30.3	3	$3/49 = .0612$	28.75, 30.35	30.05
30.4-30.9	9	$9/49 = .1837$	30.35, 30.95	30.65
31.0-31.5	12	$12/49 = .2449$	30.95, 31.55	31.25
31.6-32.1	13	$13/49 = .2653$	31.55, 32.15	31.85
32.2-32.7	9	$9/49 = .1837$	32.15, 32.75	32.45
32.8-33.3	3	$3/49 = .0612$	32.75, 33.35	33.05

In general, we can have MINITAB, Excel, or MegaStat form a histogram by using classes that we specify or by having these computer packages automatically select classes for us. Furthermore, we can define the desired classes by specifying either the class boundaries (as we did to obtain Figures 2.2 and 2.3) or the class midpoints. The midpoint of a class is the point halfway between the class limits (or boundaries). For example, the rightmost column in Table 2.2 gives the midpoints of the seven classes employed in the payment time histograms. The MINITAB output of a frequency histogram using these midpoints is shown in Figure 2.4.

When specifying classes, we usually express the lower and upper class limits to the same decimal place accuracy as the observed data values. For example, since the payment times are measured to the nearest day and thus are integers, we use integer class limits of 10-12, 13-15, and so on. As another example, consider grouping the $n = 49$ gas mileages in Table 2.1 into the classes of a frequency distribution. Since $2^1 = 32$ is less than 49 and $2^6 = 64$ is greater than 49, we should use $K = 6$ classes. The largest and smallest mileages in Table 2.1 are 33.3 and 29.8, and so we find the class length by computing $(33.3 - 29.8)/6 = .5833$. To obtain a more convenient class length, we round this value up to .6, which is expressed to the same decimal place accuracy as the mileages. To form the first class, we start with the smallest mileage—29.8—and add .5 (one-tenth less than the class length .6) to obtain the class 29.8-30.3. To form the second class, we start with the mileage 30.4 and add .5 to obtain the class 30.4-30.9. Continuing this process, we obtain the 6 classes shown in Table 2.3. Here the largest mileage—33.3—is just barely included in the last class. If we group the 49 mileages into the classes, we obtain the frequencies and relative frequencies given in Table 2.3. A MegaStat output of a relative frequency histogram of the mileages is shown in Figure 2.5. This histogram uses the class boundaries as the labels on its horizontal axis.

FIGURE 2.4 MINITAB Output of a Frequency Histogram of the 65 Payment Times

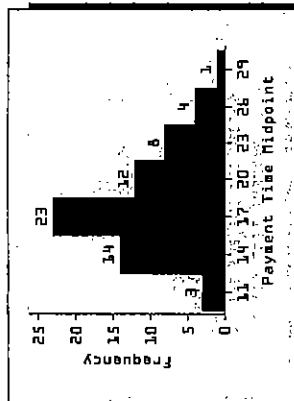
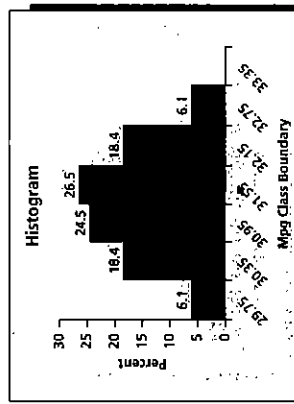


FIGURE 2.5 MegaStat Output of a Relative Frequency Histogram of the 49 Mileages



Similar to a stem-and-leaf display, a histogram describes the overall pattern of variation displayed by the data. Examining the histogram in Figure 2.5, we see that the distribution of the 49 mileages appears quite symmetrical. On the other hand, the histogram in Figure 2.4 shows that the distribution of the 65 payment times is somewhat skewed with a tail to the right. Furthermore, consider the 60 package design ratings in Table 1.3 (page 7). The Excel frequency histogram in Figure 2.6 shows that the distribution of these ratings is somewhat skewed with a tail to the left. This Excel histogram uses the upper class boundaries as the labels on its horizontal axis.

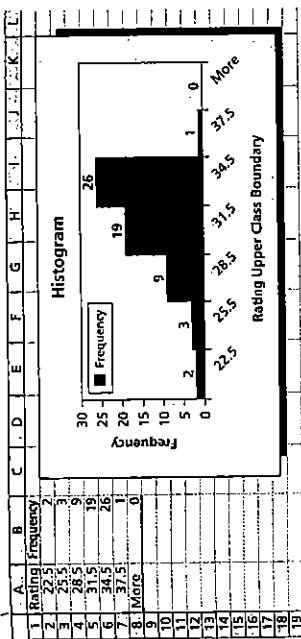
In the following box we summarize the steps needed to set up a frequency distribution and histogram:

1. Decide how many classes will be employed. Statistician William J. F. Fox has suggested that the number of classes should equal the square root of the number of observations. For example, if there are 100 observations, the number of classes should be 10. This rule is a good starting point, but it is not a hard and fast rule. The number of classes should be chosen so that the distribution of the data is clearly visible.
2. Compute the class length. The class length is the width of each class. It is computed by dividing the range of the data by the number of classes. For example, if the range is 100 and there are 10 classes, the class length is 10.
3. For each class, determine the lower and upper boundaries. The lower boundary is the smallest value in the class, and the upper boundary is the largest value in the class. For example, if the class length is 10 and the lower boundary is 20, the upper boundary is 30.
4. Compute the class midpoint. The class midpoint is the average of the lower and upper boundaries. For example, if the lower boundary is 20 and the upper boundary is 30, the class midpoint is 25.
5. Calculate the frequency for each class. The frequency is the number of observations in each class. It is computed by counting the number of observations that fall into each class.
6. Compute the relative frequency for each class. The relative frequency is the frequency divided by the total number of observations. For example, if the frequency is 10 and the total number of observations is 100, the relative frequency is 0.10.
7. Draw the histogram. The histogram is a bar chart where the x-axis represents the class boundaries and the y-axis represents the frequency or relative frequency. The height of each bar represents the frequency or relative frequency of the class.



Although we have given a procedure for determining the number of classes, it is often desirable to let the nature of the problem determine the classes. For example, to construct a histogram describing the ages of the residents of a certain city, it might be reasonable to use classes having 10-year lengths (that is, under 10 years, 10–19 years, 20–29 years, 30–39 years, and so on). In general, when constructing a histogram, the area of the rectangle positioned over a particular class should represent the relative proportion of measurements in the class. When we use

FIGURE 2.6 Excel Output of a Frequency Histogram of 60 Package Design Ratings



equal class lengths, this can be accomplished by making the height of the rectangle over a particular class represent the relative proportion of measurements in the class (as described in the previous summary box for constructing histograms). This is because the area of a rectangle is its base multiplied by its height and because, if we are using equal class lengths, then the bases of all the rectangles over the various classes are the same.

As an example, in Figure 2.2 the height of the rectangle over the second class (the payment times from 13 to 15) is the class frequency 14, and the height of the rectangle over the first class (the payment times from 10 to 12) is the class frequency 3. These different heights tell us that the proportion of payment times in the second class, which is $14/65$, is greater than the proportion of payment times in the first class, which is $3/65$. It is desirable to use equal class lengths whenever the raw data (that is, all the actual measurements) are available. However, it is sometimes necessary to draw histograms with unequal class lengths, particularly when analyzing data published in the form of a frequency distribution having unequal class lengths. Often economic and social data are published in this form. In such a case, one must vary the rectangle heights to make the areas of the rectangles represent the relative proportions of measurements in the classes. How to do this is discussed in Exercise 2.11. Also discussed in this exercise is how to deal with open-ended classes. For example, if we are constructing a histogram describing the yearly incomes of U.S. households, an open-ended class could be households earning over \$200,000 per year.

Some common population shapes Often we construct a stem-and-leaf display or histogram for a sample to make inferences about the shape of the sampled population. Sometimes it is useful to describe the shape of a population by using a smooth curve. For instance, the stem-and-leaf display and histogram for the sample mileages look quite symmetrical and bell-shaped. Therefore, it is reasonable to infer that the population of all mileages can be described by a symmetrical, bell-shaped curve. Such a curve is shown in Figure 2.7. Several different kinds of symmetrical, bell-shaped curves are used to describe populations. One such curve is particularly useful and is called the normal curve. We can graph a specific equation to obtain the normal curve. This equation will be given and discussed in Chapter 5.

To intuitively understand the normal curve, recall from our discussion of histograms that if we use classes of equal lengths, then the height of the rectangle over a given class represents the relative proportion of measurements in the class. Similarly, the height of the normal curve over a given point represents the relative proportion of population measurements that are near the given point. For example, Figure 2.8 illustrates a normal curve describing gas mileages. We see that the normal curve's height over the point 31 mpg is greater than its height over the point 33 mpg. This tells us that the proportion of mileages near 31 mpg is greater than the proportion of mileages near 33 mpg.

Many real populations are distributed according to the symmetrical, bell-shaped normal curve. We say that such populations are normally distributed. However, instead of being

FIGURE 2.7

A Symmetrical, Bell-Shaped Curve

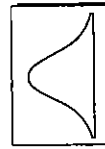


FIGURE 2.8

A Normal Curve Describing the Mileages

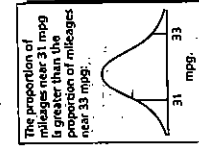


FIGURE 2.9
A Curve That Is
Skewed with a Tail
to the Right

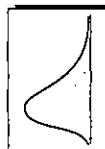


FIGURE 2.10
A Curve That Is
Skewed with a Tail
to the Left



symmetrical and bell-shaped, the overall shape of a population may be skewed with a tail to the right, as is the curve in Figure 2.9, or with a tail to the left, as is the curve in Figure 2.10. Alternatively, many other population shapes are also possible. If the stem-and-leaf display and/or histogram of a random sample of measurements looks like one of these curves, this suggests that the curve describes the overall shape of the entire population of measurements. In this case, the curve is called the *relative frequency curve* that describes the population. Said another way, the population is distributed according to the relative frequency curve. For instance, the stem-and-leaf display and histogram we constructed for the 49 mileages suggest that the population of all mileages is distributed according to a normal relative frequency curve—that is, the mileages are normally distributed. As another example, the stem-and-leaf display and histogram we constructed for the 65 payment times suggest that the population of all payment times is distributed according to a relative frequency curve that is skewed with a tail to the right. Finally, the stem-and-leaf display and histogram we constructed for the 60 package design ratings indicate that the population of all package design ratings is distributed according to a relative frequency curve that is skewed with a tail to the left. As shown in Figure 2.8, the height of any relative frequency curve over a given point represents the relative proportion of population measurements that are near the given point.

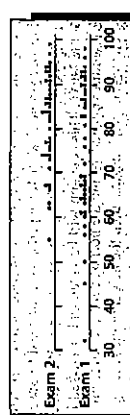
Further graphical techniques, and detecting outliers One of the authors of this book recently taught a course in business statistics to a class of 40 students. A comparison of the scores received by these students on the first two 100-point exams is given by the back-to-back stem-and-leaf display in Figure 2.11 and by the two dot plots in Figure 2.12. Note that to make each dot plot, we draw a number line on which we measure the exam scores. We then place dots above the number line to represent the exam scores. The number of dots located above a particular exam score indicates how many students received that exam score. After noticing the two-peaked appearance of the stem-and-leaf display and dot plot for Exam 1, the author investigated and found that most of the students who scored less than 70 on the exam had not been attending class regularly. Because of this, the author implemented a mandatory attendance policy. The stem-and-leaf display and dot plot for Exam 2 are single-peaked and indicate a considerable improvement in student performance. Of course, this does not prove that the attendance policy was solely responsible for the improved performance. However, many students told the author that being forced to attend class improved their test scores.

Stem-and-leaf displays and dot plots are useful for detecting outliers, which are unusually large or small observations that are well separated from the remaining observations. For example, the stem-and-leaf display and dot plot for Exam 1 indicate that the score 32 seems unusually low. How can we handle an outlier depends on its cause. If the outlier results from a measurement error or an error in recording or processing the data, it should be corrected. If the outlier cannot be corrected, it should be discarded. If an outlier is not the result of an error in measuring or recording the data, its cause may reveal important information. For example, the outlying exam score of 32 convinced the author that the student needed a tutor. After working with a tutor, the

FIGURE 2.11 A Back-to-Back Stem-and-Leaf Display

Exam 1	Exam 2
2	3*
3	3
4*	4*
5	4
6	5*
7*	5
8	6*
9	6
10	6
11	6
12	6
13	6
14	6
15	6
16	6
17	6
18	6
19	6
20	6
21	6
22	6
23	6
24	6
25	6
26	6
27	6
28	6
29	6
30	6
31	6
32	6
33	6
34	6
35	6
36	6
37	6
38	6
39	6
40	6
41	6
42	6
43	6
44	6
45	6
46	6
47	6
48	6
49	6
50	6
51	6
52	6
53	6
54	6
55	6
56	6
57	6
58	6
59	6
60	6
61	6
62	6
63	6
64	6
65	6
66	6
67	6
68	6
69	6
70	6
71	6
72	6
73	6
74	6
75	6
76	6
77	6
78	6
79	6
80	6
81	6
82	6
83	6
84	6
85	6
86	6
87	6
88	6
89	6
90	6
91	6
92	6
93	6
94	6
95	6
96	6
97	6
98	6
99	6
100	6

FIGURE 2.12 Dot Plots of the Scores on Exams 1 and 2



student showed considerable improvement on Exam 2. A more precise way to detect outliers is presented in Section 2.4.

EXERCISES FOR SECTION 2.1

CONCEPTS

- 2.1 What do each of the following population shapes look like? Explain each in words and then draw a picture to illustrate each shape.
- Symmetrical and bell-shaped.
 - Double-peaked.
 - Skewed with a tail to the left.
 - Skewed with a tail to the right.
- 2.2 Explain
- How to construct a stem-and-leaf display: a histogram; a dot plot.
 - How class limits, class boundaries, and class midpoints differ.
 - What outliers are and how they are handled.

2.3, 2.6, 2.9, 2.12

METHODS AND APPLICATIONS

- 2.3 In its January 7, 2002, issue, *Forbes* magazine published the *Forbes* Platinum List of the best-performing big companies in a variety of industry groupings, as picked by the magazine's editors. Table 2.4 gives the *Forbes* Platinum List for the retailing industry. • *Forbes*Plat

- Construct a stem-and-leaf display of the profit margin percentages for the retailers in the Platinum List. Use the whole numbers from 1 to 12 as the stems. Then describe the distribution of profit margins.
- Somewhat skewed right

TABLE 2.4 The *Forbes* Platinum List: Best-Performing Retailers as Listed in the January 7, 2002, Issue of *Forbes* Magazine • *Forbes*Plat

Company	Profit Margin (%)	Sales Growth % (Last 12 Months)	Net Income (\$ Millions) (Last 12 Months)	Return on Capital (%) (Last 12 Months)
Abercrombie & Fitch	12.5	14.7	167	39.4
American Eagle Outfitters	8.3	39.4	111	28.5
Bed Bath & Beyond	7.2	25	189	23.1
Best Buy	2.3	25.8	387	19.7
BJS Wholesale Club	1.5	11.5	81	12
Burlington Coat	2.7	8.8	65	10.3
CDW Computer Centers	4.2	11.3	169	26.5
Costco Wholesale	1.7	8.2	602	11.8
CVS	3.5	10.1	753	16.1
Dollar Tree Stores	6	21.4	112	20.3
Family Dollar Stores	5.2	17	190	22.8
Home Depot	5.5	13.7	2799	16.7
Inlight Enterprises	2.3	9.6	48	16.5
Kohl's	6.3	26.2	440	15.3
Limited	4.4	-1.2	431	17.4
Linen 'n Things	3.2	16.1	56	11.5
Lowe's Companies	4.4	18.7	946	12.6
Men's Wearhouse	4.9	3.8	65	12.7
Michaels Stores	3.3	15.1	78	10.3
Pier 1 Imports	5.9	10.7	86	15.7
RadioShack	5.7	5.3	277	23
Ross Stores	5.2	11.2	150	30.5
Sonic Automotive	1.1	7.3	70	10.3
Starbucks	6.8	22.1	181	15.4
Stein Mart	1.9	10.4	24	11.7
Target	7.7	10.3	128	20.3
Tiffany	3.2	9.6	1253	11.8
TKX Companies	10.9	-2.1	176	15.8
Wal-Mart Stores	4.7	9.6	480	32.2
Walgreen	3.1	12.9	6486	15
Williams-Sonoma	3.6	16.1	886	20.3
	2.5	17	50	10.9

Source: Reprinted by permission of *Forbes* magazine, © 2002 *Forbes*, Inc.

FIGURE 2.16 Excel Output of a Frequency Histogram of the 100 Waiting Times

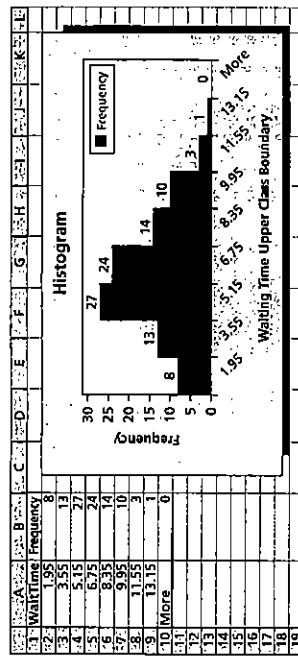


FIGURE 2.17 MINITAB Output of a Stem-and-Leaf Display of the 100 Waiting Times

Stem-and-leaf of Waittime N = 100
Leaf Unit = 0.10

```

2  0 48
   1 134588
   2 023457899
   3 12456778899
   4 00123334455567789
   (15) 5 01122334455667888
   6 11223344556778
   7 02233445789
   8 0134667
   9 123889
  10 10 279
  11 11 6

```

FIGURE 2.18 MINITAB Output of a Frequency Histogram of the 100 Waiting Times



2.7 THE TRASH BAG CASE • TrashBag

Recall that Table 1.10 (page 19) presents the breaking strengths of 40 trash bags selected during a 40-hour pilot production run. Figure 2.19 gives the MegaStat output of a relative frequency histogram and Figure 2.20 gives the MegaStat output of a stem-and-leaf display of the 40 breaking strengths. • Trashbag

- Verify that the classes and class relative frequencies given in Figure 2.19 are those obtained by using the histogram construction method discussed in this section.
- Using Figures 2.19 and 2.20, infer the shape of the relative frequency distribution describing the population of all possible trash bag breaking strengths.

2.8 Consider Table 1.9 (page 18), which gives the numbers of days needed to settle 67 homeowners' insurance claims. • ClaimSet

- Construct a stem-and-leaf display using the stems 00, 10, 20, ..., 90, 100, 110.
- Figure 2.21 gives the MINITAB output of a frequency histogram of the 67 numbers of days to settle. Verify that the classes and class frequencies used in Figure 2.21 are those obtained by using the histogram construction method discussed in this section.
- Using the stem-and-leaf display and the histogram, describe the distribution of the numbers of days to settle. What caused the distribution to look like this (see Exercise 1.16 on pages 17 and 18)?

2.9 Babe Ruth's record of 60 home runs in a single year was broken by Roger Maris, who hit 61 home runs in 1961. The yearly home run totals for Ruth in his career as a New York Yankee are (arranged in increasing order) 22, 25, 34, 35, 41, 41, 46, 46, 46, 47, 49, 54, 59, and 60. The yearly home run totals for Maris over his career in the American League are (arranged in increasing order) 8, 13, 14, 16, 21, 26, 28, 31, 39, and 61. Compare Ruth's and Maris's home run totals by constructing a back-to-back stem-and-leaf display. What would you conclude about Maris's record-breaking year?

FIGURE 2.19 MegaStat Output of a Relative Frequency Histogram of the 40 Breaking Strengths

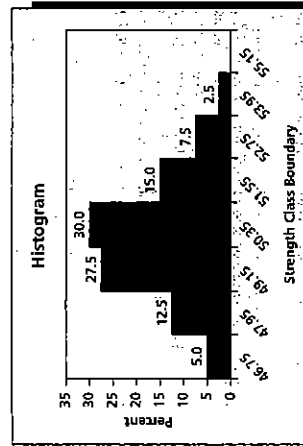
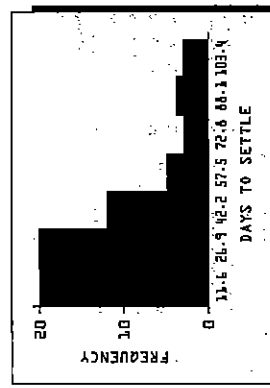


FIGURE 2.20 MegaStat Output of a Stem-and-Leaf Display of the 40 Breaking Strengths

Stem and Leaf plot for Strength
stem unit = 1 leaf unit = 0.1

Frequency	Stem	Leaf
1	46	8
0	47	5
1	47	5
2	48	23
2	48	58
4	49	0234
4	49	5689
4	50	0123
6	50	567899
5	51	01234
3	51	579
2	52	03
2	52	56
2	53	02
1	53	5
1	54	0
40		

FIGURE 2.21 MINITAB Output of a Frequency Histogram of the 67 Numbers of Days to Settle



2.10 THE CIGARETTE ADVERTISEMENT CASE • ModelAge

In an article in the *Journal of Marketing* (January 1992), Mazis, Ringold, Perry, and Denman discuss the perceived ages of models in cigarette advertisements.³ To quote the authors:

Most relevant to our study is the Cigarette Advertiser's Code, initiated by the tobacco industry in 1964. The code contains nine advertising principles related to young people, including the following provision (Advertising Age 1964): "Natural persons depicted as smokers in cigarette advertising shall be at least 25 years of age and shall not be dressed or otherwise made to appear to be less than 25 years of age."

Tobacco industry representatives have steadfastly maintained that code provisions are still being observed. A 1988 Tobacco Institute publication, "Three Decades of Initiatives by a Responsible Cigarette Industry," refers to the industry code as prohibiting advertising and promotion "directed at young people" and as "requiring that models in advertising must be, and must appear to be, at least 25 years old." John R. Nelson, Vice President of Corporate Affairs for Philip Morris, recently wrote, "We employ only adult models in our advertising who not only are but look over 25" (Nelson 1990). However, industry critics have charged that current cigarette advertising campaigns use unusually young-looking models, thereby violating the voluntary industry code.

³Source: M. B. Mazis, D. J. Ringold, E. S. Perry, and D. W. Denman, "Perceived Age and Attractiveness of Models in Cigarette Advertisements," *Journal of Marketing* 56 (January 1992), pp. 22-37.

TABLE 2.6

50 Perceived Age Estimates for a Model in a Cigarette Advertisement

Model/Age	26	30	33	27	27	32	28	19	25	29
Model/Age	31	28	24	26	29	27	28	17	28	21
Model/Age	30	28	25	31	22	29	18	27	29	23
Model/Age	28	26	24	30	27	25	26	28	20	24
Model/Age	29	32	27	17	30	27	21	29	26	28

Suppose that a sample of 50 people is randomly selected at a shopping mall. Each person in the sample is shown a typical cigarette advertisement and is asked to estimate the age of the model in the ad. Table 2.6 presents the 50 perceived ages so obtained.

a Construct (1) a stem-and-leaf display, (2) a frequency histogram, and (3) a relative frequency histogram of the 50 perceived ages.

b Using your answer in part a, infer the shape of the relative frequency distribution describing the population of the ages that would be perceived by all possible shoppers.

c What percentage of the perceived ages are below the industry's code provision (of 25 years old)? Do you think that this percentage is too high?

2.11 In this exercise we consider how to deal with class lengths that are unequal (and with open-ended classes) when setting up histograms. Often data are published in this form and we wish to construct a histogram. An example is provided by data concerning the benefits of ISO 9000 registration published by CEEM Information Services. According to CEEM:⁶

ISO 9000 is a series of international standards for quality assurance management systems. It establishes the organizational structure and processes for assuring that the production of goods or services meet a consistent and agreed-upon level of quality for a company's customers.

CEEM presents the results of a Quality Systems Update/DeLotte & Touche survey of ISO 9000-registered companies conducted in July 1993. Included in the results is a summary of the total annual savings associated with ISO 9000 implementation for surveyed companies. The findings (in the form of a frequency distribution of ISO 9000 savings) are given on the page margin. Notice that the classes in this distribution have unequal lengths and that there is an open-ended class (> \$500K).

To construct a histogram for this data, we select one of the classes as a base. It is often convenient to choose the shortest class as the base (although it is not necessary to do so). Using this choice, the 0 to \$10K class is the base. This means that we will draw a rectangle over the 0 to \$10K class having a height equal to 162 (the frequency given for this class in the published data). Because the other classes are longer than the base, the heights of the rectangles above these classes will be adjusted. Remembering that the area of a rectangle positioned over a particular class should represent the relative proportion of measurements in the class, we proceed as follows. The length of the \$10K to 25K class differs from the base class by a factor of $(25 - 10)/(10 - 0) = 3/2$, and, therefore, we make the height of the rectangle over the \$10K to 25K class equal to $(2/3)(162) = 41.333$. Similarly, the length of the \$25K to 50K class differs from the length of the base class by a factor of $(50 - 10)/(10 - 0) = 5/2$, and, therefore, we make the height of the rectangle over the \$25K to 50K class equal to $(2/5)(162) = 21.2$.

a Use the procedure just outlined to find the heights of the rectangles drawn over all the other classes (with the exception of the open-ended class, > \$500K).

b Draw the appropriate rectangles over the classes (except for > \$500K). Note that the \$250K to 500K class is a lot longer than the others. There is nothing wrong with this as long as we adjust its rectangle's height.

c We complete the histogram by placing a star (*) to the right of \$500K on the scale of measurements and by noting "37" next to the * to indicate 37 companies saved more than \$500K. Complete the histogram by doing this.

2.12 A basketball player practices free throws by taking 25 shots each day, and he records the number of shots missed each day in order to track his progress. The numbers of shots missed on days 1 through 30 are, respectively, 17, 15, 16, 13, 12, 14, 13, 11, 10, 9, 10, 9, 9, 10, 8, 10, 6, 8, 9, 7, 5, 8. Construct a stem-and-leaf display and runs plot of the numbers of missed shots. Do you think that the stem-and-leaf display is representative of the numbers of shots that the player will miss on future days? Why or why not? • Free Throw

⁶Source: CEEM Information Services, Fairfax, Virginia. © ISO 9000 for You!

2.2.1 Describing Central Tendency

The mean, median, and mode In addition to describing the shape of the distribution of a sample or population of measurements, we also describe the data set's central tendency. A measure of central tendency represents the center or middle of the data. Sometimes we think of a measure of central tendency as a typical value. However, as we will see, not all measures of central tendency are necessarily typical values.

One important measure of central tendency for a population of measurements is the population mean. We define it as follows:

The population mean, which is denoted μ and pronounced *mu*, is the average of the population measurements.

More precisely, the population mean is calculated by adding all the population measurements and then dividing the resulting sum by the number of population measurements. For instance, we consider the population of profit margins⁷ for five of the best companies in the food, drink, and tobacco industry as rated in the January 7, 2002, issue of *Forbes* magazine. The companies and profit margins (to the nearest percent) are as follows:⁸

Company	Profit Margin (Percent)
Hershey Foods	8%
PepsiCo	10
Wm. Wrigley Jr.	15
Philip Morris	12
Adolph Coors	5

The mean μ of this population of profit margins is

$$\mu = \frac{8 + 10 + 15 + 12 + 5}{5} = \frac{50}{5} = 10 \text{ percent}$$

Since this population of five profit margins is small, it is possible to compute the population mean. Often, however, a population is very large and we cannot obtain a measurement for each population unit. Therefore, we cannot compute the population mean. In such a case, we must estimate the population mean by using a sample of measurements.

In order to understand how to estimate a population mean, we must realize that the population mean is a population parameter.

A population parameter is a number calculated using the population measurements that describes some aspect of the population. That is, a population parameter is a descriptive measure of the population.

There are many population parameters, and we discuss several of them in this chapter. The simplest way to estimate a population parameter is to make what is called a point estimate.

A point estimate is a one-number estimate of the value of a population parameter.

Although a point estimate is a guess of a population parameter's value, it is not a blind guess. Rather, it is an educated guess based on sample data. One way to find a point estimate of a population parameter is to use a sample statistic.

A sample statistic is a number calculated using the sample measurements that describes some aspect of the sample. That is, a sample statistic is a descriptive measure of the sample.

The sample statistic that we use to estimate the population mean is the sample mean, which is denoted as $\bar{x}$ (pronounced *x bar*) and is the average of the sample measurements.

In order to write a formula for the sample mean, we employ the letter n to represent the number of sample measurements, and we refer to n as the sample size. Furthermore, we denote the

⁷After 12 months as of January 7, 2002.

⁸Source: "Profit Margins for Five of Best Companies," *Forbes*, January 7, 2002. Reprinted by permission of *Forbes* Magazine. © 2002 *Forbes*, Inc.

Suppose that the following rankings, arranged in increasing order, are obtained.

● VCRS₉at 1 3 5 5 7 8 8 8 8 8 9 9 9 9 10 10 10

Because the number of satisfaction ratings is even, the median of this sample is the average of the two middlemost ratings. Both of these ratings are 8—they are circled. Therefore, the median of this sample is 8, and we estimate that the median satisfaction rating of all the videocassette recorder owners is 8. This estimated median satisfaction rating seems relatively high. Note, however, that there are four rather low individual satisfaction ratings: 1, 3, 5, and 5. This suggests that some videocassette recorders may be of low quality. If the VCR manufacturer wishes to satisfy all of its customers, the company must investigate the situation.

A third measure of the central tendency of a population or sample is the mode, which is denoted M_o .

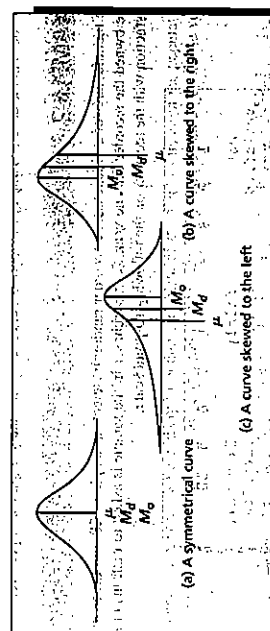
The mode, M_o , of a population or sample of measurements is the measurement that occurs most frequently.

For example, the mode of the satisfaction ratings given in Example 2.7 is 8. This is because more purchasers (six) gave the VCR a rating of 8 than any other rating. Sometimes the highest frequency occurs at two or more different measurements. When this happens, two or more modes exist. When exactly two modes exist, we say the data is *bimodal*. When more than two modes exist, we say the data is *multimodal*. Finally, when data are presented in classes (such as in a frequency histogram), the class having the highest frequency is called the *modal class*. For instance, the modal class for the histogram of mileages in Figure 2.5 (page 46) is the class from 31.55 mpg. to 32.15 mpg.

Comparing the mean, median, and mode In order to compare the mean, median, and mode, look at Figure 2.22. Part (a) of this figure depicts a population described by a symmetrical relative frequency curve. For such a population, the mean (μ), median (M_d), and mode (M_o) are all equal. Note that in this case all three of these quantities are located under the highest point of the curve. It follows that when the frequency distribution of a sample of measurements is approximately symmetrical, then the sample mean, median, and mode will be nearly the same. For instance, consider the sample of 49 mileages in Table 2.1, and note that the stem-and-leaf display of these mileages is given on the page margin. Because the number of mileages is odd, the median is the middlemost mileage, the 25th mileage. Counting 25 mileages from the top of the stem-and-leaf display, we find that the median is 31.6. Furthermore, since the stem-and-leaf display is fairly symmetrical, this sample median is approximately equal to the sample mean, which was calculated to be 31.5531.

Figure 2.22(b) depicts a population that is skewed to the right. Here the population mean is larger than the population median, and the population median is larger than the population mode

FIGURE 2.22 Relationships among the Mean μ , the Median M_d , and the Mode M_o



(the mode is located under the highest point of the relative frequency curve). In this case the population mean averages in the large values in the upper tail of the distribution. Thus the population mean is more affected by these large values than is the population median. To understand this, note that the stem-and-leaf display of the sample of 13 internists' salaries in Example 2.6 is given on the page margin and is skewed to the right. Here the mean of the 13 salaries, which is 159,769 (or \$159,769), is affected by averaging in the large salaries (192 and 241) in the right-hand tail of the distribution. Thus this mean is larger than the sample median, \$152,000. The median is said to be resistant to the large salaries 192 and 241 because the value of the median is affected only by the fact that these salaries are the two largest salaries in the sample. The value of the median is not affected by the exact sizes of the salaries 192 and 241. For example, if the largest salary were smaller—say 200—the median would remain the same but the mean would decrease. If the largest salary were larger—say 300—the median would also remain the same but the mean would increase. Therefore, the median is resistant to large values but the mean is not. Similarly, the median is resistant to values that are much smaller than most of the measurements. In general, we say that the median is resistant to extreme values.

Figure 2.22(c) depicts a population that is skewed to the left. Here the population mean is smaller than the population median, and the population median is smaller than the population mode. In this case the population mean averages in the small values in the lower tail of the distribution, and the mean is more affected by these small values than is the median. For instance, the stem-and-leaf display of the sample of purchaser satisfaction ratings in Example 2.7 is given on the page margin and is skewed to the left. In this case, the mean of these ratings, which equals 7.7, is affected by averaging in the smaller ratings (1, 3, 5, and 5) in the left-hand tail of the distribution. Thus the mean is smaller than the sample median, 8.

When a population is skewed to the right or left with a very long tail, the population mean can be substantially affected by the extreme population values in the tail of the distribution. In such a case, the population median might be better than the population mean as a measure of central tendency. For example, the yearly incomes of all people in the United States are skewed to the right with a very long tail. Furthermore, the very large incomes in this tail cause the mean yearly income to be inflated above the typical income earned by most Americans. Because of this, the median income is more representative of a typical U.S. income. The following case illustrates that the choice of the mean or the median as a measure of central tendency can depend on the purpose of the study being conducted.

EXAMPLE

In 1984 the Insurance Information Institute defended high liability insurance premiums by asserting that the mean reward in product liability cases exceeded one million dollars. Opponents of premium increases countered that the median award was only \$271,000. The mean and median differed dramatically in this case because the mean was greatly inflated by a relatively few extremely large awards. From the point of view of reporting a typical award to the public, the median is probably superior. However, from the point of view of establishing premiums, the mean is probably superior. This is because the mean helps us determine the total amount that insurance companies paid out, which is the bottom line as far as the insurance industry is concerned. For example, if the mean award of one million dollars is based on 1,000 liability cases, then the insurance industry paid out a total of one billion dollars.

When a population is symmetrical or not highly skewed, then the population mean and the population median are either equal or roughly equal, and both provide a good measure of the population central tendency. In this situation, we usually make inferences about the population mean because much of statistical theory is based on the mean rather than the median. We illustrate these ideas in the following two cases, which also show how we can obtain the mean and the median by using MINITAB, Excel, and MegaStat outputs.

EXAMPLE

The MINITAB output in Figure 2.23 tells us that the mean and the median of the sample of 65 payment times are 18.108 days and 17 days, respectively. Because the stem-and-leaf display of the payment times in Example 2.2 is not highly skewed to the right, the sample

Mileages
29 8
30 1344
30 566889
31 001233444
31 55566777889
32 0001122344
32 556788
33 3

FIGURE 2.23 MINITAB Output of Statistics Describing the 65 Payment Times

Descriptive Statistics					
Variable	N	Mean	Median	Tr Mean	St Dev
PAYTIME	65	18.108	17.000	17.963	3.961
Variable	Min	Max	Q1	Q3	
PAYTIME	10.000	29.000	15.000	21.000	

FIGURE 2.24 Excel Outputs of Statistics Describing Three Data Sets

(a) Statistics describing the 60 package design ratings		(b) Statistics describing the 13 internists' salaries		(c) Statistics describing the 20 customer satisfaction ratings	
STATISTICS		STATISTICS		STATISTICS	
Mean	30.35	Mean	150.7692	Mean	7.7
Standard Error	0.401446	Standard Error	8.498985	Standard Error	0.543381
Median	31	Median	152	Median	8
Mode	31	Mode	#N/A	Mode	8
Standard Deviation	3.107353	Standard Deviation	30.64355	Standard Deviation	2.430075
Sample Variance	9.65508	Sample Variance	939.025	Sample Variance	5.902763
Kurtosis	1.423297	Kurtosis	2.408469	Kurtosis	3.178388
Skewness	-1.17888	Skewness	1.95197	Skewness	-1.56882
Range	15	Range	114	Range	8
Minimum	20	Minimum	127	Minimum	5
Maximum	35	Maximum	241	Maximum	10
Sum	1821	Sum	2077	Sum	150
Count	60	Count	13	Count	20

mean is not much greater than the sample median. Therefore, using the mean as our measure of central tendency, we estimate that the mean payment time of all bills using the new billing system is 18.108 days. This is substantially less than the typical payment time of 39 days that had been experienced using the old billing system.

The Excel output in Figure 2.24(a) tells us that the mean and the median of the sample of 60 package design ratings are 30.35 and 31, respectively. Because the stem-and-leaf display of the package design ratings in Example 2.3 is not highly skewed to the left, the sample mean is not much less than the sample median. Therefore, using the mean as our measure of central tendency, we estimate that the mean rating of the new package design that would be given by all consumers is 30.35. This is considerably higher than the minimum standard of 25 for a successful package design.

To conclude this example, note that Figures 2.24(b), 2.24(c), and 2.25 give the Excel and MegaStat outputs of the previously discussed means and medians for the internists' salaries, customer satisfaction ratings, and mileages. Other quantities on the outputs will be discussed as we proceed through this chapter.

To conclude this section, note that the mean and the median convey useful information about a population having a relative frequency curve with a sufficiently regular shape. For instance, the mean and median would be useful in describing the mound-shaped, or single-peaked, distributions in Figure 2.22. However, these measures of central tendency do not adequately describe a double-peaked distribution. For example, the mean and the median of the exam scores in the double-peaked stem-and-leaf display of Figure 2.11 (page 48) are 75.225 and 77. Looking at the display, neither the mean nor the median represents a typical exam score. This is because the exam scores really have *no central value*. In this case the most important message conveyed by the double-peaked stem-and-leaf display is that the exam scores fall into two distinct groups.

FIGURE 2.25 MegaStat Output of Statistics Describing the 49 Mileages

Descriptive statistics		empirical rule	
count	49	mean - 1s	30.754
mean	31.553	mean + 1s	32.352
standard deviation	4.535	mean - 2s	31.516
sample variance	20.566	mean + 2s	29.895
minimum	29.8	mean - 3s	31.152
maximum	31.3	mean + 3s	29.155
range	1.5	percent in interval (99.73%)	33.951
sum	1546.100	percent in interval (99.73%)	33.951
sum of squares	48114.850		100.0%
deviation term of squares (SS)	30.562		
1st quartile	31.000	low extremes	0
median	31.600	low outliers	0
3rd quartile	32.100	high outliers	0
interquartile range	1.100	high extremes	0
mode	31.700		

EXERCISES FOR SECTION 2.2

CONCEPTS

- 2.13 Explain the difference between each of the following:
 a. A population parameter and its point estimate.
 b. A population mean and a corresponding sample mean.
- 2.14 Explain how the population mean, median, and mode compare when the population's relative frequency curve is
 a. Symmetrical.
 b. Skewed with a tail to the left.
 c. Skewed with a tail to the right.
 d. Normally distributed.

METHODS AND APPLICATIONS

- 2.15 Calculate the mean, median, and mode of each of the following populations of numbers:
 a. 9, 8, 10, 10, 12, 6, 11, 10, 12, 8
 b. 110, 120, 70, 90, 100, 80, 130, 140
- 2.16 THE CUSTOMER SATISFACTION RATING CASE • CrstSat
 Recall that Table 1.5 (page 11) presents the satisfaction ratings for the VAC-5000 vacuum cleaner that have been given by 65 randomly selected purchasers. Figures 2.26 and 2.29(a) give the MINITAB and MegaStat outputs of statistics describing the 65 satisfaction ratings.
 a. Does the sample mean $\bar{x} = 42.954$ provide evidence that the mean of the population of all possible customer satisfaction ratings for the VAC-5000 exceeds 42? (Recall that a "very satisfied" customer gives a rating that exceeds 42.) Explain your answer.
 b. Use the stem-and-leaf display in Figure 2.14 (page 51) to verify that the median of the satisfaction ratings is 43. How do the mean and median compare? What does the stem-and-leaf display tell you about why they compare this way?
- 2.17 THE BANK CUSTOMER WAITING TIME CASE • WaitTime
 Recall that Table 1.6 (page 11) presents the waiting times for teller service during peak business hours of 100 randomly selected bank customers. Figures 2.27 and 2.29(b) give the MINITAB and MegaStat outputs of statistics describing the 100 waiting times.
 a. Does the sample mean $\bar{x} = 5.46$ provide evidence that the mean of the population of all possible customer waiting times during peak business hours is less than six minutes (as is desired by the bank manager)? Explain your answer.
 b. Use the stem-and-leaf display in Figure 2.17 (page 52) to verify that the median of the waiting times is 5.25. How do the mean and median compare? What does the stem-and-leaf display tell you about why they compare this way?
- 2.18 THE TRASH BAG CASE • TrashBag
 Consider the trash bag problem. Suppose that an independent laboratory has tested 30-gallon trash bags and has found that none of the 30-gallon bags currently on the market have a mean breaking strength of 50 pounds or more. On the basis of these results, the producer of the new, improved



FIGURE 2.26 MINITAB Output of Statistics Describing the 65 Satisfaction Ratings

Variable	N	Mean	Median	Tr Mean	StDev	SE Mean
RATING	65	42.954	43.000	43.017	2.642	0.328
Variable	Min	Max	Q1	Q3		
RATING	36.000	46.000	41.000	45.000		

FIGURE 2.27 MINITAB Output of Statistics Describing the 100 Waiting Times

Variable	N	Mean	Median	Tr Mean	StDev	SE Mean
WAITTIME	100	5.460	5.250	5.419	2.475	0.248
Variable	Min	Max	Q1	Q3		
WAITTIME	0.400	11.600	3.800	7.200		

FIGURE 2.28 MINITAB Output of Statistics Describing the 40 Breaking Strengths

Variable	N	Mean	Median	Tr Mean	StDev	SE Mean
STRENGTH	40	50.575	50.650	50.589	1.644	0.260
Variable	Min	Max	Q1	Q3		
STRENGTH	46.800	54.000	49.425	51.650		

trash bag feels sure that its 30-gallon bag will be the strongest such bag on the market; if the new trash bag's mean breaking strength can be shown to be at least 50 pounds. Recall that Table 1.10 (page 19) presents the breaking strengths of 40 trash bags of the new type that were selected during a 40-hour pilot production run. Figures 2.28 and 2.29(c) give the MINITAB and MegaStat outputs of statistics describing the 40 breaking strengths.

Yes, $\bar{x} > 50$ Very similar
Little skew

- (a) Two EPS growths much larger than the others
(b) Fairly symmetrical distribution with no obvious outliers

Beck's 2000 magazine, which is a leading source of information in Table 2.9, which lists data concerning lifestyles in the United States and eight other countries. In each exercise (a) compute the appropriate mean and median; (b) compare the mean and median and explain what they say about skewness; (c) construct a dot plot and discuss what the dot plot says about skewness and whether this agrees with how the mean and median compare; (d) discuss how the United States compares to the mean and median. • LifeStyle

2.20 Analyze the data concerning voters in Table 2.7 as described above. • LifeStyle

2.21 Analyze the data concerning income tax rates in Table 2.7 as described above. • LifeStyle

2.22 Analyze the data concerning video rentals in Table 2.7 as described above. • LifeStyle

2.23 Analyze the data concerning PCs in Table 2.7 as described above. • LifeStyle

2.24 Analyze the data concerning religion in Table 2.7 as described above. • LifeStyle

2.25 Table 2.8 gives the number of unique visitors per month to the top 10 websites as given in the September 28, 1998, issue of *Fortune*. Compute the mean and median for the website data and compare them. What do they say about skewness? • WebVisit

16.91; 16.8

2.26 Miller's and Albert's are two large supermarket chains in a southern state. Each claims to charge the lowest prices in the state. The state's Department of Consumer Affairs hires a marketing

FIGURE 2.29 MegaStat Outputs of Statistics Describing Three Data Sets

(a) Satisfaction ratings statistics									
Descriptive statistics									
count	65	Waiting time	100	Breaking strength	40	count	65	Waiting time	100
mean	42.954	mean	5.460	mean	50.575	mean	50.575	mean	5.460
median	43.000	median	5.250	median	50.650	median	50.650	median	5.250
sample variance	6.98	sample variance	6.178	sample variance	2.702	sample variance	6.178	sample variance	2.702
sample standard deviation	2.64	sample standard deviation	2.475	sample standard deviation	1.644	sample standard deviation	2.475	sample standard deviation	1.644
minimum	36	minimum	0.4	minimum	46.8	minimum	36	minimum	0.4
maximum	46	maximum	11.6	maximum	54	maximum	46	maximum	11.6
range	10	range	11.2	range	7.2	range	10	range	11.2
empirical rule		empirical rule		empirical rule		empirical rule		empirical rule	
mean - 1s	40.31	mean - 1s	2.985	mean - 1s	48.931	mean - 1s	40.31	mean - 1s	2.985
mean + 1s	45.60	mean + 1s	7.935	mean + 1s	52.219	mean + 1s	45.60	mean + 1s	7.935
percent in interval (68.26%)	61.1%	percent in interval (68.26%)	66.0%	percent in interval (68.26%)	67.3%	percent in interval (68.26%)	61.1%	percent in interval (68.26%)	67.3%
mean - 2s	37.67	mean - 2s	0.509	mean - 2s	47.287	mean - 2s	37.67	mean - 2s	0.509
mean + 2s	48.24	mean + 2s	10.411	mean + 2s	53.863	mean + 2s	48.24	mean + 2s	10.411
percent in interval (95.44%)	98.5%	percent in interval (95.44%)	96.0%	percent in interval (95.44%)	95.0%	percent in interval (95.44%)	98.5%	percent in interval (95.44%)	95.0%
mean - 3s	33.03	mean - 3s	-1.566	mean - 3s	45.664	mean - 3s	33.03	mean - 3s	-1.566
mean + 3s	52.88	mean + 3s	12.886	mean + 3s	55.506	mean + 3s	52.88	mean + 3s	12.886
percent in interval (99.73%)	100.0%	percent in interval (99.73%)	100.0%	percent in interval (99.73%)	100.0%	percent in interval (99.73%)	100.0%	percent in interval (99.73%)	100.0%

TABLE 2.7 Data Comparing Lifestyles in the U.S. and Eight Other Countries • LifeStyle

	Voters	Income Tax	Religion	Websites as Given
	Percentage Who Voted in Last National Election	Highest Personal National Rate	Percentage of Households Who Attend Services Regularly	in the September 28, 1998, Issue of Fortune
U.S.	49.1%	40%	51.6%	Fortune
Germany	82.2	56	20.0	WebVista
France	68.9	54	N.A.	Unique Visitors per Month (in Millions)
Britain	71.5	40	23.6	26.6
Netherlands	78.3	60	28.9	YAHOO!
Sweden	78.6	55	10.0	AOL.COM
Italy	85.0	46	11.5	22.0
Japan	58.8	50	55.8	18.6
South Korea	63.9	44	N.A.	Microsoft

N.A.: Not available.

Source: "America vs. The New Europe," *The New York Times*, December 21, 1998, p. 156. Reprinted from the December 21, 1998, issue of *Fortune*, copyright 1998 Time, Inc. Reprinted by permission.TABLE 2.8 Top 10 Websites as Given in the September 28, 1998, Issue of *Fortune* • WebVisit

Website	Unique Visitors per Month (in Millions)
YAHOO!	26.5
AOL.COM	22.0
Microsoft	18.6
Netscape	17.6
Excite	17.3
MSN	16.3
LYCOS	14.6
GeoCities	14.2
InfoSeek	12.2
Search.nl	9.7

Source: "Midway Online," by Frank Rose, Reprinted from the September 28, 1998, issue of *Fortune*, p. 274, copyright 1998 Time, Inc. Reprinted by permission.

research firm to independently study the prices charged by the two chains. The firm, using a standardized one-week shopping plan (grocery list), makes identical purchases at 10 of each chain's stores. The stores for each chain are randomly selected, and all purchases are made during a single week. The costs of each shopping exercise are given in Table 2.9. • ShopExp

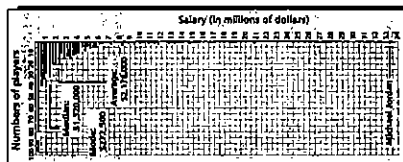
a Use the grocery data to compute a point estimate of the population mean cost of the one-week shopping plan at each chain.

b Based on these point estimates, which chain charges lower prices for the one-week shopping plan?

c Carefully define the population mean cost of the one-week shopping plan at Miller's; at Albert's.

2.27 In 1998 the National Basketball Association (NBA) experienced a labor dispute that cancelled almost half of the professional basketball season. The NBA owners, who were worried about escalating salaries because several star players had recently signed huge contracts, locked out the players and demanded that a salary cap be established. This led to discussion in the media about excessive player salaries. On October 30, 1998, an article titled "What does average salary really mean in the NBA?" by Jonathan Sills appeared in his Behind the Numbers column on the ESPN.com website. The article discussed the validity of some media claims about NBA player

FIGURE 2.30
1997–98 NBA Salaries



Source: Reprinted courtesy of ESPN.

- a Skewed right
- b About 33%
- c About 50%

TABLE 2.9
Comparison of Weekly Shopping Expenses
at Two Supermarket Chains • ShopEx

Store	Miller's Cost	Albert's Cost
1	\$119.25	\$111.99
2	\$121.32	\$114.88
3	\$122.36	\$115.11
4	\$120.14	\$117.02
5	\$122.19	\$116.89
6	\$123.71	\$116.62
7	\$121.72	\$115.38
8	\$122.42	\$114.40
9	\$123.63	\$113.91
10	\$122.44	\$111.87

salaries. Figure 2.30 shows a frequency distribution of NBA salaries as presented in the Sills article. Use the frequency distribution to do the following:

- a Compare the mean, median, and mode of the salaries and explain the relationship. Note that the minimum NBA salary at the time of the lockout was \$272,500.
- b Noting that 411 NBA players were under contract, estimate the percentage of players who earned more than the mean salary; more than the median salary.
- c Below we give three quotes from news stories cited by Sills in this article. Comment on the validity of each statement.

"Last year, the NBA middle class made an average of \$2.6 million. On that scale, I'd take the NBA lower class." —*Houston Chronicle*
 "The players make an obscene amount of money—the median salary is well over \$2 million!" —*St. Louis Post Dispatch*
 "The players want us to believe they literally can't 'survive' on \$2.6 million a year, the average salary in the NBA." —*Washington Post*

2.3 ■ Measures of Variation

Range, variance, and standard deviation In addition to estimating a population's central tendency, it is important to estimate the variation of the population's individual values. For example, Figure 2.31 shows two histograms. Each portrays the distribution of 20 repair times (in days) for personal computers at a major service center. Because the mean (and median and mode) of each distribution equals four days, the measures of central tendency do not indicate any difference between the American and National Service Centers. However, the repair times for the American Service Center are clustered quite closely together, whereas the repair times for the National Service Center are spread farther apart (the repair time might be as little as one day, but could also be as long as seven days). Therefore, we need measures of variation to express how the two distributions differ.

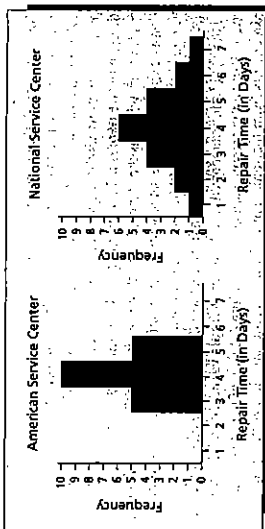
One way to measure the variation of a set of measurements is to calculate the range.

Consider a population or a sample of measurements. The range of the measurements is the largest measurement minus the smallest measurement.

In Figure 2.31, the smallest and largest repair times for the American Service Center are three days and five days; therefore, the range is $5 - 3 = 2$ days. On the other hand, the range for the National Service Center is $7 - 1 = 6$ days. The National Service Center's larger range indicates that this service center's repair times exhibit more variation.

In general, the range is not the best measure of a data set's variation. One reason is that it is based on only the smallest and largest measurements in the data set and therefore may reflect an extreme measurement that is not entirely representative of the data set's variation. For example, in the marketing research case, the smallest and largest ratings in the sample of 60 package design ratings are 20 and 35. However, to simply estimate that most package design ratings are between 20 and 35 misses the fact that 57, or 95 percent, of the 60 ratings are at least as large as the minimum rating of 25 for a successful package design. In general, to fully describe a population's variation, it is useful to estimate intervals that contain different percentages (for example,

FIGURE 2.31 Repair Times for Personal Computers at Two Service Centers



70 percent, 95 percent, or almost 100 percent) of the individual population values. To estimate such intervals, we use the population variance and the population standard deviation.

The Population Variance and Standard Deviation

The population variance σ^2 (pronounced *sigma squared*) is the average of the squared deviations of the individual population measurements from the population mean μ .

The population standard deviation σ (pronounced *sigma*) is the positive square root of the population variance.

For example, consider again the population of profit margins for five of the best companies in the food, drink, and tobacco industry as rated in the January 7, 2002, issue of *Forbes* magazine. These profit margins are 8%, 10%, 15%, 12%, and 5%. To calculate the variance and standard deviation of these profit margins, we first calculate the population mean to be

$$\mu = \frac{8 + 10 + 15 + 12 + 5}{5} = \frac{50}{5} = 10$$

Next, we calculate the deviations of the individual population measurements from the population mean $\mu = 10$ as follows:

$$(8 - 10) = -2 \quad (10 - 10) = 0 \quad (15 - 10) = 5 \quad (12 - 10) = 2 \quad (5 - 10) = -5$$

Then we compute the sum of the squares of these deviations:

$$(-2)^2 + (0)^2 + (5)^2 + (2)^2 + (-5)^2 = 4 + 0 + 25 + 4 + 25 = 58$$

Finally, we calculate the population variance σ^2 , the average of the squared deviations, by dividing the sum of the squared deviations, 58, by the number of squared deviations, 5. That is, σ^2 equals $58/5 = 11.6$. Furthermore, this implies that the population standard deviation σ —the positive square root of σ^2 —is $\sqrt{11.6} = 3.406$.

To see that the variance and standard deviation measure the variation, or spread, of the individual population measurements, suppose that the measurements are spread far apart. Then, many measurements will be far from the mean μ , many of the squared deviations from the mean will be large, and the sum of squared deviations will be large. It follows that the average of the squared deviations—the population variance—will be relatively large. On the other hand, if the population measurements are clustered close together, many measurements will be close to μ , many of the squared deviations from the mean will be small, and the average of the squared deviations—the population variance—will be small. Therefore, the more spread out the population measurements, the larger is the population variance, and the larger is the population standard deviation.

To further understand the population variance and standard deviation, note that one reason we square the deviations of the individual population measurements from the population mean is that the sum of the raw deviations themselves is zero. This is because the negative deviations cancel the positive deviations. For example, in the profit margin situation, the raw deviations are

-2, 0, 5, 2, and -5, which sum to zero. Of course, we could make the deviations positive by finding their absolute values. We square the deviations instead because the resulting population variance and standard deviation have many important interpretations that we study throughout this book. Since the population variance is an average of squared deviations of the original population values, the variance is expressed in squared units of the original population values. On the other hand, the population standard deviation—the square root of the population variance—is expressed in the same units as the original population values. For example, the previously discussed profit margins are expressed in percentages. Therefore, the variance of these profit margins is $\sigma^2 = 11.6(\%)^2$, whereas the standard deviation is $\sigma = 3.406\%$. Since the population standard deviation is expressed in the same units as the population values, it is more often used to make practical interpretations about the variation of these values.

When a population is too large to measure all the population units, we estimate the population variance and the population standard deviation by the sample variance and the sample standard deviation. We calculate the sample variance by dividing the sum of the squared deviations of the sample measurements from the sample mean by $n - 1$, the sample size minus one. Although we might intuitively think that we should divide by n rather than $n - 1$, it can be shown that dividing by n tends to produce an estimate of the population variance that is too small. On the other hand, dividing by $n - 1$ tends to produce a larger estimate that we will show in Chapter 6 is more appropriate. Therefore, we obtain:

The Sample Variance and the Sample Standard Deviation

The sample variance s^2 (pronounced *s squared*) is defined to be

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}$$

and is the point estimate of the population variance σ^2 .

The sample standard deviation s , $\sqrt{s^2}$, is the positive square root of the sample variance and is the point estimate of the population standard deviation σ .

To illustrate the calculation of the sample variance and standard deviation, we begin by considering the first five mileages in Table 2.1 (page 40): $x_1 = 30.8$, $x_2 = 31.7$, $x_3 = 30.1$, $x_4 = 31.6$, and $x_5 = 32.1$. Since the mean of these five mileages is $\bar{x} = 31.26$, it follows that

$$\begin{aligned} \sum_{i=1}^5 (x_i - \bar{x})^2 &= (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 + (x_4 - \bar{x})^2 + (x_5 - \bar{x})^2 \\ &= (30.8 - 31.26)^2 + (31.7 - 31.26)^2 + (30.1 - 31.26)^2 \\ &\quad + (31.6 - 31.26)^2 + (32.1 - 31.26)^2 \\ &= (-.46)^2 + (.44)^2 + (-1.16)^2 + (.34)^2 + (.84)^2 \\ &= 2.572 \end{aligned}$$

Therefore, the variance and the standard deviation of the sample of the first five mileages are

$$s^2 = \frac{2.572}{5 - 1} = .643 \quad \text{and} \quad s = \sqrt{.643} = .8019$$

Of course, intuitively, we are likely to obtain more accurate point estimates of the population variance and standard deviation by using all the available sample information. Recall that the mean of all 49 mileages is $\bar{x} = 31.5531$. Using this sample mean, it can be verified that

$$\begin{aligned} \sum_{i=1}^{49} (x_i - \bar{x})^2 &= (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_{49} - \bar{x})^2 \\ &= (30.8 - 31.5531)^2 + (31.7 - 31.5531)^2 + \cdots + (32.2 - 31.5531)^2 \\ &= (-.7531)^2 + (.1469)^2 + \cdots + (.6469)^2 \\ &= 30.66204 \end{aligned}$$

Therefore, the variance and the standard deviation of the sample of 49 mileages are

$$s^2 = \frac{30.66204}{48} = .638793 \quad \text{and} \quad s = \sqrt{.638793} = .7992$$

Notice that the MegaStat output in Figure 2.25 (page 61) gives these quantities (rounded to three decimal places). Here $s^2 = .6388$ and $s = .7992$ are the point estimates of the variance, σ^2 , and the standard deviation, σ , of the population of the mileages of all the cars that will be or could potentially be produced. Furthermore, the sample standard deviation is expressed in the same units as the sample values. Therefore $s = .7992$ mpg.

Before explaining how we can use s^2 and s in a practical way, we present a formula that makes it easier to compute s^2 . This formula is useful when we are using a handheld calculator that is not equipped with a statistics mode to compute s^2 .

The sample variance can be calculated using the computational formula

$$s^2 = \frac{1}{n - 1} \left[\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i \right)^2}{n} \right]$$

Consider the sample of 65 payment times in Table 1.2 (page 6). Using this data, it can be verified that

$$\begin{aligned} \sum_{i=1}^{65} x_i &= x_1 + x_2 + \cdots + x_{65} = 22 + 19 + \cdots + 21 = 1,177 \quad \text{and} \\ \sum_{i=1}^{65} x_i^2 &= x_1^2 + x_2^2 + \cdots + x_{65}^2 = (22)^2 + (19)^2 + \cdots + (21)^2 = 22,317 \end{aligned}$$

Therefore,

$$\begin{aligned} s^2 &= \frac{1}{(65 - 1)} \left[22,317 - \frac{(1,177)^2}{65} \right] \\ &= \frac{1,004.2464}{64} = 15.69135 \end{aligned}$$

and $s = \sqrt{s^2} = \sqrt{15.69135} = 3.9612$ days (see the MINITAB output in Figure 2.23 on page 60).

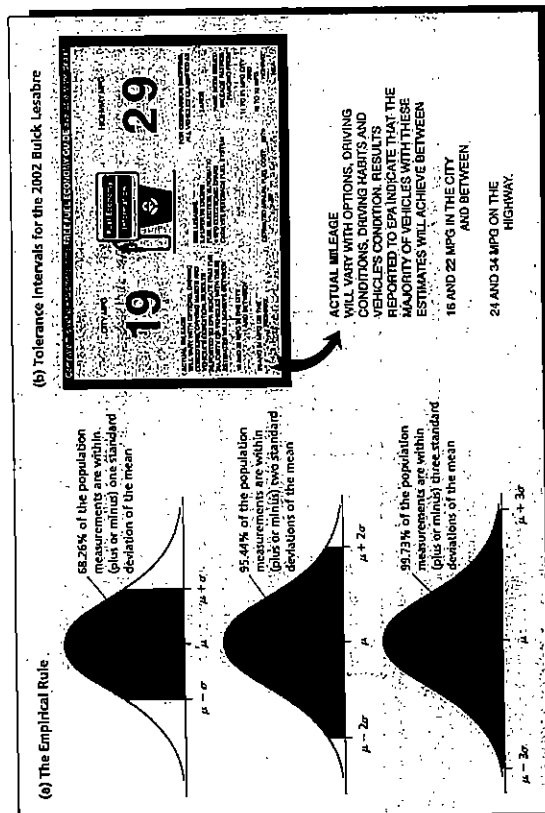
A practical interpretation: the Empirical Rule In the next box we give a practical interpretation of the population standard deviation. This interpretation is often referred to as the *Empirical Rule for a normally distributed population*.

1. If a population has mean μ and standard deviation σ , then approximately 68% of the population measurements will fall within one standard deviation of the mean, that is, between $\mu - \sigma$ and $\mu + \sigma$.

2. Approximately 95% of the population measurements will fall within two standard deviations of the mean, that is, between $\mu - 2\sigma$ and $\mu + 2\sigma$.

3. Approximately 99.7% of the population measurements will fall within three standard deviations of the mean, that is, between $\mu - 3\sigma$ and $\mu + 3\sigma$.

FIGURE 2.32 The Empirical Rule and Tolerance Intervals



In general, an interval that contains a specified percentage of the individual measurements in a population is called a **tolerance interval**. It follows that the one, two, and three standard deviation intervals around μ given in (1), (2), and (3) are tolerance intervals containing, respectively, 68.26 percent, 95.44 percent, and 99.73 percent of the measurements in a normally distributed population. Often we interpret the *three-sigma interval* [$\mu \pm 3\sigma$] to be a tolerance interval that contains *almost all* of the measurements in a normally distributed population. Of course, we usually do not know the true values of μ and σ . Therefore, we must estimate the tolerance intervals by replacing μ and σ in these intervals by the mean $\bar{x}$ and standard deviation s of a sample that has been randomly selected from the normally distributed population.

Again consider the sample of 49 mileages. We have seen that $\bar{x} = 31.5531$ and $s = .7992$ for this sample are the point estimates of the mean μ and the standard deviation σ of the population of all mileages. Furthermore, the stem-and-leaf display and histogram of the 49 mileages suggest that the population of all mileages is normally distributed. To more simply illustrate the Empirical Rule, we will round $\bar{x}$ to 31.6 and s to .8. It follows that, using the interval

- 1 $[\bar{x} \pm s] = [31.6 \pm .8] = [30.8, 32.4]$, we estimate that 68.26 percent of all individual cars will obtain mileages between 30.8 mpg and 32.4 mpg.
- 2 $[\bar{x} \pm 2s] = [31.6 \pm 1.6] = [30.0, 33.2]$, we estimate that 95.44 percent of all individual cars will obtain mileages between 30.0 mpg and 33.2 mpg.
- 3 $[\bar{x} \pm 3s] = [31.6 \pm 2.4] = [29.2, 34.0]$, we estimate that 99.73 percent of all individual cars will obtain mileages between 29.2 mpg and 34.0 mpg.

Since the difference between the upper and lower limits of each tolerance interval is fairly small, we might conclude that the variability of the individual car mileages around the estimated mean mileage of 31.6 mpg is fairly small. Furthermore, the interval $[\bar{x} \pm 3s] = [29.2, 34.0]$ implies

that almost any individual car that a customer might purchase this year will obtain a mileage between 29.2 mpg and 34.0 mpg.

Before continuing, recall that we have rounded $\bar{x}$ and s to one decimal point accuracy in order to simplify our initial example of the Empirical Rule. If, instead, we calculate the Empirical Rule intervals by using $\bar{x} = 31.5531$ and $s = .7992$ and then round the interval endpoints to one decimal place accuracy at the end of the calculations, we obtain the same intervals as obtained above. In general, however, rounding intermediate calculated results can lead to inaccurate final results. Because of this, throughout this book we will avoid greatly rounding intermediate results.

We next note that if we actually count the number of the 49 mileages in Table 2.1 that are contained in each of the intervals $[\bar{x} \pm s] = [30.8, 32.4]$, $[\bar{x} \pm 2s] = [30.0, 33.2]$, and $[\bar{x} \pm 3s] = [29.2, 34.0]$, we find that these intervals contain, respectively, 33, 47, and 49 of the 49 mileages. The corresponding sample percentages—67.35 percent, 95.92 percent, and 100 percent—are close to the theoretical percentages—68.26 percent, 95.44 percent, and 99.73 percent—that apply to a normally distributed population. This is further evidence that the population of all mileages is (approximately) normally distributed and thus that the Empirical Rule holds for this population.

To conclude this example, we note that the automaker has studied the combined city and highway mileages of the new model, because the federal tax credit is based on these combined mileages. When reporting fuel economy estimates for a particular car model to the public, however, the EPA realizes that the proportions of city and highway driving vary from purchaser to purchaser. Therefore, the EPA requires automakers to perform separate studies of city and highway mileages. Figure 2.32(b) presents a window sticker that summarizes the city and highway mileage estimates for the 2002 Buick Lesabre equipped with a six-cylinder engine and an automatic transmission. The city mpg of 19 and the highway mpg of 29 given at the top of the sticker are point estimates of, respectively, the mean city mileage and the mean highway mileage that would be obtained by all such 2002 Lesabres. The intervals "between 16 and 22 mpg in the city" and "between 24 and 34 mpg on the highway" given at the bottom left of the sticker are estimates of tolerance intervals for a "majority" of these Lesabres. Here, although *The Fuel Economy Guide—2002 Model Year* does not specify the exact percentage that defines a majority, it does say that a majority should be interpreted to mean "most." Therefore, the first interval estimates that most Lesabres will obtain between 16 mpg and 22 mpg in city driving. Similarly, the second interval estimates that most Lesabres will obtain between 24 mpg and 34 mpg in highway driving.

A quality improvement application: meeting customer requirements Tolerance intervals are often used to determine whether customer requirements are being met. Customer requirements often specify that a quality characteristic must be inside an acceptable range of values called *specifications*. Specifications are written for *individual measurements*. For example, suppose that marketing research done by the fast-food restaurant of Example 1.3 (page 12) suggests that coffee tastes best if its temperature is between 153° and 167°. Therefore, the customer requirements (specifications) would say that the temperature of each individual cup of coffee must be between 153° and 167° (this specification would typically be written as $160^\circ \pm 7^\circ$).

If a process is able to consistently produce output that meets customer requirements (specifications), we say that the process is *capable* (of meeting the requirements). From a practical standpoint this means that *almost all* of the individual measurements must be within the specification limits. Furthermore, if the population of all process measurements is approximately normally distributed, it is common practice to conclude that a process that is in statistical control is capable of meeting customer requirements if the three-sigma tolerance interval estimate $[\bar{x} \pm 3s]$ is within the specification limits. We say this because, if this interval is within the specification limits, then we estimate that almost all (99.73 percent) of the process measurements are within the specification limits.

For example, recall that the runs plot of the 24 coffee temperatures in Figure 1.3 (page 14) indicates that the fast-food restaurant's coffee-making process is in statistical control. The mean and the standard deviation of these temperatures are $\bar{x} = 160.0833$ and $s = 5.3724$, and a stem-and-leaf display indicates that the temperatures are approximately normally distributed. It follows that we estimate that the interval

$$[\bar{x} \pm 3s] = [160.0833 \pm 3(5.3724)] \\ = [143.97, 176.20]$$



contains 99.73 percent of all coffee temperatures. Since this interval tells us that some coffee temperatures are outside the customer specifications of 153° to 167° , the coffee-making process is not capable of meeting customer requirements. Here, although the process is exhibiting a constant amount of variation around a constant level (or mean), the constant amount of variation—as indicated by the standard deviation of $s = 5.3724$ —is too large. Suppose that to reduce the standard deviation of the coffee temperatures, the restaurant tests a new coffeemaker. A sample of 24 coffee temperatures is in control and approximately normally distributed with a mean of $\bar{x} = 160.0174$ and a reduced standard deviation of $s = 2.1215$. It follows that we estimate that the interval

$$\begin{aligned} [\bar{x} \pm 3s] &= [160.0174 \pm 3(2.1215)] \\ &= [153.65, 166.38] \end{aligned}$$

contains 99.73 percent of all coffee temperatures. We infer that almost all coffee temperatures produced by the new coffee-making process are within the customer specifications of 153° to 167° . Therefore, the improved process is capable of meeting customer requirements.

Skewness and the Empirical Rule The Empirical Rule holds for normally distributed populations. In addition, this rule also approximately holds for populations having mound-shaped (single-peaked) distributions that are not very skewed to the right or left. For example, recall that the distribution of 65 payment times is somewhat but not highly skewed to the right. Moreover, the mean and the standard deviation of the payment times can be calculated to be $\bar{x} = 18.1077$ and $s = 3.9612$. If we actually count the number of payment times that are contained in each of the intervals $[\bar{x} \pm s] = [14.1, 22.1]$, $[\bar{x} \pm 2s] = [10.2, 26.0]$, and $[\bar{x} \pm 3s] = [6.2, 30.0]$, we find that these intervals contain, respectively, 45, 62, and 65 of the 65 payment times. The corresponding sample percentages—69.23 percent, 95.38 percent, and 100 percent—are close to the theoretical percentages—68.26 percent, 95.44 percent, and 99.73 percent—given by the Empirical Rule. Therefore, we have evidence that the Empirical Rule approximately holds for the population of all bill payment times. It follows that we estimate that (1) 68.26 percent of all bill payment times will be between 14 days and 22 days, (2) 95.44 percent will be between 10 days and 26 days, and (3) 99.73 percent (that is, almost all) will be between 6 days and 30 days. These results indicate that the variability of the individual bill payment times around the estimated mean of 18.1077 days is reasonably small (note that, in making this conclusion, we take into account the fact that any population of bill payment times will exhibit some variability). Furthermore, because almost all bill payment times will be substantially less than the 39 days typical of the old billing system, we conclude that the new, computer-based electronic billing system is a substantial improvement.

In some situations, the skewness of a mound-shaped distribution of population measurements can make it tricky to know whether and how to use the Empirical Rule. For example, we previously concluded that the distribution of 60 package design ratings is somewhat but not highly skewed to the left. The mean and the standard deviation of the 60 package design ratings are $\bar{x} = 30.35$ and $s = 3.1073$. If we actually count the number of ratings contained in each of the intervals $[\bar{x} \pm s] = [27.2, 33.5]$, $[\bar{x} \pm 2s] = [24.1, 36.6]$, and $[\bar{x} \pm 3s] = [21, 39.7]$, we find that these intervals contain, respectively, 44, 57, and 59 of the 60 ratings. The corresponding sample percentages—73.33 percent, 95 percent, and 98.33 percent—are, respectively, greater than, approximately equal to, and less than the theoretical percentages—68.26 percent, 95.44 percent, and 99.73 percent—given by the Empirical Rule. Therefore, if we consider the population of all consumer ratings of the package design, we might estimate that (1) at least 68.26 percent of all ratings will be between 27 and 34, (2) approximately 95.44 percent of the ratings will be between 24 and 35 (a rating cannot exceed 35), and (3) fewer than 99.73 percent of the ratings will be between 21 and 35. Result (3), and the low ratings 20 and 22 in the left-hand tail of the stem-and-leaf display in Example 2.3 (page 42), suggest that the package design ratings distribution is too skewed to the left to use the Empirical Rule to make conclusions about almost all ratings. However, we are not necessarily concerned about almost all ratings, because the package design will help sell cookies if it appeals to a large percentage of consumers. Results (1) and (2), which describe 68.26 percent and 95.44 percent of all consumer

ratings, imply that large percentages of consumer ratings will exhibit reasonably small variability around the estimated mean rating of 30.35. This, and the fact that 57, or 95 percent, of the 60 ratings are at least as large as the minimum rating of 25 for a successful package design, suggest that the package design will be successful.

Chebyshev's Theorem If we fear that the Empirical Rule does not hold for a particular population, we can consider using Chebyshev's Theorem to find an interval that contains a specified percentage of the individual measurements in the population.

Consider a population that has mean μ and standard deviation σ . Then, for any value k of 2 or greater, at least $100(1 - 1/k^2)\%$ of the population measurements lie in the interval $[\mu - k\sigma, \mu + k\sigma]$.

For example, if we choose k equal to 2, then at least $100(1 - 1/2^2)\% = 100(3/4)\% = 75\%$ of the population measurements lie in the interval $[\mu \pm 2\sigma]$. As another example, if we choose k equal to 3, then at least $100(1 - 1/3^2)\% = 100(8/9)\% = 88.89\%$ of the population measurements lie in the interval $[\mu \pm 3\sigma]$. As yet a third example, suppose that we wish to find an interval containing at least 99.73 percent of all population measurements. Here we would set $100(1 - 1/k^2)\%$ equal to 99.73%, which implies that $(1 - 1/k^2) = .9973$. If we solve for k , we find that $k = 19.25$. This says that at least 99.73 percent of all population measurements lie in the interval $[\mu \pm 19.25\sigma]$. Unless σ is extremely small, this interval will be so long that it will tell us very little about where the population measurements lie. We conclude that Chebyshev's Theorem can help us find an interval that contains a reasonably high percentage (such as 75 percent or 88.89 percent) of all population measurements. However, unless σ is extremely small, Chebyshev's Theorem will not provide a useful interval that contains almost all (say, 99.73 percent) of the population measurements.

Although Chebyshev's Theorem technically applies to any population, it is only of practical use when analyzing a non-mound-shaped (for example, a double-peaked) population that is not extremely skewed to the right or left. Why is this? First, we would not use Chebyshev's Theorem to describe a mound-shaped population that is not very skewed because we can use the Empirical Rule to do this. In fact, the Empirical Rule is better for such a population because it gives us a shorter interval that will contain a given percentage of measurements. For example, if the Empirical Rule can be used to describe a population, the interval $[\mu \pm 3\sigma]$ will contain 99.73 percent of all measurements. On the other hand, if we use Chebyshev's Theorem, the interval $[\mu \pm 19.25\sigma]$ is needed. As another example, the Empirical Rule tells us that 95.44 percent of all measurements lie in the interval $[\mu \pm 2\sigma]$, whereas Chebyshev's Theorem tells us only that at least 75 percent of all measurements lie in this interval.

It is also not appropriate to use Chebyshev's Theorem—or any other result making use of the population standard deviation σ —to describe a population that is extremely skewed. This is because, if a population is extremely skewed, the measurements in the long tail to the left or right will greatly inflate σ . This implies that tolerance intervals calculated using σ will be so long that they are of little use. In this case, it is best to measure variation by using percentiles, which are discussed in the next section.

z-scores We can determine the relative location of any value in a population or sample by using the mean and standard deviation to compute the value's z-score. For any value x in a population or sample, the z-score corresponding to x is defined as follows:

z-score:

$$z = \frac{x - \text{mean}}{\text{standard deviation}}$$

The z-score, which is also called the *standardized value*, is the number of standard deviations that x is from the mean. A positive z-score says that x is above (greater than) the mean, while a

negative z-score says that x is below (less than) the mean. For instance, a z-score equal to 2.3 says that x is 2.3 standard deviations above the mean. Similarly, a z-score equal to -1.68 says that x is 1.68 standard deviations below the mean. A z-score equal to zero says that x equals the mean.

A z-score indicates the relative location of a value within a population or sample. For example, below we calculate the z-scores for each of the profit margins for five of the best companies in the food, drink, and tobacco industry as rated in the January 7, 2002, issue of *Forbes* magazine. Recall that for these five companies, the mean profit margin is 10% and the standard deviation is 3.406%.

Company	Profit margin, x	$x - \text{mean}$	z-score
Hershey Foods	8%	$8 - 10 = -2$	$-2/3.406 = -.59$
PepsiCo	10	$10 - 10 = 0$	$0/3.406 = 0$
Wm. Wrigley Jr.	15	$15 - 10 = 5$	$5/3.406 = 1.47$
Philip Morris	12	$12 - 10 = 2$	$2/3.406 = .59$
Adolph Coors	5	$5 - 10 = -5$	$-5/3.406 = -1.47$

These z-scores tell us that the profit margin for Wm. Wrigley Jr. is the farthest above the mean. More specifically, this profit margin is 1.47 standard deviations above the mean. The profit margin for Adolph Coors is the farthest below the mean—it is 1.47 standard deviations below the mean. Since the z-score for PepsiCo equals zero, its profit margin equals the mean.

Values in two different populations or samples having the same z-score are the same number of standard deviations from their respective means and, therefore, have the same relative locations. For example, suppose that the mean score on the midterm exam for students in Section A of a statistics course is 65 and the standard deviation of the scores is 10. Meanwhile, the mean score on the same exam for students in Section B is 80 and the standard deviation is 5. A student in Section A who scores an 85 and a student in Section B who scores a 90 have the same relative locations within their respective sections because their z-scores, $(85 - 65)/10 = 2$ and $(90 - 80)/5 = 2$, are equal.

The coefficient of variation Sometimes we need to measure the size of the standard deviation of a population or sample relative to the size of the population or sample mean. The coefficient of variation, which makes this comparison, is defined for a population or sample as follows:

$$\text{Coefficient of variation} = \frac{\text{standard deviation}}{\text{mean}} \times 100$$

The coefficient of variation compares populations or samples having different means and different standard deviations. For example, an article² in the August 17, 1998, issue of *Fortune* compares the annual returns for U.S. stock funds. According to the article, the mean return for the Legg Mason Value Primary fund is 29.4 percent with a standard deviation of 17.3 percent, while the mean return for the Reynolds Blue Chip Growth fund is 23.7 percent with a standard deviation of 17.5 percent. It follows that the coefficient of variation for the Legg Mason fund is $(17.3/29.4) \times 100 = 58.84$, and that the coefficient of variation for the Reynolds fund is $(17.5/23.7) \times 100 = 73.84$. This tells us that, for the Legg Mason fund, the standard deviation is 58.84 percent of the value of its mean return. For the Reynolds fund, the standard deviation is 73.84 percent of the value of its mean return.

In the context of situations like the stock fund comparison, the coefficient of variation is often used as a measure of risk because it measures the variation of the returns (the standard deviation) relative to the size of the mean return. For instance, although the Reynolds fund and the Legg Mason fund have comparable standard deviations (17.5 percent versus 17.3 percent), the Reynolds fund has a higher coefficient of variation than does the Legg Mason fund (73.84 versus 58.84). This says that, relative to the mean return, the variation in returns for the Reynolds fund is higher. That is, we would conclude that investing in the Reynolds fund is riskier than investing in the Legg Mason fund.

²Source: A. Kover, "The Best Mutual Funds," *Fortune*, August 17, 1998, pp. 87-98.

Exercises for Section 2.3

CONCEPTS

- 2.28 What is the range?
2.29 Discuss how the variance and the standard deviation measure variation.
2.30 Why are the variance and standard deviation usually considered more effective measures of variation than the range?
2.31 The Empirical Rule for a normally distributed population and Chebyshev's Theorem have the same basic purpose. In your own words, explain what this purpose is.
2.32 When is a process capable, and what are process specification limits? Give an example of a situation in which process capability is important.

METHODS AND APPLICATIONS

- 2.33 Consider the following population of five numbers: 5, 8, 10, 12, 15. Calculate the range, variance, and standard deviation of this population.
2.34 Table 2.10 gives the percentage of homes sold during the second quarter of 2001 that were within reach of the median income household at the prevailing mortgage interest rate for six Texas metropolitan areas. The data were compiled by the National Association of Home Builders. Calculate the range, variance, and standard deviation of this population of affordability percentages. • HouseAff
2.35 Table 2.11 gives data concerning the top 10 U.S. airlines (ranked by revenue) as listed on the *Fortune* magazine website on February 14, 2002. • AirRev
a Calculate the population range, variance, and standard deviation of the 10 revenues and of the 10 profits (note that values in parentheses are negative—that is, losses rather than profits). Profits: 1690, 243587.61, 493.55
b Using the population of profits, compute and interpret the z-score for each airline. American $z = 1.11$
2.36 In order to control costs, a company wishes to study the amount of money its sales force spends entertaining clients. The following is a random sample of six entertainment expenses (dinner costs for four people) from expense reports submitted by members of the sales force.
Alaska $z = -.68$
\$157 \$132 \$109 \$145 \$125 \$139
a Calculate $\bar{x}$, s^2 , and s for the expense data. In addition, show that the two different formulas for calculating s^2 give the same result. 134.5, 276.7, 16.63
b Assuming that the distribution of entertainment expenses is approximately normally distributed, calculate estimates of tolerance intervals containing 68.26 percent, 95.44 percent, and 99.73 percent of all entertainment expenses by the sales force. [117.87, 151.13] [101.24, 167.76] [84.61, 184.39]
c If a member of the sales force submits an entertainment expense (dinner cost for four) of \$190, should this expense be considered unusually high (and possibly worthy of investigation by the company)? Explain your answer. Yes
d Compute and interpret the z-score for each of the six entertainment expenses.

TABLE 2.10 Housing Affordability in Texas

Metro Area	Percentage of Homes Sold within Reach of Median Income Household
Austin-San Marcos	60.8%
Dallas	67.0
El Paso	63.3
Houston	64.4
San Antonio	67.5
Fort Worth-Arlington	75.3

Data compiled by National Association of Home Builders.
Source: World Wide Web, http://www.nahb.com/factsheet0201_03/regional_alpha03.htm, National Association of Home Builders, February 12, 2002.

TABLE 2.11 The Top 10 Airlines (Ranked by Revenue)

Company	Revenue (\$ Billions)	Profits (\$ Millions)
American Airlines	20.2	813
United Airlines	19.4	50
Delta Airlines	15.9	1,303
Northwest Airlines	11.4	256
Continental Airlines	9.9	342
US Airways Group	9.3	(269)
Southwest Airlines	5.6	603
Trans World Airlines	3.5	(387)
America West Holdings	2.3	2
Alaska Air Group	2.2	(70)

Sources: "Top Ten Airlines in 2001," *Fortune* 502, April 16, 2002. Copyright © 2002 Time, Inc. All rights reserved.

2.37 THE TRASH BAG CASE • TrashBag

The mean and the standard deviation of the sample of 40 trash bag breaking strengths are $\bar{x} = 50.575$ and $s = 1.6438$.

- What do the stem-and-leaf display and histogram in Figures 2.19 and 2.20 (page 53) say about whether the Empirical Rule should be used to describe the trash bag breaking strengths?
- Use the Empirical Rule to calculate estimates of tolerance intervals containing 68.26 percent, 95.44 percent, and 99.73 percent of all possible trash bag breaking strengths.
- Does the estimate of a tolerance interval containing 99.73 percent of all breaking strengths provide evidence that almost any bag a customer might purchase will have a breaking strength that exceeds 45 pounds? Explain your answer.
- How do the percentages of the 40 breaking strengths in Table 1.10 (page 19) that actually fall into the intervals $[\bar{x} \pm s]$, $[\bar{x} \pm 2s]$, and $[\bar{x} \pm 3s]$ compare to those given by the Empirical Rule? Do these comparisons indicate that the statistical inferences you made in parts b and c are reasonably valid?

Appropriate

[48.9, 52.2], [47.3, 53.9], [45.6, 55.5]
Yes67.5%, 95%, 100%
Yes

2.38 THE BANK CUSTOMER WAITING TIME CASE • WaitTime

The mean and the standard deviation of the sample of 100 bank customer waiting times are $\bar{x} = 5.46$ and $s = 2.475$.

- What do the stem-and-leaf display and histogram in Figures 2.16 and 2.17 (page 52) say about whether the Empirical Rule should be used to describe the bank customer waiting times?
- Use the Empirical Rule to calculate estimates of tolerance intervals containing 68.26 percent, 95.44 percent, and 99.73 percent of all possible bank customer waiting times.
- Does the estimate of a tolerance interval containing 68.26 percent of all waiting times provide evidence that at least two-thirds of all customers will have to wait less than eight minutes for service? Explain your answer.
- How do the percentages of the 100 waiting times in Table 1.6 (page 11) that actually fall into the intervals $[\bar{x} \pm s]$, $[\bar{x} \pm 2s]$, and $[\bar{x} \pm 3s]$ compare to those given by the Empirical Rule? Do these comparisons indicate that the statistical inferences you made in parts b and c are reasonably valid?

Somewhat reasonable

[2.99, 7.94], [5.1, 10.41], [-1.97, 12.89]
Yes66%, 96%, 100%
Yes

2.39 THE CUSTOMER SATISFACTION RATING CASE • CustSat

The mean and the standard deviation of the sample of 65 customer satisfaction ratings are $\bar{x} = 42.95$ and $s = 2.6424$.

- What do the stem-and-leaf display and histogram in Figures 2.14 and 2.15 (page 51) say about whether the Empirical Rule should be used to describe the customer satisfaction ratings?
- Use the Empirical Rule to calculate estimates of tolerance intervals containing 68.26 percent, 95.44 percent, and 99.73 percent of all possible customer satisfaction ratings.
- Does the estimate of a tolerance interval containing 95.44 percent of all satisfaction ratings provide evidence that at least 95 percent of all customers will give a satisfaction rating for the VACS-5000 that exceeds 35? Explain your answer.
- How do the percentages of the 65 customer satisfaction ratings in Table 1.5 (page 11) that actually fall into the intervals $[\bar{x} \pm s]$, $[\bar{x} \pm 2s]$, and $[\bar{x} \pm 3s]$ compare to those given by the Empirical Rule? Do these comparisons indicate that the statistical inferences you made in parts b and c are reasonably valid?

Somewhat reasonable

[40.3, 45.6], [37.7, 48.2], [35, 50.9]
Yes63%, 98.46%, 100%
Yes

Skewed right

[32.09, 41.04]: 73%
[27.61, 45.51]: 95.24%
[23.14, 49.99]: 96.83%
Yes

Transaction times skewed

is 17.3 percent with a standard deviation of 10 percent. She also reports that the mean yearly return for the Green Century Balanced fund is 9.38 percent with a standard deviation of 22.4 percent.

[-2.7, 37.3]
[-35.42, 54.18]

- For each mutual fund, find an interval in which you would expect 95.44 percent of all yearly returns to fall. Assume returns are normally distributed.
- Using the intervals you computed in part a, compare the two mutual funds with respect to average yearly returns and with respect to variability of returns.
- Calculate the coefficient of variation for each mutual fund, and use your results to compare the funds with respect to risk. Which fund is riskier?

57.80, 238.81
Green Century
Balanced is riskier

Consider the data concerning a critical dimension of an auto part given in Exercise 1.15 (page 17), and assume the process producing this part is in statistical control.

- When we compute the sample mean and sample standard deviation of the 18 dimensions, we obtain $\bar{x} = 3.0028$ and $s = .01437$. Assuming the dimensions are mound-shaped, use these values to compute an estimated tolerance interval that you would expect to contain almost all (99.73 percent) of the auto part's dimensions. Based on this interval, can we conclude that the process is capable of meeting specifications of $3.00 \pm .03$ —that is, 2.97 to 3.03? Explain your answer.
- After a research and development program is carried out to improve the manufacturing process that produces the auto part, the following 18 measurements of the dimension are obtained (they are given in time order). Does the process appear to be in statistical control? Justify your answer. • AutoPart2

Yes, in control

Hour	Measurement (inches)	Hour	Measurement (inches)
1	3.010	10	3.005
2	3.005	11	2.995
3	2.990	12	2.995
4	3.010	13	3.010
5	2.995	14	3.000
6	2.990	15	2.990
7	3.000	16	2.985
8	2.990	17	3.010
9	3.010	18	3.000

- When we compute the sample mean and sample standard deviation of the 18 observed dimensions from the improved process, we obtain $\bar{x} = 3$ and $s = .00786$. Assuming the measurements are normally distributed, use these results to compute an estimated tolerance interval that you would expect to contain almost all (99.73 percent) of the dimensions produced by the improved process. Based on this interval, can we conclude that the improved process is capable of meeting specifications of $3.00 \pm .03$ —that is, 2.97 to 3.03? Explain your answer.

[2.976, 3.024]

Capable

2.4 Percentiles, Quartiles, and Box-and-Whiskers Displays

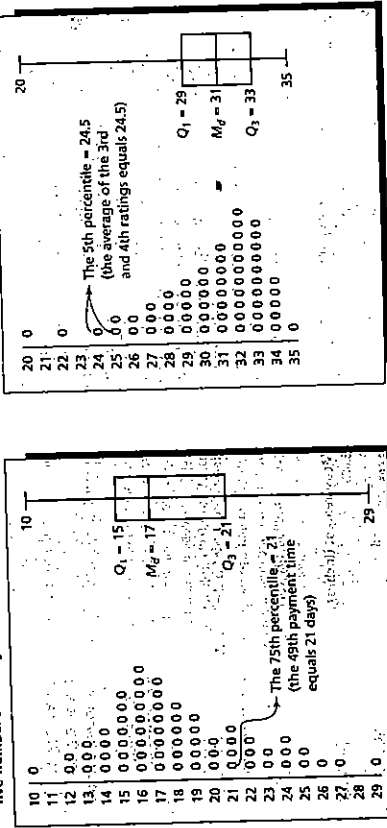
Percentiles, quartiles, and five-number displays In this section we consider percentiles and their applications. We begin by defining the p th percentile.

For a set of measurements arranged in increasing order, the p th percentile is a value such that p percent of the measurements fall at or below the value, and $(100 - p)$ percent of the measurements fall at or above the value.

There are various procedures for calculating percentiles. One procedure is as follows: To calculate the p th percentile for a set of n measurements, we first arrange the measurements in increasing order (by, for example, constructing a stem-and-leaf display). Then we calculate the index $i = (p/100)n$. If i is not an integer, the next integer greater than i denotes the position of the p th percentile in the ordered arrangement. If i is an integer, then the p th percentile is the average of the measurements in positions i and $i + 1$ in the ordered arrangement. For example, Figure 2.33(a) presents the stem-and-leaf display of the 65 payment times. In order to calculate the 75th percentile of these 65 payment times, we calculate the index $i = (75/100)65 = 48.75$. Because $i = 48.75$ is not an integer, the 75th percentile is the 49th payment time in the stem-and-leaf display. Counting up to the 49th payment time in this display, we find that the 75th percentile

- Explain why the Empirical Rule does not describe the transaction times extremely well.
- In the article "The Best Mutual Funds for a Volatile Future" in the December 21, 1998, issue of *Fortune*, Julie Creswell reports that the mean yearly return for the Westwood Balanced mutual fund

FIGURE 2.33 Using Stem-and-Leaf Displays to Find Percentiles and Five-Number Summaries
(a) The 75th percentile of the 65 payment times, and a five-number summary
(b) The 5th percentile of the 60 package design ratings, and a five-number summary



is 21 days (see Figure 2.33(a)). This implies that approximately 75 percent of all payment times are less than or equal to 21 days. As another example, Figure 2.33(b) presents the stem-and-leaf display of the 60 package design ratings. In order to calculate the fifth percentile of these 60 package design ratings, we calculate the index $i = (5/100)60 = 3$. Because $i = 3$ is an integer, the fifth percentile is the average of the third and fourth ratings in the stem-and-leaf display. Counting up to these ratings in this display, we find that the fifth percentile is $(24 + 25)/2 = 24.5$ (see Figure 2.33(b)). Since any rating is a whole number, we estimate that approximately 5 percent of all ratings are 24 or less and that approximately 95 percent of all ratings are 25 or more.

In general, unless percentiles correspond to very high or very low percentages, they are resistant (like the median) to extreme values. For example, the 75th percentile of the payment times would remain 21 days even if the three largest payment times—26, 27, and 29—were, instead, 35, 56, and 84. On the other hand, the standard deviation in this situation would increase from 3.9612 to 10.2119. In general, if a population is highly skewed to the right or left, it can be best to describe the variation of the population by using various percentiles. For example, we might describe the variation of the yearly incomes of all people in the United States by using the 10th, 25th, 50th, 75th, and 90th percentiles of these incomes.

One appealing way to describe the variation of a set of measurements is to divide the data into four parts, each containing approximately 25 percent of the measurements. This can be done by defining the *first*, *second*, and *third quartiles* as follows:

- The *first quartile*, denoted Q_1 , is the 25th percentile.
- The *second quartile* (or *median*), denoted M_d , is the 50th percentile.
- The *third quartile*, denoted Q_3 , is the 75th percentile.

Note that the second quartile is simply another name for the median. Furthermore, the procedure we have described here that would be used to find the 50th percentile (second quartile) will always give the same result as the previously described (see Section 2.2) procedure for finding the median. For example, to find the second quartile of the 65 payment times, we find the 50th percentile by calculating $i = (50/100)65 = 32.5$, which says that the second quartile is the 33rd payment time in the stem-and-leaf display of Figure 2.33(a). Counting up to the 33rd payment time in the display, we find that the second quartile equals 17, which is the result we obtained for the median of the payment times in Example 2.9 (page 59).

EXAMPLE 2.4 The Average Receivable Days

Since $i = (25/100)65 = 16.25$, the first quartile (25th percentile) of the 65 payment times is the 17th payment time in the stem-and-leaf display of Figure 2.33(a). Therefore, $Q_1 = 15$. Remembering that the median of the payment times is 17, and that the 75th percentile of the payment times is 21, the quartiles are $Q_1 = 15$, $M_d = 17$, and $Q_3 = 21$.

We often describe a set of measurements by using a *five-number summary*. The summary consists of (1) the smallest measurement; (2) the first quartile, Q_1 ; (3) the median, M_d ; (4) the third quartile, Q_3 ; and (5) the largest measurement. It is easy to graphically depict a five-number summary; we have done this for the 65 payment times alongside the stem-and-leaf display of Figure 2.33(a). Notice that we have drawn a vertical line extending from the smallest payment time to the largest payment time. In addition, a rectangle is drawn that extends from Q_1 to Q_3 , and a horizontal line is drawn to indicate the location of the median. The summary divides the payment times into four parts, with the middle 50 percent of the payment times depicted by the rectangle. The summary indicates that the largest 25 percent of the payment times is more spread out than the smallest 25 percent of the payment times, and that the second-largest 25 percent of the payment times is more spread out than the second-smallest 25 percent of the payment times.

Overall, the summary indicates that the payment times are somewhat skewed to the right. As another example, for the 60 package design ratings, $Q_1 = 29$, $M_d = 31$, and $Q_3 = 33$. The graphical five-number summary of the ratings is shown alongside the stem-and-leaf display of the package design ratings in Figure 2.33(b). The summary shows that the smallest 25 percent of the ratings is more spread out than any of the other quarters of the ratings, and that the other three quarters are equally spread out. Overall, the summary shows that the package design ratings are skewed to the left.

Using the first and third quartiles, we define the *interquartile range* to be $IQR = Q_3 - Q_1$. This quantity can be interpreted as the length of the interval that contains the *middle 50 percent* of the measurements. For instance, Figure 2.33(a) tells us that the interquartile range of the sample of 65 payment times is $Q_3 - Q_1 = 21 - 15 = 6$. This says that we estimate that the middle 50 percent of all payment times fall within a range that is six days long.

The procedure we have presented for calculating the first and third quartiles is not the only procedure for computing these quantities. In fact, several procedures exist, and, for example, different statistical computer packages use several somewhat different methods for computing the quartiles. One procedure calculates what are called *lower* and *upper hinges* and then defines the first quartile to be the lower hinge and defines the third quartile to be the upper hinge. We explain how to calculate the lower and upper hinges in Exercise 2.50, and we will see that these quantities can sometimes differ from the values obtained for the 25th and 75th percentiles. In general, the different methods for computing the first and third quartiles sometimes produce somewhat different results. However, no matter what procedure is used to compute the quartiles, the objective is to divide the data into four parts, each containing approximately 25 percent of the measurements.

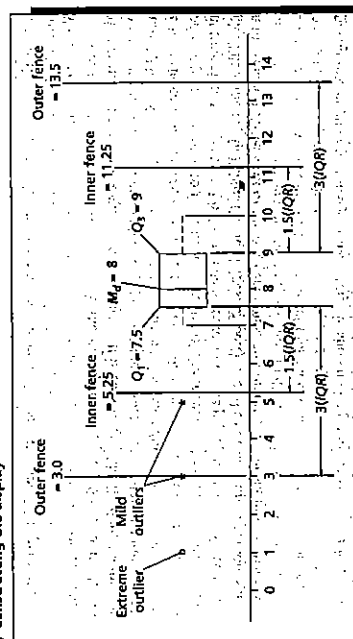
Box-and-whiskers displays (box plots) A more sophisticated modification of the graphical five-number summary is called a *box-and-whiskers display* (sometimes called a *box plot*). Such a display is constructed by using Q_1 , M_d , Q_3 , and the interquartile range. As an example, again consider the 20 customer satisfaction ratings

1 3 5 5 7 8 8 8 8 8 8 8 9 9 9 9 10 10 10

for which it can be shown that $Q_1 = 7.5$, $M_d = 8$, $Q_3 = 9$, and $IQR = Q_3 - Q_1 = 9 - 7.5 = 1.5$. To construct a box-and-whiskers display, we first draw a box that extends from Q_1 to Q_3 . As shown in Figure 2.34(a), for the satisfaction ratings data this box extends from $Q_1 = 7.5$ to $Q_3 = 9$. The box contains the middle 50 percent of the data set. Next a vertical line is drawn through the box at the value of the median M_d (sometimes a plus sign $+$) is plotted at the median instead of a vertical line). This line divides the data set into two roughly equal parts. We next define what we call *inner* and *outer fences*. The inner fences are located $1.5 \times IQR$ below Q_1

FIGURE 2.34 A Box-and-Whiskers Display of the Satisfaction Ratings

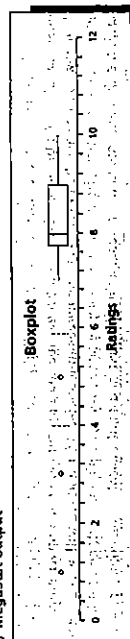
(a) Constructing the display



(b) MINITAB output



(c) MegaStat output

and $1.5 \times IQR$ above Q_3 . For the satisfaction ratings data, the inner fences are

$$Q_1 - 1.5(IQR) = 7.5 - 1.5(1.5) = 5.25 \quad \text{and} \quad Q_3 + 1.5(IQR) = 11.25 + 1.5(1.5) = 13.5$$

(again see Figure 2.34(a)). The outer fences are located $3 \times IQR$ below Q_1 and $3 \times IQR$ above Q_3 . For the satisfaction ratings data, the outer fences are

$$Q_1 - 3(IQR) = 7.5 - 3(1.5) = 3.0 \quad \text{and} \quad Q_3 + 3(IQR) = 11.25 + 3(1.5) = 15.0$$

(these are also shown in Figure 2.34(a)). The inner and outer fences help us to draw the plot's whiskers: dashed lines extending below Q_1 and above Q_3 (as in Figure 2.34(a)). One whisker is drawn from Q_1 to the smallest measurement between the inner fences. For the satisfaction ratings data, this whisker extends from $Q_1 = 7.5$ down to 7, because 7 is the smallest rating between the inner fences 5.25 and 11.25. The other whisker is drawn from Q_3 to the largest measurement between the inner fences. For the satisfaction ratings data, this whisker extends from $Q_3 = 11.25$ up to 10, because 10 is the largest rating between the inner fences 5.25 and 11.25. The inner and outer

fences are also used to identify outliers. An outlier is a measurement that is separated from (that is, different from) most of the other measurements in the data set. Measurements that are located between the inner and outer fences are considered to be **mild outliers**, whereas measurements that are located outside the outer fences are considered to be **extreme outliers**. We indicate the locations of mild outliers by plotting these measurements with the symbol $\circ$, and we indicate the locations of extreme outliers by plotting these measurements with the symbol $\bullet$. For the satisfaction ratings data, the ratings 3 and 5 are mild outliers ($\circ$) because these ratings are between the inner fence of 5.25 and the outer fence of 3.0. The rating 1 is an extreme outlier ($\bullet$) because this rating is outside the outer fence 3.0. These outliers are plotted in Figure 2.34(a). Parts (b) and (c) of Figure 2.34 give MINITAB and MegaStat outputs of the box-and-whiskers plot. Notice that MINITAB identifies the median by using a plus sign (+), while MegaStat uses a vertical line. In addition, MegaStat plots all outliers using the same symbol and marks the inner and outer fences using vertical dashed lines. Note here that MegaStat computes the quartiles Q_1 and Q_3 and the inner and outer fences using methods that differ slightly from the methods we have described. The MegaStat Help menus describe how the calculations are done. We now summarize how to construct a box-and-whiskers plot.

1. Draw a box that extends from the first quartile Q_1 to the third quartile Q_3 . Also draw a vertical line through the box located at the median M_d .
2. Determine the values of the inner fences and outer fences. The inner fences are located $1.5 \times IQR$ below Q_1 and $1.5 \times IQR$ above Q_3 . That is, the inner fences are $Q_1 - 1.5(IQR)$ and $Q_3 + 1.5(IQR)$. The outer fences are located $3 \times IQR$ below Q_1 and $3 \times IQR$ above Q_3 . That is, the outer fences are $Q_1 - 3(IQR)$ and $Q_3 + 3(IQR)$.
3. Draw whiskers as dashed lines that extend below Q_1 and above Q_3 . Draw one whisker from Q_1 to the smallest measurement that is between the inner fences. Draw the other whisker from Q_3 to the largest measurement that is between the inner fences.
4. Measurements that are located between the inner and outer fences are called **mild outliers** and are plotted using the symbol $\circ$. Measurements that are located outside the outer fences are called **extreme outliers** and are plotted using the symbol $\bullet$.

When interpreting a box-and-whiskers display, keep several points in mind. First, the box (between Q_1 and Q_3) contains the middle 50 percent of the data. Second, the median (which is inside the box) divides the data into two roughly equal parts. Third, if one of the whiskers is longer than the other, the data set is probably skewed in the direction of the longer whisker. Last, observations designated as outliers should be investigated. Understanding the root causes behind the underlying observations will often provide useful information. For instance, understanding why several of the satisfaction ratings in the box plot of Figure 2.34 are substantially lower than the great majority of the ratings may suggest actions that can improve the videocassette manufacturer's product and/or service. Outliers can also be caused by inaccurate measuring, reporting, or plotting of the data. Such possibilities should be investigated, and incorrect data should be adjusted or eliminated.

Generally, a box plot clearly depicts the central tendency, variability, and overall range of a set of measurements. A box plot also portrays whether the measurements are symmetrically distributed. However, the exact shape of the distribution is better portrayed by a stem-and-leaf display and/or a histogram. For instance, Figure 2.35 shows the MegaStat output of the stem-and-leaf display and box plot of the scores on the 100-point statistics exam of Figure 2.11 (page 48) that was given before an attendance policy was begun. We see that, although the box plot in Figure 2.35 tells us that the exam scores are somewhat skewed with a tail to the left, it does not reveal the double-peaked nature of the exam score distribution. On the other hand, the stem-and-leaf display clearly shows that this distribution is double-peaked.

Graphical five-number summaries and box-and-whiskers displays are perhaps best used to compare different sets of measurements. We demonstrate this use of such displays in the following example.

FIGURE 2.35 MegaStat Output of a Stem-and-Leaf Display and Box Plot of the Exam Scores

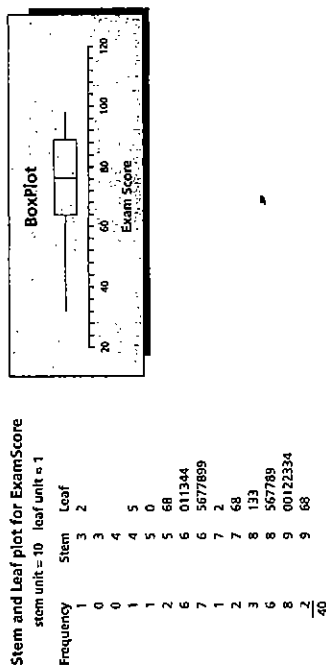
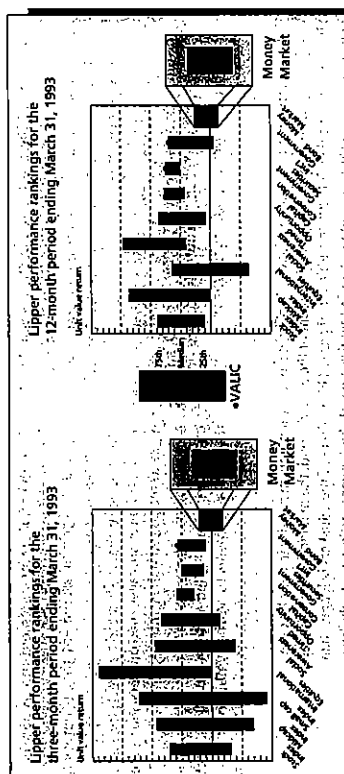


FIGURE 2.36 Graphical Comparison of the Performance of Mutual Funds by Using Five-Number Summaries



Source: Reprinted by permission of Lipet, Inc.

In July of 1993, the Variable Annuity Life Insurance Company (VALIC) sent its investors an analysis of the performance of its variable account mutual fund options relative to other variable annuity fund options in various categories (stock index, money market, and so forth). VALIC used the graphical five-number summaries in Figure 2.36 to summarize and compare performances. The dot within each five-number summary represents the return of the VALIC mutual fund option for that category. In explaining the plots, VALIC said:

The data show that all of VALIC's mutual fund options ranked at or above their respective category median return for the three-month period ending March 31, 1993. Also, eight of VALIC's mutual fund options ranked above their respective category median return for the 12-month period ending March 31, 1993.

Notice that the lengths of the five-number summaries indicate performance variability for the various funds. For example, while the median three-month returns for Midcap Index funds and Small cap Index funds are similar, the returns for Small cap funds are more variable. Also, in general, three-month returns for funds of all types are more variable than 12-month returns.

EXERCISES SECTION 2.4

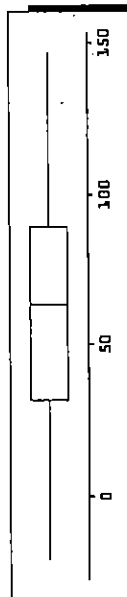
CONCEPTS

- 2.43 Explain each of the following in your own words: a percentile; the first quartile, Q_1 ; the third quartile, Q_3 ; and the interquartile range, IQR .
- 2.44 Suppose that we are using a box-and-whiskers plot to depict a population or sample of measurements. How would you interpret each of the following?
- The whisker to the right is much longer than the whisker to the left.
 - The interquartile range is much longer than either of the whiskers and there are no outliers.
 - The distance between Q_1 and the median is far less than the distance between the median and Q_3 .
 - The interquartile range is very short.

METHODS AND APPLICATIONS

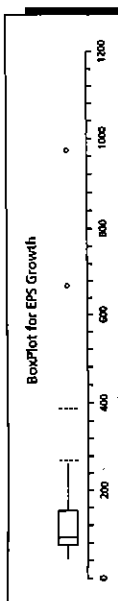
- 2.45 Consider the 65 customer satisfaction ratings, the 100 bank customer waiting times, and the 40 trash bag breaking strengths. Using the stem-and-leaf displays of these data sets in Figures 2.14 (page 51), 2.17 (page 52), and 2.20 (page 53), find for each data set:
- The 50th percentile.
 - The median.
 - The first quartile.
 - The third quartile.
 - The 10th percentile.
 - The interquartile range.
- 2.46 Construct a five-number summary and a box-and-whiskers display for each data set.
- 2.47 Consider Table 2.5 (page 50), which gives data concerning the 30 fastest-growing companies as listed on February 17, 2002, at the *Fortune* magazine website. • Fast(Grow)
- Below we present a MINITAB output of descriptive statistics and a box-and-whiskers display for the total return percentages of the 30 fastest-growing companies.

Variable	N	Mean	Median	StDev	Minimum	Maximum	Q1	Q3
TotReturn	30	65.20	63.50	41.51	-23.00	147.00	32.00	89.75



- Using the MINITAB output, describe the distribution of the total return percentages with respect to central tendency, variability, skewness, and outliers.
- b Below we present a MegaStat output of descriptive statistics and a box-and-whiskers display for the EPS (earnings per share) growth percentages for the 30 fastest-growing companies.

	EPS Growth	count	30
mean	160.73	43	75.00
sample variance	37,445.03	973	92.50
sample standard deviation	193.51	950	153.00

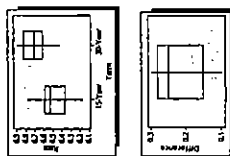


Using the MegaStat output, describe the distribution of the EPS growth percentages with respect to central tendency, variability, skewness, and outliers.

- 2.47 On its website, the *Statesman Journal* newspaper (Salem, Oregon, 1999) reports mortgage loan interest rates for 30-year and 15-year fixed-rate mortgage loans for a number of Willamette Valley lending institutions. Of interest is whether there is any systematic difference between 30-year rates and 15-year rates (expressed as annual percentage rate or APR) and, if there is, what is the size of that difference. The table on the next page displays the 30-year rate and the 15-year rate for each of eight randomly selected lending institutions. Also given is the difference between the 30-year rate

Very skewed right
Two outliers
30 year rates higher
Variability about
equal

Average of differences is .2237



Lending Institution	30-Year	15-Year	Difference
American Mortgage N.W. Inc.	6.715	6.599	0.116
City and Country Mortgage	6.648	6.367	0.281
Commercial Bank	6.740	6.550	0.190
Landmark Mortgage Co.	6.597	6.362	0.235
Liberty Mortgage, Inc.	6.425	6.162	0.263
MAPS Credit Union	6.880	6.583	0.297
Mortgage Brokers, Inc.	6.900	6.800	0.100
Mortgage First Corp.	6.675	6.394	0.281
Silver Eagle Mortgage	6.790	6.540	0.250

Source: Via World Wide Web, Salem Homeplace Mortgage Rates Directory, <http://www.salemhomeplace.com/mortgage/rates/index.html>, January 4, 1998. Reproduced by permission of the Statesman Journal newspaper, Salem, Oregon.

2.48 In the book *Business Research Methods*, Donald R. Cooper and C. William Emory present box-and-whiskers plots comparing the net profits of firms in five different industry sectors. Each plot (for a sector) was constructed using net profit figures for a sample of firms from the *Forbes* 500s. Figure 2.37 gives the five box-and-whiskers plots.

a Using the plots in Figure 2.37, write an analysis comparing net profits for the five sectors. Compare central tendency, variability, skewness, and outliers.

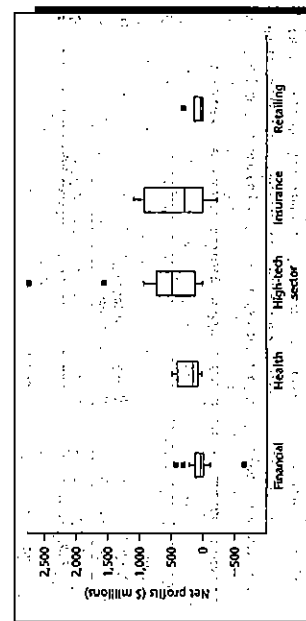
b For which sectors are net profits most variable? Least variable?

c Which sectors provide opportunities for the highest net profit?

2.49 Figure 2.38 gives seven pairs of five-number summaries presented in an article in the January 1995 issue of *Quality Progress*. In the article, authors Dale H. Myers and Jeffrey Heller discuss how AT&T has employed a quality award process (called the Chairman's Quality Award or CQA) to improve quality. To quote Myers and Heller:

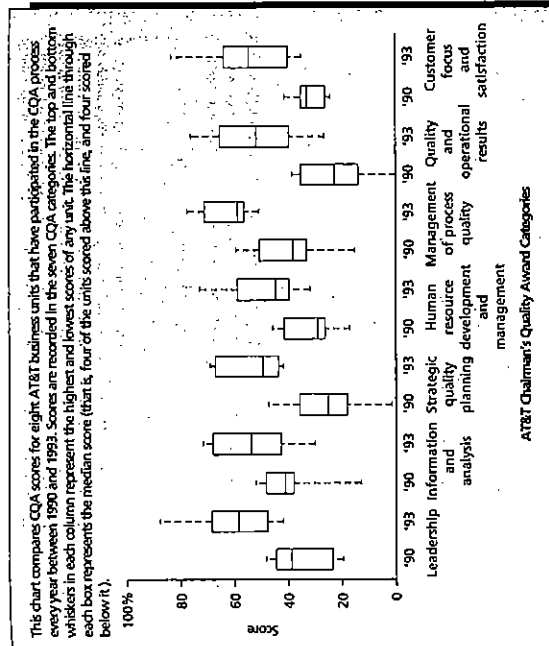
In 1989, AT&T began searching for a systematic process to achieve two major goals: aligning its business management systems more closely with customers' needs and integrating quality principles into every business practice. AT&T wanted this new process to be based on clear and quantifiable standards so that its key building blocks—the business units and divisions—could objectively assess the strengths and shortcomings of their operations.

FIGURE 2.37 Box-and-Whiskers Plots Comparing Net Profits for Five Industry Sectors



Data from: "The *Forbes* 500s Annual Directory," *Forbes*, April 30, 1990, pp. 221-434. Source: Copyright © 1995. Reprinted by permission of McGraw-Hill Companies, Inc.

FIGURE 2.38 Comparison of AT&T Chairman's Quality Award Scores from 1990 to 1993 for Eight Business Units



This chart compares CQA scores for eight AT&T business units that have participated in the CQA process every year between 1990 and 1993. Scores are recorded in the seven CQA categories. The top and bottom whiskers in each column represent the highest and lowest scores of any unit. The horizontal line through each box represents the median score (that is, four of the units scored above the line, and four scored below it).

- a All categories
b Most: strategic quality planning; quality and operational results. Least: Info. and analysis; human resource development and management.
c Increase: leadership, human resource development, customer focus and satisfaction. Decrease: strategic quality planning, management of process quality. Others stayed about the same.
d Leadership: strategic quality planning; human resource development; quality and operational results.

Source: D. H. Myers and J. Heller, "The Dual Role of AT&T Self-Assessment Process," *Quality Progress*, January 1995, pp. 75-83. Copyright © 1995. American Society for Quality Control. Used with permission.

Within a year, AT&T took its first major step into the world of objective self-assessment. The New Jersey-based telecommunications giant created a Chairman's Quality Award (CQA) process modeled after the Malcolm Baldrige National Quality Award process.

Using clear and objective criteria, the CQA process helps units and divisions assess their business performance and share their most successful practices with each other. It also provides feedback that helps them identify their strengths and opportunities for improvement. A business unit (department, division, etc.) that chooses to participate in the award program is examined and scored in seven categories—leadership, information and analysis, strategic quality planning, human resource development and management, management of process quality, quality and operational results, and customer focus and satisfaction.

In order to track AT&T's improvement from 1990 to 1993, the company identified eight business units that participated in the award process every year from 1990 to 1993. For each award category (leadership and so on), a five-number display of the eight business units' 1990 scores was compared to a five-number display of their 1993 scores. The two five-number displays (1990 and 1993) are given for all seven categories in Figure 2.38. Use this figure to answer the following:

- a Based on central tendency, which categories showed the most improvement from 1990 to 1993?
b Based on central tendency, which categories showed the least improvement?
c In which categories did the variability of the CQA scores increase from 1990 to 1993? In which categories did the variability remain about the same from 1990 to 1993?
d In which categories did the nature of the skewness of the CQA scores change from 1990 to 1993? Interpret these changes.

*We discuss the Malcolm Baldrige National Quality Award process in Chapter 14.

2.50 In Section 2.4 we presented a commonly accepted way to compute the first, second, and third quartiles. Some statisticians, however, advocate an alternative method for computing Q_1 and Q_3 . This method defines the first quartile, Q_1 , as what is called the *lower hinge* and defines the third quartile, Q_3 , as the *upper hinge*. In order to calculate these quantities for a set of n measurements, we first arrange the measurements in increasing order. Then, if n is even, the *lower hinge* is the median of the smallest $n/2$ measurements, and the *upper hinge* is the median of the largest $n/2$ measurements. If n is odd, we insert M_n into the data set to obtain a set of $n + 1$ measurements. Then the *lower hinge* is the median of the smallest $(n + 1)/2$ measurements, and the *upper hinge* is the median of the largest $(n + 1)/2$ measurements.

$Q_1 = 7.5$, $Q_2 = 9$
Lower hinge = 7.5
Upper hinge = 9

$Q_1 = 138$
 $Q_2 = 177$
Lower hinge = 139.5
Upper hinge = 174

a Consider the sample of $n = 20$ customer satisfaction ratings:
1 3 5 5 7 8 8 8 8 8 9 9 9 9 9 10 10 10 10 10
The smallest 10 ratings The largest 10 ratings

Using the method presented in Section 2.4 (pages 75 to 77), find Q_1 and Q_3 . Then find the lower hinge and the upper hinge for the satisfaction ratings. How do your results compare?

b Consider the following sample of $n = 11$ doctors' salaries (in thousands of dollars):

127 132 138 141 146 152 154 171 177 192 241

Using the method presented in Section 2.4, find Q_1 and Q_3 . The median of the 11 salaries is $M_n = 152$. If we insert this median into the data set, we obtain the following set of $n + 1 = 12$ salaries:

127 132 138 141 146 152 152 154 171 177 192 241

The smallest 6 salaries The largest 6 salaries

Find the lower hinge and the upper hinge for the salaries. Compare your values of Q_1 and Q_3 with the lower and upper hinges.

c For the 11 doctors' salaries, which quantities (Q_1 , M_n , and Q_3 as defined in Section 2.4) or (the lower hinge, M_n , and the upper hinge) in your opinion best divide the salaries into four parts?

2.5 ■ Describing Qualitative Data

Bar charts and pie charts Recall that when we employ a qualitative or categorical variable we simply record into which of several categories a population element falls. For example, for each automobile produced in the United States we might record the manufacturer—Chrysler, Ford, General Motors, a Japanese manufacturer, or some other manufacturer. We often display such data graphically. For instance, Figure 2.39 gives the Excel output of a bar chart of the percentages of automobiles sold in the United States in 1997 by these manufacturers. Figure 2.40 gives the Excel output of a pie chart of these percentages, while Figure 2.41 gives a MINITAB bar chart comparing the percentages sold by the manufacturers in 1970 with the percentages sold in 1997. Figure 2.42 makes the same comparison by using two pie charts. Notice the declining percentages of automobiles sold by Chrysler, Ford, and General Motors, and the increased percentage of automobiles sold by Japanese manufacturers. We have emphasized the increase in the percentage sold by Japanese manufacturers by *exploding* their *pie slices* in Figure 2.42.

In general, bar charts and pie charts are convenient ways to summarize the percentages of population units that are contained in several different categories.

Estimating proportions Suppose that a population unit can fall into one of several categories. Often we are interested in a specific category, and, in such cases, we often wish to estimate

p = the proportion of all population elements that are contained in the category of interest
In order to estimate this proportion, we can randomly select a sample from the population. Then the **sample proportion**

$\hat{p}$ = the proportion of the sample elements that are contained in the category of interest
is a reasonable point estimate of the population proportion p .

FIGURE 2.39 An Excel Bar Chart of U.S. Automobile Sales in 1997

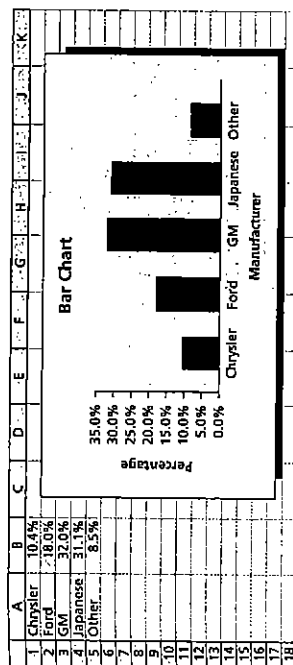


FIGURE 2.40 An Excel Pie Chart of U.S. Automobile Sales in 1997

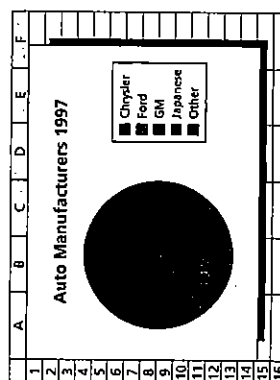


FIGURE 2.41 A MINITAB Bar Chart Comparing 1997 U.S. Automobile Sales with Those in 1970

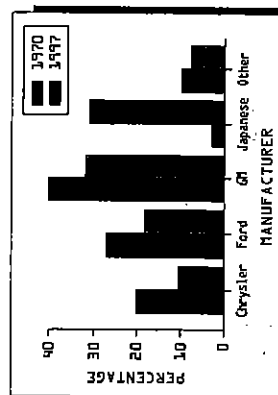
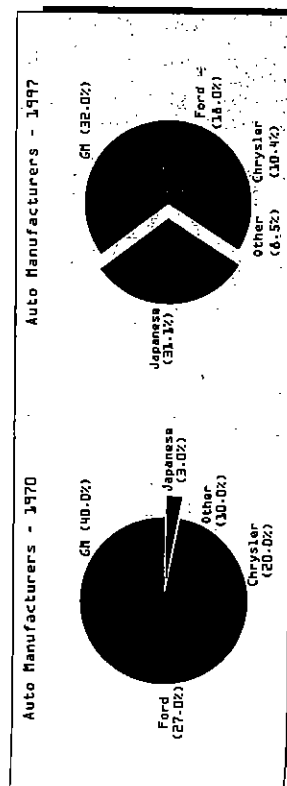


FIGURE 2.42 Two MINITAB Pie Charts Comparing the Percentages of Automobiles Sold by Different Manufacturers in 1970 with Those in 1997



In the following examples, we introduce two new cases that illustrate estimating a population proportion p .

In the book *Essentials of Marketing Research* (1993), William R. Dillon, Thomas J. Madden, and Neil H. Fildes discuss a survey of marketing professionals, the results of which were originally published by Ishmael P. Aksoy and Edward A. Riordan in the *Journal of Marketing Research* (1989). In the study, marketing researchers were presented with various scenarios involving ethical issues such as confidentiality, conflict of interest, and social acceptability. The marketing researchers were asked to indicate whether they approved or disapproved of the actions described in each scenario. For instance, one scenario that involved the issue of confidentiality was described as follows:

Use of ultraviolet ink A project director went to the marketing research director's office and requested permission to use an ultraviolet ink to precode a questionnaire for a mail survey. The project director pointed out that although the cover letter promised confidentiality, respondent identification was needed to permit adequate cross-tabulations of the data. The marketing research director gave approval.

Of the 205 marketing researchers who participated in the survey, 117 said they disapproved of the actions taken in the scenario.

In this situation we would like to make an inference about the population of all marketing researchers. Specifically, we wish to estimate the *population proportion*

$$p = \text{the proportion of all marketing researchers who disapprove of the actions taken in the ultraviolet ink scenario}$$

Since 117 of the 205 surveyed marketing researchers said they disapproved, the *sample proportion*

$$\hat{p} = \frac{117}{205} = .57$$

is the point estimate of p . This point estimate says we estimate that p , the proportion of all marketing researchers who disapprove, is .57. That is, we estimate that 57 percent of all marketing researchers disapprove of the actions taken in the ultraviolet ink scenario.

In an article titled "Consumer Responses to Electronic Article Surveillance Alarms" in the Fall 1993 volume of the *Journal of Retailing*, Scott Dawson studies the unintended effects of false electronic article surveillance alarms. As Dawson explains in the article, in 1990 estimated annual losses from shoplifting were \$50 billion. Electronic article surveillance (EAS), an important weapon used by retailers to combat shoplifting, places a small sensor on an item of merchandise. If a shoplifter attempts to exit the store with the item, an electronic alarm is set off by the sensor. When an item is legitimately purchased, the sales clerk removes the sensor to prevent the alarm from sounding when the customer exits the store. Sometimes, however, the clerk forgets to remove the sensor during the purchase. This results in a false alarm when the customer exits—an embarrassing situation for everyone involved (especially the customer). Such false alarms occur quite frequently. In fact, according to Dawson's article "nearly half of all consumers have experienced an accidental EAS alarm."

Dawson conducted a survey to study consumer reaction to such false alarms. Based on a systematic random sample of 250 consumers, he found that 40 of these consumers said that, if they were to set off an EAS alarm because store personnel did not deactivate the merchandise, then they would never shop at the store again.

Suppose we wish to estimate p , the population proportion of all consumers who would say they would never shop at the store again if subjected to an EAS false alarm. Since 40 of the 250 sampled consumers said they would never shop at the store again, the sample proportion

$\hat{p} = 40/250 = .16$ is the point estimate of p . This point estimate says we estimate that 16 percent of all consumers would say they would never shop at the store again if subjected to an EAS false alarm.

Finally, suppose a retailer is considering the installation of an EAS system. In an attempt to convince the retailer to purchase the system, a company that markets EAS systems claims that no more than 5 percent of consumers would say that they would never shop at a store again if they were subjected to an EAS false alarm. Based on Dawson's survey results, the retailer would have a hard time believing this claim. That is, the sample proportion $\hat{p} = .16$ suggests that more than 5 percent of all consumers would say that they would never shop at the store again. But is the evidence here conclusive? We will address this question in later chapters.

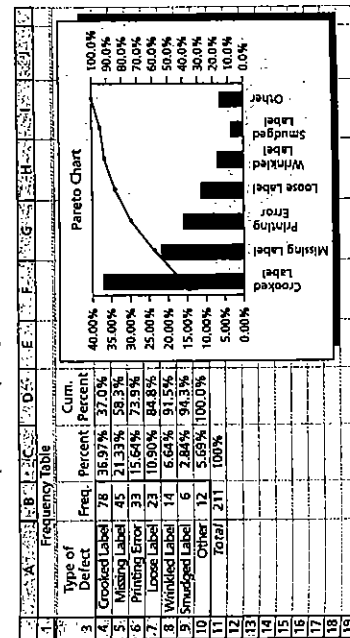
The Pareto chart Pareto charts are used to help identify important quality problems and opportunities for process improvement. By using these charts we can prioritize problem-solving activities. The Pareto chart is named for Vilfredo Pareto (1848–1923), an Italian economist. Pareto suggested that, in many economies, most of the wealth is held by a small minority of the population. It has been found that the "Pareto principle" often applies to defects. That is, only a few defect types account for most of a product's quality problems.

Here, defects can be divided into two categories—the "vital few" and the "trivial many." The vital few are the small number of defects that account for a large percentage of the total, while the trivial many are the large number of defects that account for the small remaining percentage of the total. If the vital few defects are very costly to an organization, it may wish to work on eliminating their causes before working to solve other problems.

To illustrate the use of Pareto charts, suppose that a jelly producer wishes to evaluate the labels being placed on 16-ounce jars of grape jelly. Every day for two weeks, all defective labels found on inspection are classified by type of defect. If a label has more than one defect, we will record the type of defect that is most noticeable. The Excel output in Figure 2.43 presents the frequencies and percentages of the types of defects observed over the two-week period.

In general, the first step in setting up a Pareto chart summarizing data concerning types of defects (or categories) is to construct a frequency table like the one in Figure 2.43. Defects or categories should be listed at the left of the table in *decreasing order by frequencies*—the defect with the highest frequency will be at the top of the table, the defect with the second-highest frequency below the first, and so forth. If an "other" category is employed, it should be placed at the bottom of the table. The "other" category should not make up 50 percent or more of the total of the frequencies, and the frequency for the "other" category should not exceed the frequency for the defect at the top of the table. If the frequency for the "other" category is too high, data

FIGURE 2.43 Excel Output of a Frequency Table and a Pareto Chart of Labeling Defects



should be collected so that the "other" category can be broken down into new categories. Once the frequency and the percentage for each category are determined, a cumulative percentage for each category is computed. As illustrated in Figure 2.43, the cumulative percentage for a particular category is the sum of the percentages corresponding to the particular category and the categories that are above that category in the table.

The Pareto chart is a bar chart. Different kinds of defects or problems are listed on the horizontal scale. The heights of the bars on the vertical scale typically represent the frequency of occurrence (or the percentage of occurrence) for each defect or problem. The bars are arranged in decreasing height from left to right. Thus, the most frequent defect will be at the far left, the next most frequent defect to its right, and so forth. If an "other" category is employed, it is placed at the far right. The Pareto chart for the labeling defects data is given in Figure 2.43. Here the heights of the bars represent the percentages of occurrences for the different labeling defects, and the vertical scale on the far left corresponds to these percentages. The chart graphically illustrates that crooked labels, missing labels, and printing errors are the most frequent labeling defects.

As is also illustrated in Figure 2.43, a Pareto chart is sometimes augmented by plotting a cumulative percentage point for each bar in the Pareto chart. The vertical coordinate of this cumulative percentage point equals the cumulative percentage in the frequency table corresponding to the bar. The cumulative percentage points corresponding to the different bars are connected by a line segment, and a vertical scale corresponding to the cumulative percentages is placed on the far right. Examining the cumulative percentage points in Figure 2.43, we see that crooked and missing labels make up 58.3 percent of the labeling defects and that crooked labels, missing labels, and printing errors make up 73.9 percent of the labeling defects.

2.5

CONCEPTS

- 2.51 Find an example of a pie chart or bar chart in a newspaper or magazine. Copy it, and hand it in with a written analysis of the information conveyed by the chart.
- 2.52 What is a population proportion? Give an example of a population proportion that might interest you in the profession you intend to enter.

2.54, 2.55

Yes, baseball attendance decreased in 1995



METHODS AND APPLICATIONS

- 2.53 On August 5, 1994, the *Journal News* (Hamilton, Ohio) published the results of an Associated Press poll of baseball fans' attitudes regarding the impending baseball strike (which began shortly thereafter). The AP poll was a national telephone survey of 665 randomly selected adults who considered themselves baseball fans, taken July 27–August 2, 1994, by ICR Survey Research Group of Media, Pennsylvania, which is part of AVS Consultants. The survey asked whether a strike would cause the respondents to watch or attend more or fewer games after the strike was over. The results of the survey are summarized in the Excel output of a bar chart and a pie chart shown in Figure 2.44. Based on fan reaction in 1995 (after the strike ended), are the poll results consistent with what actually happened? Why? • Baseball

- 2.54 a In a poll released by the Gallup Organization on January 4, 2000, 622 randomly selected adults were asked the following question on December 28, 1999.

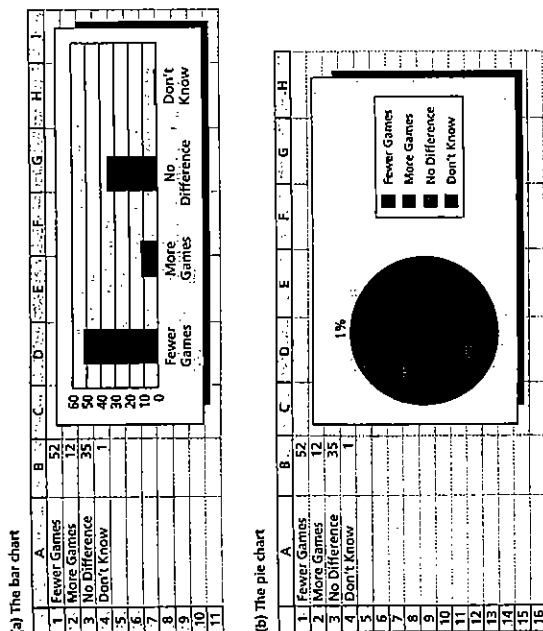
As you may know, most computer systems around the world have to be reprogrammed so that they can accurately recognize the date once we reach the Year 2000. How likely do you think it is that there will be major problems in the United States this New Year's as a direct or indirect result of the Y2K computer problem—very likely, somewhat likely, somewhat unlikely, or very unlikely?

The poll's results were as follows: Very likely (7%), Somewhat likely (30%), Somewhat unlikely (32%), Very unlikely (30%). No opinion (1%). Use these data to construct a bar chart and a pie chart. What actually happened?

- b Of the 100 fastest-growing companies in America as given in the September 28, 1998, issue of *Fortune*, there were 22 energy companies, 20 industrial companies, 10 retail companies, 8 health care companies, 5 financial service companies, 26 technology companies, and 9 other companies. Use these data to construct a pie chart and a bar chart.

Source: R9891230 APD, The Gallup Poll, December 30, 1999. © 1999 The Gallup Organization. All rights reserved.

FIGURE 2.44 Excel Bar Chart and Pie Chart of Baseball Fans' Attitudes Regarding the Impending Baseball Strike in 1994 • Baseball



2.55 THE MARKETING ETHICS CASE: CONFLICT OF INTEREST

Consider the marketing ethics case described in Example 2.16. One of the scenarios presented to the 205 marketing researchers is as follows:

A marketing testing firm to which X company gives most of its business recently went public. The marketing research director of X company had been looking for a good investment and proceeded to buy some \$20,000 of their stock. The firm continues as X company's leading supplier for testing.

Of the 205 marketing researchers who participated in the ethics survey, 111 said that they disapproved of the actions taken in the scenario. Use this sample result to compute a point estimate of the proportion of all marketing researchers who disapprove of the actions taken in this conflict of interest scenario.

- 2.56 In an article in the November 1993 issue of *Quality Progress*, Barbara A. Cleary reports on improvements made in a software supplier's responses to customer calls. In this article, the author states:

In an effort to improve its response time for these important customer-support calls, an inbound telephone inquiry team was formed at PQ Systems, Inc., a software and training organization in Dayton, Ohio. The team found that 88 percent of the customers' calls were already being answered immediately by the technical support group, but those who had to be called back had to wait an average of 56.6 minutes. No customer complaints had been registered, but the team believed that this response rate could be improved.

As part of its improvement process, the company studied the disposition of complete and incomplete calls to its technical support analysts. A call is considered complete if the customer's problem has been resolved; otherwise the call is incomplete. Figure 2.45 shows a Pareto chart analysis for the incomplete customer calls.

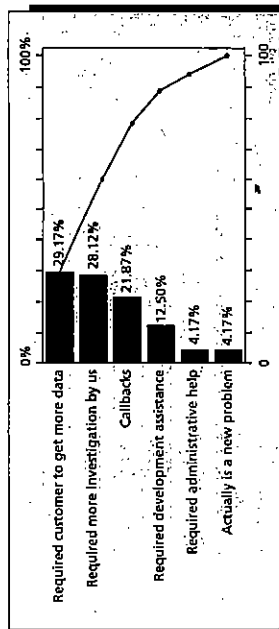
- a What percentage of incomplete calls required "more investigation" by the analyst or "administrative help"?
- b What percentage of incomplete calls actually presented a "new problem"?
- c In light of your answers to a and b, can you make a suggestion?

32.29%

4.17%

$\hat{p} = .541$

FIGURE 2.45 A Pareto Chart for Incomplete Customer Calls to Technical Support Analysts



Source: B. A. Gray, "Company Cries about Customer Calls," *Quality Progress* (November 1993), pp. 66-73. Copyright © 1993 American Society for Quality Control. Used with permission.

2.57 In the article "Despite Asia's Woes, U.S. Banks Are Standing Tall" in the February 16, 1998, issue of *Fortune*, the following data concerning what Asia owes (bank loans to Indonesia, South Korea, Malaysia, Philippines, and Thailand as of June 1997) are given: Japan—\$97.2 billion; Germany—\$31.7 billion; France—\$24.6 billion; U.S.—\$23.8 billion; Britain—\$16.3 billion; other European and Canada—\$80.9 billion. Use these data to construct a bar chart and a pie chart.

2.58 In the Fall 1993 issue of *VALIC Investment Digest*, the Variable Annuity Life Insurance Company used pie charts to illustrate an investment strategy called rebalancing. This strategy involves reviewing an investment portfolio annually to return the asset mix (stocks, bonds, Treasury bills, and so on) to a prespecified allocation mix. VALIC describes rebalancing as follows (refer to the pie charts in Figure 2.46):

Rebalancing—A Strategy to Keep Your Allocation on Track

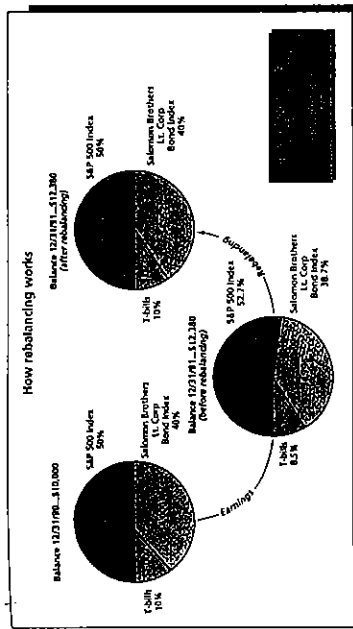
Once you've established your ideal asset allocation mix, many experts recommend that you review your portfolio at least once a year to make sure your portfolio remains consistent with your prespecified asset allocation mix. This practice is referred to as *rebalancing*.

For example, let's assume a moderate asset allocation mix of 50 percent equities funds, 40 percent bond funds, and 10 percent cash equivalent funds. The chart (see Figure 2.46) based on data provided by Ibbotson, a major investment and consulting firm, illustrates how rebalancing works. Using the Standard & Poor's 500 Index, the Salomon Brothers Long-Term High-Grade Corporate Bond Index, and the U.S. 30-day Treasury bill average as a cash equivalent rate, our hypothetical portfolio balance on 12/31/90 is \$10,000. One year later the account had grown to \$12,380. By the end of 1991, the allocation had changed to 52.74%/38.74%/8.5%. The third pie chart illustrates how the account was once again rebalanced to return to a 50%/40%/10% asset allocation mix.

Rebalancing has the potential for more than merely helping diversify your portfolio. By continually returning to your original asset allocation, it is possible to avoid exposure to more risk than you previously decided you were willing to assume.

- Suppose you control a \$100,000 portfolio and have decided to maintain an asset allocation mix of 60 percent stock funds, 30 percent bond funds, and 10 percent government securities. Draw a pie chart illustrating your portfolio (like the ones in Figure 2.46).
- Over the next year your stock funds earn a return of 30 percent, your bond funds earn a return of 15 percent, and your government securities earn a return of 6 percent. Calculate the end-of-year values of your stock funds, bond funds, and government securities. After calculating the end-of-year value of your entire portfolio, determine the asset allocation mix (percent stock funds, percent bond funds, and percent government securities) of your portfolio before rebalancing. Finally, draw an end-of-year pie chart of your portfolio before rebalancing.
- Rebalance your portfolio. That is, determine how much of the portfolio's end-of-year value must be invested in stock funds, bond funds, and government securities in order to restore your original asset allocation mix of 60 percent stock funds, 30 percent bond funds, and 10 percent government securities. Draw a pie chart of your portfolio after rebalancing.

FIGURE 2.46 Using Pie Charts to Illustrate Portfolio Rebalancing



Source: The Variable Annuity Life Insurance Company, VALIC 6, no. 4 (Fall 1993).

*2.6 Using Scatter Plots to Study Relationships between Variables

Statistical methods are often used to study and quantify relationships between variables. The purpose of studying such relationships is often to describe, predict, or control a variable of interest called the *dependent variable* (which is denoted y). We accomplish this by relating y to one or more other variables that are called *independent variables*. One way to relate variables is to perform *regression analysis*. This technique allows us to find an *equation* that relates y to the independent variable(s). Then, for instance, we might use the regression equation to predict y on the basis of the independent variable(s). We explain regression analysis in detail in Chapters 11 and 12. A simpler way to relate variables is to *graphically* study relationships between the variables. We discuss the graphical approach in this section.

One way to explore the relationship between a dependent variable y and an independent variable (denoted x) is to make a *scatter diagram*, or *scatter plot*, of y versus x . First, data concerning the two variables are observed in pairs. To construct the scatter plot, each value of y is plotted against its corresponding value of x . If y and x are related, the plot shows us the direction of the relationship. That is, y could be positively related to x (y increases as x increases) or y could be negatively related to x (y decreases as x increases).

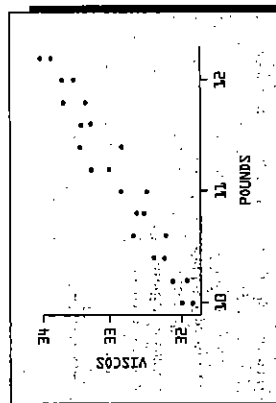
EXAMPLE

Suppose that a manufacturer produces a bulk chemical product. One important quality characteristic of this product is its viscosity. It is felt that the amount of a certain chemical, chemical XB-135, used in producing the product may be related to viscosity. In order to verify and quantify this relationship, 24 batches of the product are produced. The amount (x) of chemical XB-135 (in pounds) is varied from batch to batch and the viscosity (y) obtained for each batch is measured. Table 2.12 gives (in time order) the values of x and the corresponding values of y obtained for the 24 batches. The MINITAB output of a scatter plot of y (labeled VISCOS on the output) versus x (labeled POUNDS) is given in Figure 2.47. The scatter plot indicates a strong positive relationship between y and x —that is, as the amount of chemical XB-135 used is increased, the viscosity of the product increases. We must now be careful. It would be tempting to conclude that increases in the amount of chemical XB-135 *cause* increases in viscosity. However, this is not necessarily the case. Perhaps some other factor could be causing the apparent

TABLE 2.12 Viscosity Data for 24 Batches of a Chemical Product Produced on August 1, 2002

Batch	Pounds of Chemical		Viscosity	
	XB-135 (x)	Batch	XB-135 (x)	Viscosity (y)
1	10.0	13	11.2	32.93
2	10.0	14	11.2	33.19
3	10.2	15	11.4	33.35
4	10.2	16	11.4	32.76
5	10.4	17	11.6	33.33
6	10.4	18	11.6	33.19
7	10.6	19	11.8	33.28
8	10.6	20	11.8	33.57
9	10.8	21	12.0	33.60
10	10.8	22	12.0	33.43
11	11.0	23	12.2	33.91
12	11.0	24	12.2	33.76

FIGURE 2.47 MINITAB Output of a Scatter Plot of Viscosity (VISCOS) versus Amount of Chemical XB-135 (POUNDS)



relationship. For instance, the 24 batches were produced in time order. If some other variable that affects viscosity (such as temperature or pressure in the reaction chamber, or the composition of a raw material) is changing over time, this change could be responsible for the observed increases in viscosity. Assuming that we have held other variables that may affect viscosity constant, the evidence supporting a cause-and-effect relationship may be quite strong. This is because the manufacturer has purposely varied the amount of chemical XB-135 that is used. However, it is really up to the scientific community to establish and understand any cause-and-effect relationship that may exist.

If we are convinced that we can control viscosity by changing the amount of chemical XB-135 that we use, we may wish to quantify the relationship between y and x . In other words, we would like to develop an equation relating y to x . This can be done by using regression analysis (see Chapter 11). With such an equation, we can predict y on the basis of x , and we can determine the amount of chemical XB-135 to use in order to achieve a specified viscosity.

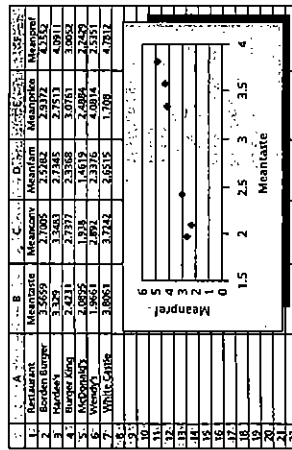
Because the plot points in Figure 2.47 seem to fluctuate around a straight line, we say that there is a straight line (or linear) relationship between y and x . However, not all relationships are linear. For example, demand for a product, y , might increase at an increasing or decreasing rate as advertising expenditure to promote the product, x , increases. In this case, we say that there is a curved relationship between y and x . We discuss curved relationships in Chapter 12.

FIGURE 2.48 A Scatter Plot of Units Sold versus Shelf Space



Source: W. R. Dillon, T. I. Madden, and N. H. Firtle, *Essentials of Marketing Research* (Burr Ridge, IL: Richard D. Irwin, Inc., 1993), p. 452. Copyright © 1992. Reprinted by permission of McGraw-Hill Companies, Inc.

FIGURE 2.49 Excel Output of the Mean Restaurant Ratings and a Scatter Plot of Mean Preference versus Mean Taste • FastFood



Source: The Ohio State University

Exercises for Section 2.6

CONCEPTS

- 2.59 Draw a scatter plot of y versus x in which y increases in a linear (straight-line) fashion as x increases.
- 2.60 Draw a scatter plot of y versus x in which y decreases linearly as x increases.
- 2.61 What is the difference between a scatter plot and a runs plot?

METHODS AND APPLICATIONS

- 2.62 In the book *Essentials of Marketing Research*, William R. Dillon, Thomas J. Madden, and Neil H. Firtle present a scatter plot of the number of units sold of 20 varieties of a canned soup versus the amount of shelf space allocated to each variety. The scatter plot is shown in Figure 2.48.

- Does there appear to be a relationship between y (units sold) and x (shelf space)? Does the relationship appear to be straight line (linear) or curved? How does y (units sold) change as x (shelf space) increases?
- If you were told that a variety of soup is allotted a small amount of shelf space, what would you guess about sales?
- Do you think that the amount of shelf space allocated to a variety causes sales to be higher or lower? Give an alternative explanation for the appearance of the scatter plot.

2.63 THE FAST-FOOD RESTAURANT RATING CASE • FastFood

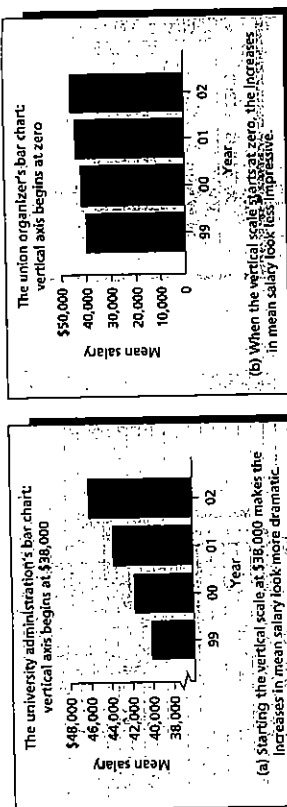
In the early 1990s researchers at The Ohio State University studied consumer ratings of fast-food restaurants. Each of 406 randomly selected individuals rated the six fast-food restaurants shown in the Excel output of Figure 2.49. Each individual gave each restaurant a rating of 1, 2, 3, 4, 5, or 6 on the basis of taste, convenience, familiarity, and price, and then ranked the restaurants from 1 through 6 on the basis of overall preference. In each case, 1 is the best rating and 6 the worst. The mean ratings given by the 406 individuals are given in Figure 2.49 along with a scatter plot of mean preference versus mean taste. Construct scatter plots of mean preference versus each of mean convenience, mean familiarity, and mean price. Then interpret all the scatter plots.

*2.7 Misleading Graphs and Charts

The statistical analyst's goal should be to present the most accurate and truthful portrayal of a data set that is possible. Such a presentation allows managers using the analysis to make informed decisions. However, it is possible to construct statistical summaries that are misleading. Although we do not advocate using misleading statistics, you should be aware of some of the ways statistical graphs and charts can be manipulated in order to distort the truth. By knowing



FIGURE 2.50 Two Bar Charts of the Mean Salaries at a Major University from 1999 to 2002



what to look for, you can avoid being misled by a (we hope) small number of unscrupulous practitioners.

As an example, suppose that the faculty at a major university will soon vote on a proposal to join a union. Both the union organizers and the university administration plan to distribute recent salary statistics to the entire faculty. Suppose that the mean faculty salary at the university and the mean salary increase at the university (expressed as a percentage) for each of the years 1999 through 2002 are as follows:

Year	Mean Salary (All Ranks)	Mean Salary Increase (Percent)
1999	\$40,000	3.0%
2000	41,500	4.0
2001	43,500	4.5
2002	46,100	6.0

The university administration does not want the faculty to unionize and, therefore, hopes to convince the faculty that substantial progress has been made to increase salaries without a union. On the other hand, the union organizers wish to portray the salary increases as minimal so that the faculty will feel the need to unionize.

Figure 2.50 gives two bar charts of the mean salaries at the university for each year from 1999 to 2002. Notice that in Figure 2.50(a) the administration has started the vertical scale of the bar chart at a salary of \$38,000 by using a *scale break* (\$). Alternatively, the chart could be set up without the scale break by simply starting the vertical scale at \$38,000. Starting the vertical scale at a value far above zero makes the salary increases look more dramatic. Notice that when the union organizers present the bar chart in Figure 2.50(b), which has a vertical scale starting at zero, the salary increases look far less impressive.

Figure 2.51 presents two bar charts of the mean salary increases (in percentages) at the university for each year from 1999 to 2002. In Figure 2.51(a), the administration has made the widths of the bars representing the percentage increases proportional to their heights. This makes the upward movement in the mean salary increases look more dramatic because the observer's eye tends to compare the areas of the bars, while the improvements in the mean salary increases are really only proportional to the heights of the bars. When the union organizers present the bar chart of Figure 2.51(b), the improvements in the mean salary increases look less impressive because each bar has the same width.

Figure 2.52 gives two runs plots (also called time series plots) of the mean salary increases at the university from 1999 to 2002. In Figure 2.52(a) the administration has stretched the vertical axis of the graph. That is, the vertical axis is set up so that the distances between the percentages are large. This makes the upward trend of the mean salary increases appear to be steep. In Figure 2.52(b) the union organizers have compressed the vertical axis (that is, the distances between

FIGURE 2.51 Two Bar Charts of the Mean Salary Increases at a Major University from 1999 to 2002

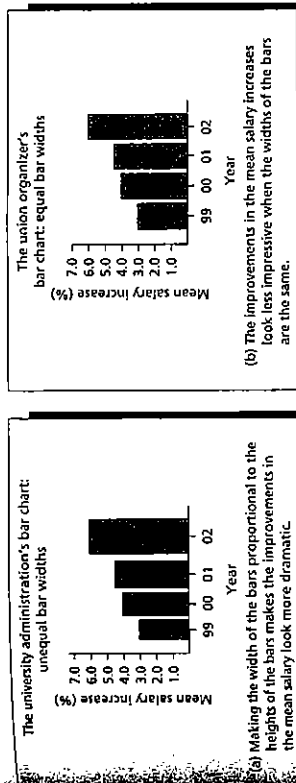
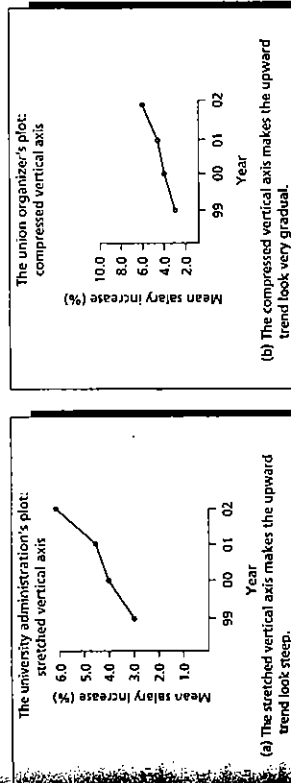


FIGURE 2.52 Two Graphs of the Mean Salary Increases at a Major University from 1999 to 2002



percentages are small). This makes the upward trend of the mean salary increases appear to be gradual. As we will see in the exercises, stretching and compressing the horizontal axis in a line plot can also greatly affect the impression given by the plot.

It is also possible to create totally different interpretations of the same statistical summary by simply using different labeling or captions. For example, consider the bar chart of mean salary increases in Figure 2.51(b). To create a favorable interpretation, the university administration might use the caption "Salary Increase Is Higher for the Fourth Year in a Row." On the other hand, the union organizers might create a negative impression by using the caption "Salary Increase Fails to Reach 10% for Fourth Straight Year."

In summary, we do not approve of using statistics to mislead and distort reality. Statistics should be used to present the most truthful and informative summary of the data that is possible. However, it is important to carefully study any statistical summary so that you will not be misled. Look for manipulations such as stretched or compressed axes on graphs, axes that do not start at zero, and bar charts with bars of varying widths. Also, carefully think about assumptions, and make your own conclusions about the meaning of any statistical summary rather than relying on captions written by others. Doing these things will help you to see the truth and to make well-informed decisions.

Introduction for Section 2.7

CONCEPTS

- 2.64 When we construct a bar chart or graph, what is the effect of starting the vertical axis at a value that is far above zero? Explain.
- 2.65 Find an example of a misleading use of statistics in a newspaper, magazine, corporate annual report, or other source. Then explain why your example is misleading.

METHODS AND APPLICATIONS

- 2.66 Figure 2.53 gives two more time series plots of the previously discussed salary increases. In Figure 2.53(a) the administration has compressed the horizontal axis. In Figure 2.53(b) the union organizers have stretched the horizontal axis. Discuss the different impressions given by the two time series plots.
- 2.67 In the article "How to Display Data Badly" in the May 1984 issue of *The American Statistician*, Howard Wainer presents a *stacked bar chart* of the number of public and private elementary schools (1929–1970). This bar chart is given in Figure 2.54. Wainer also gives a line graph of the number of private elementary schools (1930–1970). This graph is shown in Figure 2.55.

FIGURE 2.53 Two Graphs of the Mean Salary Increases at a Major University from 1999 to 2002

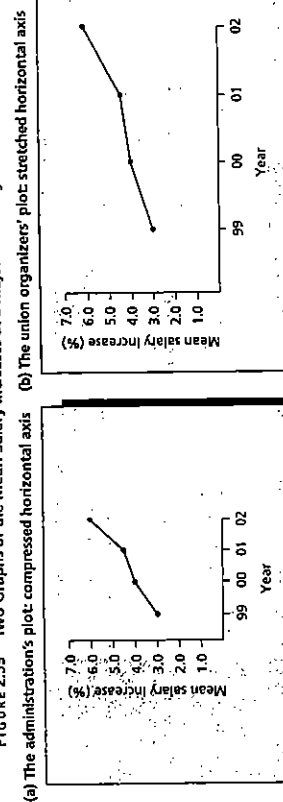


FIGURE 2.54 Wainer's Stacked Bar Chart

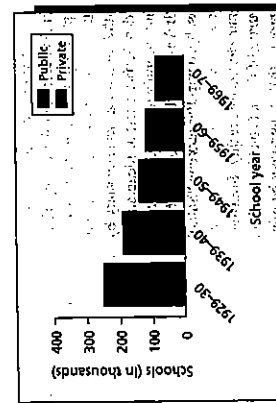
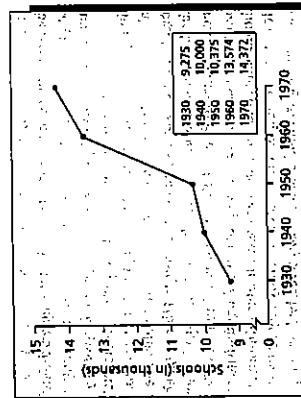


FIGURE 2.55 Wainer's Line Graph



Source: H. Wainer, "How to Display Data Badly," *The American Statistician*, May 1984, pp. 135–40. Copyright © 1984 American Statistical Association. Used with permission.

- a Looking at the bar chart of Figure 2.54, does there appear to be an increasing trend in the number of private elementary schools from 1930 to 1970?
- b Looking at the line graph of Figure 2.55, does there appear to be an increasing trend in the number of private elementary schools from 1930 to 1970?
- c Which portrayal of the data do you think is more appropriate? Explain why.
- d Is either portrayal of the data entirely appropriate? Explain.

No, very slight (if any)
Yes, strong trend
Line graph
Probably not

*2.8 Weighted Means and Grouped Data

Weighted means In Section 2.2 we studied the mean, which is an important measure of central tendency. In order to calculate a mean, we sum the population (or sample) measurements, and then divide this sum by the number of measurements in the population (or sample). When we do this, each measurement counts equally. That is, each measurement is given the same importance or weight.

Sometimes it makes sense to give different measurements unequal weights. In such a case, a measurement's weight reflects its importance, and the mean calculated using the unequal weights is called a **weighted mean**.

We calculate a weighted mean by multiplying each measurement by its weight, summing the resulting products, and dividing the resulting sum by the sum of the weights:

Weighted Mean

The weighted mean equals

$$\frac{\sum w_i x_i}{\sum w_i}$$

where

x_i = the value of the i th measurement

w_i = the weight applied to the i th measurement

Such a quantity can be computed for a population of measurements or for a sample of measurements.

In order to illustrate the need for a weighted mean and the required calculations, consider the June 2001 unemployment rates for various regions in the United States.¹⁰

Census Region	Civilian Labor Force (Millions)	Unemployment Rate
Northeast	26.9	4.1%
South	50.6	4.7%
Midwest	34.7	4.4%
West	32.5	5.0%

If we wish to compute a mean unemployment rate for the entire United States, we should use a weighted mean. This is because each of the four regional unemployment rates applies to a different number of workers in the labor force. For example, the 4.7 percent unemployed for the South applies to a labor force of 50.6 million workers and thus should count more heavily than the 5.0 percent unemployed for the West, which applies to a smaller labor force of 32.5 million workers.

The unemployment rate measurements are $x_1 = 4.1$ percent, $x_2 = 4.7$ percent, $x_3 = 4.4$ percent, and $x_4 = 5.0$ percent, and the weights applied to these measurements are $w_1 = 26.9$, $w_2 = 50.6$, $w_3 = 34.7$, and $w_4 = 32.5$. That is, we are weighting the unemployment rates by the

¹⁰Source: U.S. Bureau of Labor Statistics, <http://stats.bls.gov/news.release/flus.t01.htm>, August 7, 2001.

regional labor force sizes. The weighted mean is computed as follows:

$$\begin{aligned}\mu &= \frac{26.9(4.1) + 50.6(4.7) + 34.7(4.4) + 32.5(5.0)}{26.9 + 50.6 + 34.7 + 32.5} \\ &= \frac{663.29}{144.7} = 4.58\%\end{aligned}$$

In this case the unweighted mean of the four regional unemployment rates equals 4.55 percent. Therefore, the unweighted mean understates the U.S. unemployment rate by .03 percent (or understates U.S. unemployment by .0003(144.7 million) = 43,410 workers).

The weights chosen for calculating a weighted mean will vary depending on the situation. For example, in order to compute the mean percentage return for a portfolio of investments, the percentage returns for various investments might be weighted by the dollar amounts invested in each. Or in order to compute a mean profit margin for a company consisting of several divisions, the profit margins for the different divisions might be weighted by the sales volumes of the divisions. Again, the idea is to choose weights that represent the relative importance of the measurements in the population or sample.

Descriptive statistics for grouped data We usually calculate measures of central tendency and variability using the individual measurements in a population or sample. However, sometimes the only data available are in the form of a frequency distribution or a histogram. For example, newspapers and magazines often summarize data using frequency distributions and histograms without giving the individual measurements in a data set. Data summarized in frequency distribution or histogram form are often called **grouped data**. In this section we show how to compute descriptive statistics for such data.

Suppose we are given a frequency distribution summarizing a sample of 65 customer satisfaction ratings for a consumer product.

Satisfaction Rating	Frequency
36-38	4
39-41	15
42-44	25
45-47	19
48-50	2

Because we do not know each of the 65 individual satisfaction ratings, we cannot compute an exact value for the mean satisfaction rating. However, we can calculate an approximation of this mean. In order to do this, we use the midpoint of each class to represent the measurements in each class. When we do this, we are really assuming that the average of the measurements in each class equals the class midpoint. Letting M_i denote the midpoint of class i , and letting f_i denote the frequency of class i , we compute the mean by calculating a weighted mean of the class midpoints using the class frequencies as the weights. The logic here is that if f_i measurements are included in class i , then the midpoint of class i should count f_i times in the weighted mean. In this case, the sum of the weights equals the sum of the class frequencies, which equals the sample size. Therefore, we obtain the following equation for the sample mean of grouped data:

Sample Mean for Grouped Data

$$\bar{x} = \frac{\sum f_i M_i}{\sum f_i} = \frac{\sum f_i M_i}{n}$$

where
 f_i = the frequency for class i
 M_i = the midpoint for class i
 $n = \sum f_i$ = the sample size

TABLE 2.13 Calculating the Sample Mean Satisfaction Rating

Satisfaction Rating	Frequency (f_i)	Class Midpoint (M_i)	$f_i M_i$
36-38	4	37	148
39-41	15	40	600
42-44	25	43	1,075
45-47	19	46	874
48-50	2	49	98
	$n = 65$		2,795

$$\bar{x} = \frac{\sum f_i M_i}{n} = \frac{2,795}{65} = 43$$

TABLE 2.14 Calculating the Sample Variance of the Satisfaction Ratings

Satisfaction Rating	Frequency (f_i)	Class Midpoint (M_i)	Deviation ($M_i - \bar{x}$)	Squared Deviation ($(M_i - \bar{x})^2$)	$f_i(M_i - \bar{x})^2$
36-38	4	37	37 - 43 = -6	36	144
39-41	15	40	40 - 43 = -3	9	135
42-44	25	43	43 - 43 = 0	0	0
45-47	19	46	46 - 43 = 3	9	171
48-50	2	49	49 - 43 = 6	36	72
	$\sum f_i = 65$				$\sum f_i(M_i - \bar{x})^2 = 522$

$$s^2 = \text{sample variance} = \frac{\sum f_i(M_i - \bar{x})^2}{n - 1} = \frac{522}{65 - 1} = 8.15625$$

Table 2.13 summarizes the calculation of the mean satisfaction rating for the previously given frequency distribution of satisfaction ratings. Note that in this table each midpoint is halfway between its corresponding class limits. For example, for the first class $M_1 = (36 + 38)/2 = 37$. We find that the sample mean satisfaction rating is 43.

We can also compute an approximation of the sample variance for grouped data. Recall that when we compute the sample variance using individual measurements, we compute the squared deviation from the sample mean $(x_i - \bar{x})^2$ for each individual measurement x_i and then sum the squared deviations. For grouped data, we do not know each of the x_i values. Because of this, we again let the class midpoint M_i represent each measurement in class i . It follows that we compute the squared deviation $(M_i - \bar{x})^2$ for each class and then sum these squares, weighting each squared deviation by its corresponding class frequency f_i . That is, we approximate $\sum (x_i - \bar{x})^2$ by using $\sum f_i(M_i - \bar{x})^2$. Finally, we obtain the sample variance for the grouped data by dividing this quantity by the sample size minus 1. We summarize this calculation in the following box:

Sample Variance for Grouped Data

$$s^2 = \frac{\sum f_i(M_i - \bar{x})^2}{n - 1}$$

where $\bar{x}$ is the sample mean for the grouped data.

Table 2.14 illustrates calculating the sample variance of the previously given frequency distribution of satisfaction ratings. We find that the sample variance is $s^2 = 8.15625$ and, therefore, that the sample standard deviation is $s = \sqrt{8.15625} = 2.8559$.

Finally, although we have illustrated calculating the mean and variance for grouped data in the context of a sample, similar calculations can be done for a population of measurements. If we let

N be the size of the population, the grouped data formulas for the population mean and variance are given in the following box:

Population Mean for Grouped Data

$$\mu = \frac{\sum fM_i}{N}$$

Population Variance for Grouped Data

$$\sigma^2 = \frac{\sum f(M_i - \mu)^2}{N}$$

2.8

CONCEPTS

- 2.68 Consider calculating a student's grade point average using a scale where 4.0 represents an A and 0.0 represents an F. Explain why the grade point average is a weighted mean. What are the x_i values? What are the weights?
- 2.69 When we perform grouped data calculations, we represent the measurements in a class by using the midpoint of the class. Explain the assumption that is being made when we do this.
- 2.70 When we compute the mean, variance, and standard deviation using grouped data, the results obtained are approximations of the population (or sample) mean, variance, and standard deviation. Explain why this is true.

METHODS AND APPLICATIONS

- 2.71 According to the Morningstar.com website, the year-to-date total return percentages on August 8, 2001, for several popular funds were as follows:

Fund	Total Return % (ytd)
Vanguard 500 Index	-9.76
Windsor Core Growth	25.46
Fidelity Low-Priced Stock	16.52
Fidelity Dividend Growth	-2.37
Janus Worldwide	-18.84

Source: <http://quote.morningstar.com/jopreports.html>, August 9, 2001.

Suppose that an investor has \$100,000 invested in the Vanguard 500 Index fund, \$500,000 invested in the Windsor Core Growth fund, \$500,000 invested in the Fidelity Low-Priced Stock fund, \$200,000 invested in the Fidelity Dividend Growth fund, and \$50,000 invested in the Janus Worldwide fund.

- a Compute a weighted mean that measures the average total return for the investor's portfolio.
b Compare your weighted mean with the unweighted mean of the five total return percentages. Explain why they differ.

weighted: 13.716%
unweighted: 2.202%

- 2.72 The following are the May 2001 unemployment rates and civilian labor force sizes for five states in the Midwest.

State	Size of Civilian Labor Force (Millions)	Unemployment Rate (%)
Iowa	1.57	2.8
Michigan	5.24	5.0
Illinois	6.41	5.2
Indiana	3.10	3.1
Wisconsin	3.05	4.1

Source: United States Bureau of Labor Statistics, <http://stats.bls.gov>, August 9, 2001.

- a Using a weighted mean, compute an average unemployment rate for the five state region.
b Calculate the unweighted mean for the five unemployment rates. Explain why the weighted and unweighted means differ.

4.44%
4.04%

2.9

The Geometric Mean

- 2.73 The following frequency distribution summarizes the weights of 195 fish caught by anglers participating in a professional bass fishing tournament.

Weight (Pounds)	Frequency
1-3	53
4-6	118
7-9	21
10-12	3

- a Calculate the (approximate) sample mean for this data.
b Calculate the (approximate) sample variance for this data.
- 2.74 The following is a frequency distribution summarizing revenue growth data for the 20 fastest-growing firms as given on *Fortune* magazine's website (as of August 7, 2001).

Revenue Growth (Percent)	Frequency
0-49	2
50-99	10
100-149	7
150-199	0
200-249	1

Source: <http://www.fortune.com>, August 7, 2001.

94.5
1850
43.01

- Calculate the (approximate) population mean, variance, and standard deviation for these data.
- 2.75 The Data and Story Library website (a website devoted to applications of statistics) gives a histogram of the ages of a sample of 60 CEOs taken in 1993. We present the data in the form of a frequency distribution below.

Age (Years)	Frequency
28-32	1
33-37	3
38-42	3
43-47	13
48-52	14
53-57	12
58-62	9
63-67	1
68-72	3
73-77	1

Source: <http://lib.stat.cmu.edu/DASIS/story/ceos.html>, August 7, 2001.

51.5
81.61
9.03

- Calculate the (approximate) sample mean, variance, and standard deviation of this data.

*2.9 The Geometric Mean

In Section 2.2 we defined the mean to be the average of a set of population or sample measurements. This mean is sometimes referred to as the arithmetic mean. While very useful, the arithmetic mean is not a good measure of the rate of change exhibited by a variable over time. To see this, consider the rate at which the value of an investment changes—its rate of return. Suppose that an initial investment of \$10,000 increases in value to \$20,000 at the end of one year and then decreases in value to its original \$10,000 value after two years. The rate of return for the first year, R_1 , is

$$R_1 = \left(\frac{20,000 - 10,000}{10,000} \right) \times 100\% = 100\%$$

and the rate of return for the second year, R_2 , is

$$R_2 = \left(\frac{10,000 - 20,000}{20,000} \right) \times 100\% = -50\%$$

Although the value of the investment at the beginning and end of the two-year period is the same, the arithmetic mean of the yearly rates of return is $(R_1 + R_2)/2 = (100\% + (-50\%))/2 = 25\%$. This arithmetic mean does not communicate the fact that the value of the investment is unchanged at the end of the two years.

To remedy this situation, we define the geometric mean of the returns to be the constant return R_g that yields the same wealth at the end of the investment period as do the actual returns. In our example, this says that if we express R_1 , R_2 , and R_3 as decimal fractions (here $R_1 = 1$ and $R_2 = -0.5$),

$$(1 + R_1)^2 \times 10,000 = (1 + R_1)(1 + R_2) \times 10,000$$

or

$$\begin{aligned} R_g &= \sqrt{(1 + R_1)(1 + R_2)} - 1 \\ &= \sqrt{(1 + 1)(1 + (-0.5))} - 1 \\ &= \sqrt{1} - 1 = 0 \end{aligned}$$

Therefore, the geometric mean R_g expresses the fact that the value of the investment is unchanged after two years.

In general, if $R_1, R_2, \dots, R_n$ are returns (expressed in decimal form) over n time periods:

The geometric mean of the returns $R_1, R_2, \dots, R_n$ is

$$R_g = \sqrt[n]{(1 + R_1)(1 + R_2) \cdots (1 + R_n)} - 1$$

and the ending value of an initial investment I experiencing returns $R_1, R_2, \dots, R_n$ is $I(1 + R_g)^n$. As another example, suppose that in year 3 our investment's value increases to \$25,000, which says that the rate of return for year 3 (expressed as a percentage) is

$$\begin{aligned} R_3 &= \left(\frac{25,000 - 10,000}{10,000} \right) \times 100\% \\ &= 150\% \end{aligned}$$

Since (expressed as decimals) $R_1 = 1$, $R_2 = -0.5$, and $R_3 = 1.5$, the geometric mean return at the end of year 3 is

$$\begin{aligned} R_g &= \sqrt[3]{(1 + 1)(1 + (-0.5))(1 + 1.5)} - 1 \\ &= \sqrt[3]{1.575} - 1 \\ &= .3572 \end{aligned}$$

and the value of the investment after 3 years is

$$10,000(1 + .3572)^3 = \$25,000$$

CONCEPTS

2.76 In words, explain the interpretation of the geometric mean return for an investment.

2.77 If we know the initial value of an investment and its geometric mean return over a period of years, can we compute the ending value of the investment? If so, how?

METHODS AND APPLICATIONS

- 2.78 Suppose that a company's sales were \$5,000,000 three years ago. Since that time sales have grown at annual rates of 10 percent, -10 percent, and 25 percent.
- Find the geometric mean growth rate of sales over this three-year period. .0736
 - Find the ending value of sales after this three-year period. \$6,187,248
- 2.79 Suppose that a company's sales were \$1,000,000 four years ago and are \$4,000,000 at the end of the four years. Find the geometric mean growth rate of sales. .4142
- 2.80 The Standard and Poor's 500 stock index is a commonly used measure of stock market performance in the United States. In the table below, we give the value of the S & P 500 index on the first day of market trading for each year from 1997 to 2002.

Year	S&P 500 Index
1997	737.01
1998	975.04
1999	1,228.10
2000	1,455.22
2001	1,283.27
2002	1,154.67

Source: <http://table.finance.yahoo.com>.

- Show that the percentage changes (rates of return) for the S&P 500 index for the years from 1997 to 1998 and from 1998 to 1999 are, respectively, 32.3 percent and 26.0 percent (that is, 323 and 260 expressed to decimal fractions). .185
 - Find the rates of return for the S&P 500 index for each of the years: from 1999 to 2000; from 2000 to 2001; from 2001 to 2002. -.118
-.100
.094
 - Calculate the geometric mean return for the S&P 500 index over the period from 1997 to 2002. \$1,567,063
 - Suppose that an investment of \$1,000,000 is made in 1997 and that the portfolio performs with returns equal to those of the S&P 500 index. What is the investment portfolio worth in 2002?
- 2.81 According to the USA Statistics in Brief summary of U.S. census data, the amount of consumer credit outstanding (in billions of dollars) is as follows¹¹:
- | | | |
|--------------|----------------|----------------|
| 1990 : \$789 | 1995 : \$1,096 | 2000 : \$1,534 |
|--------------|----------------|----------------|
- Find the geometric mean five-year rate of increase in consumer credit outstanding. .39434
 - Use the geometric mean rate of increase to project the amount of consumer credit outstanding in 2005. 2,139

Chapter Summary

We began this chapter by studying how to depict the shape of the distribution of a data set. We learned that stem-and-leaf displays and histograms are useful graphics for portraying a data set's distribution. We also learned about some common population shapes. We saw that data sets often have shapes that are symmetrical, skewed with a tail to the right, or skewed with a tail to the left.

Next we presented and compared several measures of central tendency. We defined the population mean and we saw how to estimate the population mean by using a sample mean. We also defined the median and mode, and we compared the mean, median, and mode for symmetrical distributions and for distributions that are skewed to the right or left. We then studied

measures of variation (or spread). We defined the range, variance, and standard deviation, and we saw how to estimate a population variance and standard deviation by using a sample. We learned that a good way to interpret the standard deviation when a population is (approximately) normally distributed is to use the empirical rule, and we applied this rule to assess process capability. We next studied Chebyshev's Theorem, which gives us intervals containing reasonably large fractions of the population units no matter what the population's shape might be. We also saw that, when a data set is highly skewed, it is best to use percentiles and quartiles to measure variation, and we learned how to construct a box-and-whiskers plot by using the quartiles.

¹¹Source: <http://www.census.gov/statlab/www/part5.html>.

After learning how to measure and depict central tendency and variability, we presented several methods for portraying qualitative data. In particular, we used bar charts and pie charts for this purpose. We also discussed using a sample to estimate the proportion of population units that fall into a category of interest. We concluded this chapter with several optional sections. First, we studied using scatter plots to examine relationships

Glossary of Terms

bar chart: A graphical display of categorical data (data in categories) made up of vertical or horizontal bars. (page 84)

box-and-whiskers display (box plot): A graphical portrayal of a data set that depicts both the central tendency and variability of the data. It is constructed using Q_1 , M_e , and Q_3 . (page 77)

capable process: A process that is able to consistently produce output that meets (or conforms to) specifications (requirements). (page 69)

central tendency: A term referring to the middle of a population or sample of measurements. (page 55)

Chebyshev's Theorem: A theorem that (for any population) allows us to find an interval that contains a specified percentage of the individual measurements in the population. (page 71)

coefficient of variation: A quantity that measures the variation of a population or sample relative to its mean. (page 72)

dependent variable (denoted y): A variable that we wish to describe, predict, or control. (page 91)

Empirical Rule: For a normally distributed population, this rule tells us that 68.26 percent, 95.44 percent, and 99.73 percent, respectively, of the population measurements are within one, two, and three standard deviations of the population mean. (page 67)

extreme outlier (in a box-and-whiskers display): Measurements located outside the outer fences. (page 79)

extreme value: A measurement in a population or sample that is different from most of the other measurements. (page 59)

first quartile (denoted Q_1): A value below which approximately 25 percent of the measurements lie; the 25th percentile. (page 76)

frequency: The count of the number of measurements in a class or of the number of measurements having a particular value. (page 43)

frequency distribution: A numerical summary that divides the values of a variable into classes and gives the number of values in each class. (page 43)

geometric mean: The constant return (or rate of change) that yields the same wealth at the end of several time periods as do actual returns. (page 102)

grouped data: Data presented in the form of a frequency distribution or a histogram. (page 98)

histogram: A graphical portrayal of a data set that shows the data set's distribution. It divides the data into classes and gives the frequency for each class. Histograms are particularly useful for summarizing large data sets. (page 43)

independent variable (denoted x): A predictor variable that can be used to describe, predict, or control a dependent variable. (page 91)

inner fences (in a box-and-whiskers display): Points located $1.5 \times IQR$ below Q_1 and $1.5 \times IQR$ above Q_3 . (page 77)

interquartile range (denoted IQR): The difference between the third quartile and the first quartile (that is, $Q_3 - Q_1$). (page 77)

measure of variation: A descriptive measure of the spread of the values in a population or sample. (page 64)

median (denoted M_e): A measure of central tendency that divides a population or sample into two roughly equal parts. (page 57)

mild outlier (in a box-and-whiskers display): Measurements located between the inner and outer fences. (page 79)

mode (denoted M_d): The measurement in a sample or a population that occurs most frequently. (page 58)

inward-shaped: Description of a relative frequency curve that is "piled up in the middle." (page 70)

normal curve: A bell-shaped, symmetrical relative frequency curve. We will present the exact equation that gives this curve in Chapter 5. (page 47)

outlier: An unusually large or small observation that is well separated from the remaining observations. (page 48)

Pareto chart: A bar chart of the frequencies or percentages for various types of defects. These are used to identify opportunities for improvement. (page 78)

percentile: The value such that a specified percentage of the measurements in a population or sample fall at or below it. (page 75)

pie chart: A graphical display of categorical data (data in categories) made up of "pie slices." (page 84)

point estimate: A one-number estimate for the value of a population parameter. (page 55)

population mean (denoted μ): The average of a population of measurements. (page 55)

population parameter: A descriptive measure of a population. It is calculated using the population measurements. (page 55)

population proportion (denoted p): The proportion of population units that are contained in a category of interest. (page 84)

population standard deviation (denoted σ): The positive square root of the population variance. It is a measure of the variation of the population measurements. (page 65)

population variance (denoted σ^2): The average of the squared deviations of the individual population measurements from the population mean. It is a measure of the variation of the population measurements. (page 65)

range: The difference between the largest and smallest measurements in a population or sample. It is a simple measure of variation. (page 64)

relative frequency: The frequency of a class divided by the total number of measurements. (page 44)

relative frequency curve: A curve that describes the shape of a population of measurements. (page 48)

relative frequency histogram: A graphical portrayal of a data set that shows the data set's distribution. It divides the data into classes, gives the relative frequency for each class, and is particularly useful for summarizing large data sets. (page 44)

sample mean (denoted $\bar{x}$): The average of the measurements in a sample. It is the point estimate of the population mean. (page 56)

sample proportion (denoted $\hat{p}$): The proportion of sample elements that are contained in a category of interest. (page 84)

sample size (denoted n): The number of measurements in a sample. (page 55)

sample standard deviation (denoted s): The positive square root of the sample variance. It is the point estimate of the population standard deviation. (page 66)

sample statistic: A descriptive measure of a sample. It is calculated from the measurements in the sample. (page 55)

sample variance (denoted s^2): A measure of the variation of the sample measurements. It is the point estimate of the population variance. (page 66)

Important Formulas

The population mean, μ : page 55
 The sample mean, $\bar{x}$: page 56
 The median: page 57
 The mode: page 58
 The population range: page 64
 The population variance, σ^2 : page 65
 The population standard deviation, σ : page 65
 The sample variance, s^2 : page 66
 The sample standard deviation, s : page 66
 Computational formula for $\bar{x}$: page 67
 The Empirical Rule: page 67

Chebyshev's Theorem: page 71
 z-score: page 71
 The coefficient of variation: page 72
 The p th percentile: page 75
 The quartiles: page 76
 The weighted mean: page 97
 Sample mean for grouped data: page 98
 Sample variance for grouped data: page 99
 Population mean for grouped data: page 100
 Population variance for grouped data: page 100
 The geometric mean: page 102

Supplementary Exercises

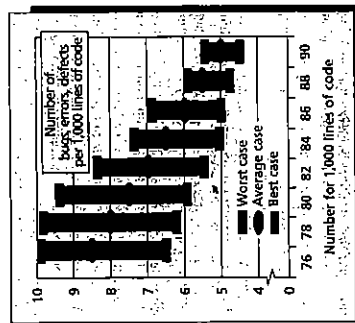
- 2.82 In the book *Modern Statistical Quality Control and Improvement*, Nicholas R. Parnum presents data concerning the elapsed times from the completion of medical lab tests until the results are recorded on patients' charts. Table 2.15 gives the times it took (in hours) to deliver and chart the results of 84 lab tests over one week.  Lab Test
- Construct a frequency histogram and a relative frequency histogram for the lab test waiting time data.
 - Looking at the histogram, are most of the test results delivered and charted within several hours?
 - Are there some deliveries with excessively long waiting times? Which deliveries might be investigated in order to discover reasons behind unusually long delays?
- 2.83 Figure 2.56 depicts data for a study of 80 software projects at NASA's Goddard Space Center. The figure shows the number of bugs per 1,000 lines of code from 1976 to 1990. Write a short paragraph describing how the reliability of the software has improved. Explain how the data indicate improvement.

TABLE 2.15 Elapsed Time (in Hours) for Completing and Delivering Medical Lab Tests
• LabTest

6.1	8.7	1.1	4.0
2.1	3.9	2.2	5.0
2.1	7.1	4.3	8.8
3.5	1.2	3.2	1.3
1.3	9.3	4.2	7.3
5.7	6.5	4.4	16.2
1.3	1.3	3.0	2.7
15.7	4.9	2.0	5.2
3.9	13.9	1.8	2.2
8.4	5.2	11.9	3.0
24.0	24.5	24.8	24.0
1.7	4.4	2.5	16.2
17.8	2.9	4.0	6.7
5.3	8.3	2.8	5.2
17.5	1.1	3.0	8.3
1.2	1.1	4.5	4.4
5.0	2.6	12.7	5.7
4.7	5.1	2.6	1.6
3.4	8.1	2.4	16.7
4.8	1.7	1.9	12.1
9.1	5.6	13.0	6.4

Source: N. R. Farum, *Modern Statistical Quality Control and Improvement*, p. 55. Reprinted by permission of Brooks/Cole, an imprint of the Wadsworth Group, a division of Thompson Learning, Fax 800-730-2215.

FIGURE 2.56 Software Performance at NASA's Goddard Space Center, 1976-1990



Source: Reprinted from the January 15, 1992, issue of *Business Week* by special permission. Copyright © 1992 by The McGraw-Hill Companies.

2.84 THE INVESTMENT CASE • InvestRet

The Fall 1995 issue of *Investment Digest*, a publication of The Variable Annuity Life Insurance Company of Houston, Texas, discusses the importance of portfolio diversification for long-term investors. The article states:

"While it is true that investment experts generally advise long-term investors to invest in 'viable' investments, they also agree that the key to any sound investment portfolio is diversification. That is, investing in a variety of investments with differing levels of historical return and risk.

Investment risk is often measured in terms of the volatility of an investment over time. When volatility, sometimes referred to as *standard deviation*, increases, so too does the level of return. Conversely, as risk (standard deviation) declines, so too do returns.

In order to explain the relationship between the return on an investment and its risk, *Investment Digest* presents a graph of mean return versus standard deviation (risk) for nine investment classes over the period from 1970 to 1994. This graph, which *Investment Digest* calls the "risk/return trade-off," is shown in Figure 2.57. The article says that this graph

... illustrates the historical risk/return trade-off for a variety of investment classes over the 24-year period between 1970 and 1994.

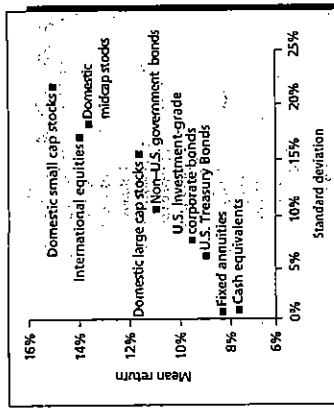
In the chart, cash equivalents and fixed annuities, for instance, had a standard deviation of 0.81% and 0.54% respectively, while posting returns of just over 7.73% and 8.31%. At the other end of the spectrum, domestic small-company stocks were quite volatile—with a standard deviation of 21.82%—but compensated for that increased volatility with a return of 14.93%.

The answer seems to lie in asset allocation. Investment experts know the importance of asset allocation. In a nutshell, asset allocation is a method of creating a diversified portfolio of investments that minimize historical risk and maximize potential returns to help you meet your retirement goals and needs.

Suppose that, by reading off the graph of Figure 2.57, we obtain the mean return and standard deviation combinations for the various investment classes as shown in Table 2.16.

Further suppose that future returns in each investment class will behave as they have from 1970 to 1994. That is, for each investment class, regard the mean return and standard deviation in

FIGURE 2.57 The Risk/Return Trade-Off



Source: The Variable Annuity Life Insurance Company, *VAUC*, (1995), no. 3.

Table 2.16 as the population mean and the population standard deviation of all possible future returns. Then do the following:

a. Assuming that future returns for the various investment classes are mound-shaped, for each investment class compute intervals that will contain approximately 68.26 percent and 99.73 percent of all future returns.

b. Making no assumptions about the population shapes of future returns, for each investment class compute intervals that will contain at least 75 percent and 88.89 percent of all future returns.

c. Assuming that future returns are mound-shaped, find
(1) An estimate of the maximum return that might be realized for each investment class.
(2) An estimate of the minimum return (or maximum loss) that might be realized for each investment class.

d. Assuming that future returns are mound-shaped, which two investment classes have the highest estimated maximum returns? What are the estimated minimum returns (maximum losses) for these investment classes?

e. Assuming that future returns are mound-shaped, which two investment classes have the smallest estimated maximum returns? What are the estimated minimum returns for these investment classes?

f. Calculate the coefficient of variation for each investment class and compare the investment classes with respect to risk. Which class is riskiest? Least risky?

2.85 Table 2.17 gives data concerning consumer complaints against U.S. airlines (1993-2000). Depict the data graphically. Then summarize what the data tell you about improvements in airline service from 1993 to 2000. • AirComplaints

2.86 A filling process is supposed to fill jars with 16 ounces of grape jelly. Specifications state that each jar must contain between 15.95 and 16.05 ounces. A jar is selected from the process every half hour until a sample of 100 jars is obtained. When the fills of the jars are measured, we find $\bar{x} = 16.0024$ and $s = .02454$.

a. Assume that the fill process is in statistical control, and also assume that the population of all jar fills is normally distributed. Determine whether the process is capable of meeting specifications by using an interval that you would expect to contain 99.73 percent of all fills.

b. When the process is improved, it is found to be in statistical control. Moreover, a sample of 100 fills yields $\bar{x} = 16.0011$ and $s = .01328$. Is the improved process capable of meeting specifications? Justify your answer.

2.87 THE INTERNATIONAL BUSINESS TRAVEL EXPENSE CASE

Suppose that a large international corporation wishes to obtain its own "benchmark" for one-day travel expenses in Moscow. To do this, it records the one-day travel expenses for a random sample of 35 executives visiting Moscow. The mean and the standard deviation of these expenses

Answers for fixed annuities:

a. 68.26% [7.77, 8.85] 99.73% [6.69, 9.93]

b. At least 75%: [7.23, 9.39]

At least 88.89%: [6.69, 9.93]

c. 9.93

d. Domestic small cap stocks, -50.53

e. Domestic midcap stocks, -40.93

f. Fixed annuities, 6.69

Cash equivalents, 5.30

Riskiest: Domestic small cap stocks

Least risky: Fixed annuities

Not capable

Yes, capable

TABLE 2.17 Consumer Complaints against U.S. Airlines, 1993-2000

Complaint Category	1993	1994	1995	1996	1997	1998	1999	2000
Total	4,438	5,179	4,629	5,782	6,394	7,980	17,345	20,564
Flight problems ¹	1,211	1,586	1,133	1,628	1,659	2,270	6,449	8,698
Customer service ²	599	805	667	959	1,418	1,716	3,557	4,074
Baggage	627	761	628	882	826	1,105	2,351	2,753
Ticketing/boarding ³	577	598	666	857	904	805	1,329	1,405
Refunds	482	393	576	521	531	601	935	803
Overseas ⁴	257	301	263	353	414	387	673	759
Fares ⁵	398	267	185	180	195	276	584	708
Disability ⁶	(NA)	(NA)	(NA)	(NA)	(NA)	331	520	612
Advertising	51	94	66	61	57	39	57	42
Tours	16	127	18	16	13	23	28	25
Animals	(NA)	(NA)	(NA)	(NA)	(NA)	(NA)	(NA)	(NA)
Smoking	30	20	15	13	5	(?)	(?)	(?)
Credit	4	2	4	3	1	(?)	(?)	(?)
Other	186	225	408	269	331	427	759	675

NA—Not available. ¹Cancellations, delays, etc. from schedule. ²Unhelpful employees, inadequate service or cabin service. ³Arrival of delayed passengers, lost or damaged baggage, lost or damaged luggage, lost or damaged baggage, lost or damaged baggage. ⁴Lost or damaged baggage, lost or damaged baggage, lost or damaged baggage, lost or damaged baggage. ⁵Lost or damaged baggage, lost or damaged baggage, lost or damaged baggage, lost or damaged baggage. ⁶Lost or damaged baggage, lost or damaged baggage, lost or damaged baggage, lost or damaged baggage.

Source: U.S. Dept. of Transportation, Office of Consumer Affairs, Air Travel Consumer Report, monthly.

are calculated to be $\bar{x} = \$538$ and $s = \$41$. Furthermore, a histogram shows that the expenses are approximately normally distributed.

- a. Find an interval that we estimate contains 99.73 percent of all one-day travel expenses in Moscow.
b. If an executive submits an expense of \$720 for a one-day stay in Moscow, should this expense be considered unusually high? Why or why not?

2.88 THE UNITED KINGDOM INSURANCE CASE

Figure 2.58 summarizes information concerning insurance expenditures of households in the United Kingdom in 1993.

- a. Approximately what percentage of households spent on life insurance?
b. What is the approximate average expenditure (in UKL) per household on life insurance? Note: the averages given in Figure 2.58 are for households that spend in the class.

2.89 Figure 2.59 was used in various Chevrolet magazine advertisements in 1997 to compare the overall resale values of Chevrolet, Dodge, and Ford trucks in the years from 1990 to 1997. What is somewhat misleading about this graph?

FIGURE 2.58 Insurance Expenditures of Households in the United Kingdom (1993)

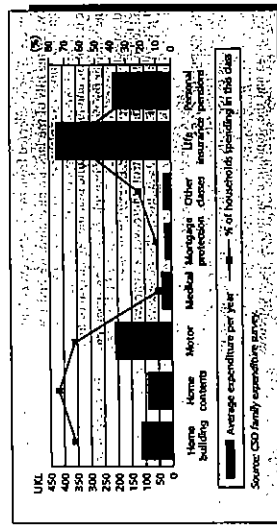
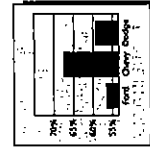


FIGURE 2.59 A Graph Comparing the Resale Values of Chevy, Dodge, and Ford Trucks



Source: Reprinted courtesy of General Motors Corporation.

2.90 In its June 4, 1995, issue, the *Cincinnati Enquirer* published a report on the profitability of banks in the Cincinnati Metropolitan Statistical Area (MSA). A summary of the data presented is given in Table 2.18. The *Enquirer* explained its report as follows: BankReport

About the Survey

- Return on equity—a profitability ratio measuring how well equity capital is used. The 1994 industry average: 14.82 percent.
- Net interest margin—often referred to as “spread,” it is the difference between what banks pay on deposits and what they charge on loans. The 1994 industry average: 4.15 percent.
- Equity capital—A key measurement of capital that can be used for several purposes, such as cushioning against operating losses like bad debt or protecting depositors’ money. It also can be used to evaluate the net worth of an institution.
- Loan loss reserves—cash set aside to cover problem loans.

TABLE 2.18 A Report of Bank Profitability for Greater Cincinnati, Ohio, Banks

BankReport

Greater Cincinnati banks for the most part continued their profitability in 1994. The figures reflect banks with headquarters in Ohio, Kentucky, or Indiana and with operations in the Cincinnati Metropolitan Statistical Area (MSA).

Bank	City	Assets (Millions)	Change from 1993 (Millions)	Net Income (Millions)	Change from 1993 on Assets on Equity	Return on Equity	Net Interest Margin	Loan Loss	Equity Capital (Millions)
Fifth Third Bank	Cincinnati	\$8,469.11	21,454	\$136.92	1.81%	24.68%	3.47%	1.00	\$592.54
Star Bank	Cincinnati	8,241.90	24,351	99.22	1.36	19.27	4.13	1.55	541.70
Provident Bank	Cincinnati	5,100.28	17,200	55.98	1.20	19.39	3.65	1.30	290.46
PKC Bank Ohio	Cincinnati	4,462.27	1,066	46.80	0.62	15.11	3.66	4.22	317.30
Bank One	Cincinnati	595.09	13,200	3.75	-54.13	0.38	4.98	1.00	88.15
Society National Bank	Cincinnati	942.00	10,000	11.80	16.00	1.25	18.96	0.70	62.30
Fifth Third Bank	Covington	764.04	21,311	10.98	2.63	1.57	21.38	3.70	54.91
Northern Kentucky Bank	Covington	747.99	3,554	9.92	24.75	1.34	13.61	4.04	61.32
PKC Bank Northern Kentucky	Covington	693.00	51,755	6.80	13.68	1.36	15.52	4.03	70.70
Liberty National Bank of Northern Kentucky	Fort Mitchell	375.14	-0.17	2.05	-40.24	0.55	5.48	3.47	36.56
Lebanon Citizens	Erlanger	369.46	-5.48	3.83	2.49	1.04	11.64	4.06	33.18
President Bank of Northern Kentucky	Lebanon	328.12	6.69	4.27	3.04	1.37	15.50	3.77	29.05
Northern Kentucky Bank	Alexandria	287.10	3.03	2.06	-17.38	0.77	9.39	3.39	22.01
North Side BRTC	Cincinnati	226.11	2.40	3.22	20.09	1.45	10.81	4.41	30.40
Bank of Boone City	Florence	126.64	17.66	1.47	9.06	1.29	14.71	4.37	10.67
Peoples Bank of Northern Kentucky	Crestview Hills	89.04	40.86	0.44	-14.97	0.61	6.30	3.55	7.20
Eagle Bank	Williamstown	86.26	3.08	1.23	-9.51	1.46	11.87	4.63	10.76
Hebron Deposit Bank	Hebron	82.05	3.15	0.84	9.40	1.04	12.14	3.82	7.23
Spirit of America NB	Milford	75.08	129.14	20.71	34.88	46.75	122.28	0.41	N/A
Merchants BRTC	West Harrison	70.34	6.34	0.21	N/A	0.31	5.43	4.99	2.46
First National Bank	Lebanon	57.98	-6.98	0.90	9.40	1.50	13.16	4.59	0.64
Community National Bank	Franklin	56.07	0.37	0.41	0.90	0.72	8.72	3.37	1.09
Grant County	Williamstown	51.89	-1.21	0.49	-22.64	0.93	9.38	4.32	1.51
Deposit Bank	Burlington	50.00	37.34	0.45	74.90	1.05	5.65	4.39	0.92
Heritage Bank	Fort Mitchell	44.44	45.54	0.24	1178.90	0.66	7.33	4.76	1.02

Source: Reprinted from the *Cincinnati Enquirer* of June 4, 1995, by permission of the *Cincinnati Enquirer* and Jeff McKinney.

Each is symmetrical with one outlier.

13, 9, 9

d None of the banks

e 1309, 267

f Strongly skewed right

g 17.89, 6.69

h Skewed right

a Construct dot plots for each of the following:

(1) Return on assets.

(2) Return on equity.

(3) Net interest margin.

b Describe the shape of each distribution.

c Mark the 1994 industry average (see About the Survey on page 109) on each dot plot. How many banks exceed the industry average (1) return on assets, (2) return on equity, and (3) net interest margin?

d Label the dots on each dot plot by bank. How many banks (and which ones) exceed the industry averages for all three criteria?

e Compute the mean and median assets for the 25 banks. What do these values say about skewness?

f Compute the mean and median for the percentage change in assets from 1993. What do these values say about skewness?

g Construct (1) a graphical five-number summary and (2) a box-and-whiskers plot for the net income and the changes in net income from 1993.

- 2.91 Table 2.19 gives February 1999 mortgage loan rates at financial institutions in Butler County, Ohio, as obtained in a survey conducted by the authors. Rates for 15-year and 30-year mortgages are given, and rates for both fixed- and variable-rate loans are given. A fixed-rate (F) loan has an interest rate that stays the same for the term of the mortgage, whereas a variable-rate (V) loan's interest rate changes with the market during the term of the mortgage. By constructing appropriate box-and-whiskers plots and by placing them side by side for ready comparison, compare the following mortgage rate distributions on the basis of central tendency, variability, skewness, and outliers.
- a Compare rates for 15-year mortgages versus rates for 30-year mortgages.
- b Compare rates for fixed-rate mortgages versus rates for variable-rate mortgages.
- c Compare rates for four loan types: fixed 15-year mortgages, fixed 30-year mortgages, variable 15-year mortgages, and variable 30-year mortgages.

2.92 Internet Exercise

Overview: The Data and Story Library (DASL) houses a rich collection of data sets useful for teaching and learning statistics, from a variety of sources, contributed primarily by university faculty members. DASL can be reached through the BSC by clicking on the Data Bases button in the BSC home screen and by then clicking on the Data and Story Library link. The screen image to the right shows the BSC Data Base page. The DASL can also be reached directly using the url <http://lib.stat.cmu.edu/DASL>. The objective of this exercise is to retrieve a data set of chief executive officer salaries and to construct selected graphical and numerical statistical summaries of the data.

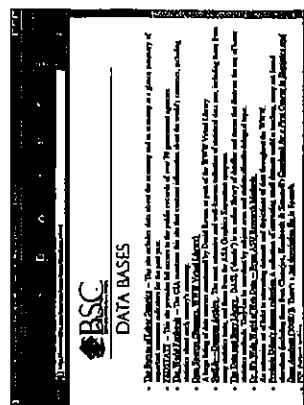


TABLE 2.19 Mortgage Loan Rates at Butler County, Ohio, Financial Institutions (February 1999)

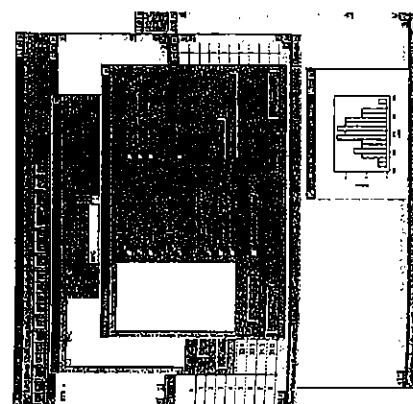
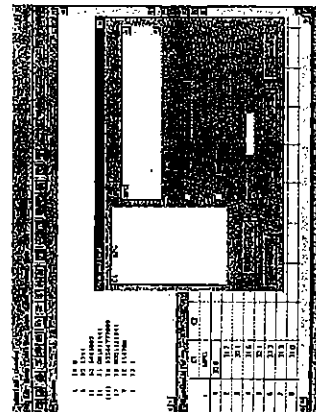
Institution	15-Year	30-Year
First Third Bank	6.60 f	6.875 f
First Southwestern	5.50 v	6.375 v
LCNB	6.625 f	7.0 f
Oxford Bank	6.0 v	6.75 v
Union County National	6.75 f	7.125 f
Bank One	6.5 v	6.5 v
	6.75 f	7.25 f
	6.875 f	7.375 v
	6.59 v	7.0 f
	6.75 f	7.25 f
	6.9 v	6.75 v

- a From the McGraw-Hill/Irwin Business Statistics Center Data Bases page, go to the DASL website and select "List all topics." From the Stories by Topic page, select Economics, then CEO Salaries to reach the CEO Salaries story. From the CEO Salaries story page, select the Datafile Name: CEO Salaries to reach the data set page. The data set includes the ages and salaries (save for a single missing observation) for a sample of 60 CEOs. Capture these observations and copy them into an Excel or MINITAB worksheet. This data capture can be accomplished in a number of ways. One simple approach is to use simple copy and paste procedures from the DASL

- data set to Excel or MINITAB (data sets CEOs.xls, CEOs.mtw).
- b Use your choice of statistical software to create graphical and numerical summaries of the CEO Salary data and use these summaries to describe the data. In Excel, create a histogram of salaries and generate descriptive statistics. In MINITAB, create a histogram, stem-and-leaf display, box plot, and descriptive statistics; or generate a MINITAB Graphical Data Summary. Offer your observations about typical salary level, the variation in salaries, and the shape of the distribution of CEO salaries.

Appendix 2.1 Descriptive Statistics Using MINITAB

The instructions in this section begin by describing the entry of data into the MINITAB Data window. Alternatively, the data may be loaded directly from the data disk included with the text. The appropriate data file



Stem and leaf plot in Figure 2.1 on page 41 (data file: GasMiles.mtw):

- Enter the mileage data from Table 2.1 (page 40) into column C1 with variable name MPG.
- Select Graph : Stem-and-Leaf (or Graph : Character Graphs : Stem-and-Leaf).
- In the Stem-and-Leaf dialog box, select the variable MPG into the Variables box by highlighting MPG in the variables list and by then clicking the Select button (or by double-clicking on MPG in the variables list or by typing MPG into the Variables box).
- Click OK in the Stem-and-Leaf dialog box.
- The stem-and-leaf plot appears in the Session window and can be selected for printing or copied and pasted to a word processor document. (See Appendix 1.1.)

Relative frequency histogram of the gas mileages (data file: GasMiles.mtw):

- Enter the mileage data from Table 2.1 (page 40) into column C1 with variable name MPG.
- Select Graph : Histogram (or, for a text-based histogram in the Session window, select Graph : Character Graphs : Histogram).
- In the Histogram dialog box, select the variable MPG into the Graph Variables box.
- Click on the Options button.
- In order to request a relative frequency histogram, click "Percent" for "Type of histogram." We would click "Frequency" to request a frequency histogram.
- In order to request that midpoints be used on the scale of the horizontal axis, click "Mid-Point" for "Type of intervals." We would click "CutPoint" to request class boundaries on the scale of the horizontal axis.

- Click "Automatic" to request that MINITAB automatically calculate the class midpoints to be used.
- Click OK in the Histogram Options dialog box.
- Click OK in the Histogram dialog box.
- The histogram appears in a graphics window. Note that the labels on the horizontal scale are midpoints that have been automatically calculated by MINITAB.
- The histogram can be selected for printing or can be copied and pasted into a word processor document. (See Appendix 1.1.)

To specify desired class boundaries:

- Click on the Options button in the Histogram dialog box.
- Click "CutPoint" for "Type of Intervals" in the Histogram Options dialog box to indicate that class boundaries are to be specified (rather than class midpoints).
- Click "Midpoint/Cutpoint positions" and enter the desired class boundaries (cut-points) in the box provided. Enter the cut-points from smallest to largest.
- Click OK in the Histogram Options dialog box.

To shade the histogram bars:

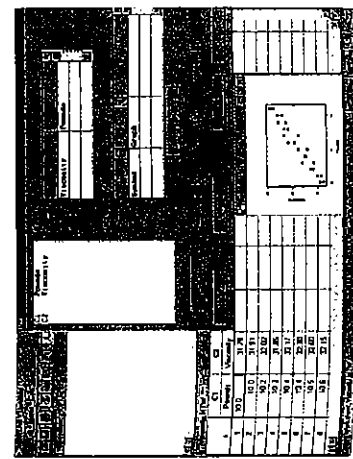
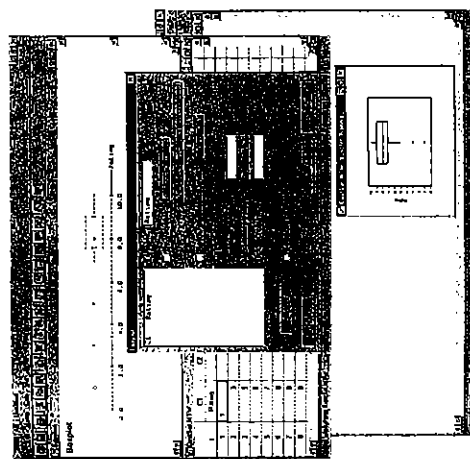
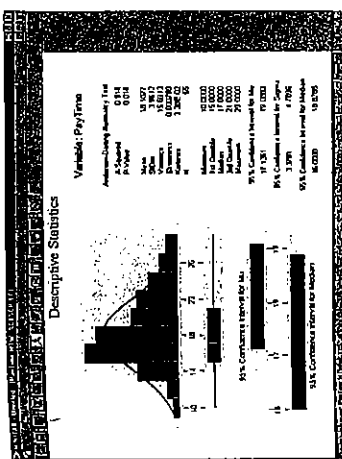
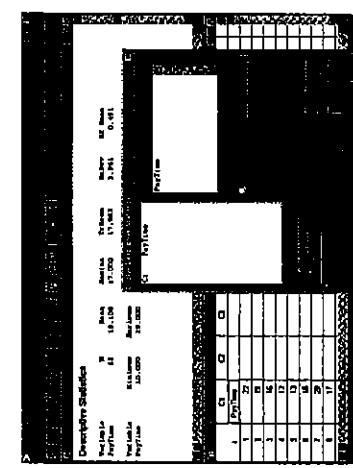
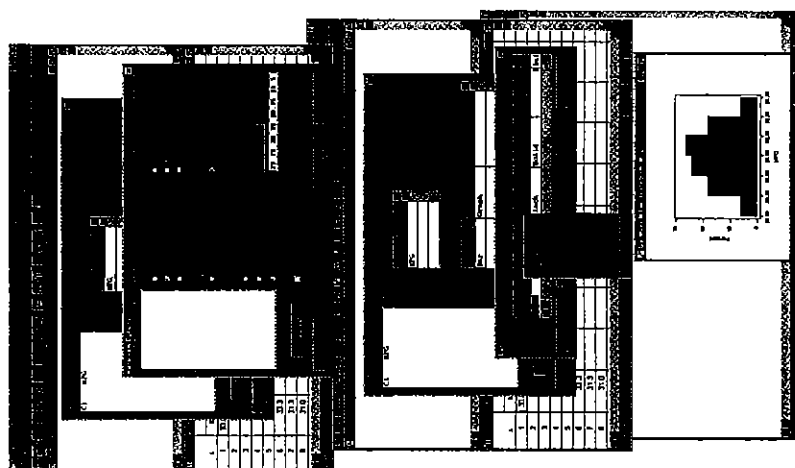
- Click the Edit Attributes button in the Histogram dialog box.
- Select Black (or some other desired color) from the "Back Color" pull-down menu.
- Select Solid from the "Fill Type" pull-down menu.
- Click OK in the Bar dialog box.
- Click OK in the Histogram dialog box.

There are many other options available through use of the Histogram Options and Edit Attributes buttons.

The histogram appears in a graphics window and can be selected for printing or can be copied and pasted into a word processor document. (See Appendix 1.1.)

Numerical descriptive statistics in Figure 2.23 on page 60 (data file: PayTime.mtw):

- Enter the payment time data from Table 1.2 (page 6) into column C1 with variable name PayTime.
- Select Stat : Basic Statistics : Display Descriptive Statistics.
- In the Display Descriptive Statistics dialog box, select the variable PayTime into the Variables box.



- Click OK in the Display Descriptive Statistics dialog box.
 - Descriptive statistics appear in the Session window and can be selected for printing or can be copied and pasted into a word processor document. (See Appendix 1.1.)
- MINITAB also offers a unique and informative graphical data summary that includes a histogram, a box plot, and numerical descriptive statistics as well as confidence intervals (discussed in Chapter 7) for the mean and the median. To obtain the Graphical summary:
- After selecting the PayTime variable into the Variables box in the Display Descriptive Statistics dialog box, click the Graphs... button.
 - In the Display Descriptive Statistics—Graph dialog box, click the Graphical Summary check box and click OK.
 - Click OK in the Display Descriptive Statistics dialog box.

- The graphical data summary will appear in a high-resolution graphics window and can be selected for printing or can be copied and pasted to a word processor document. (See Appendix 1.1.)

Box plot in Figure 2.34(b) on page 78 (data file: VCSat.mtw):

- Enter the satisfaction rating data from Page 58 into column C1 with variable name Rating.
- Select Graph : Character Graphs : Boxplot.
- In the Boxplot dialog box, select the variable Rating into the Variable box.
- Click OK in the Boxplot dialog box.
- The Boxplot appears in the Session window and can be selected for printing or can be copied and pasted into a word processor document. (See Appendix 1.1.) (Note: To create a high-resolution boxplot, select Graph : Boxplot.)

Scatter plot in Figure 2.47 on page 92 (data file: Viscosity.mtw):

- In the Data window, enter the viscosity data—pounds of chemical XB-135 from Table 2.12 on page 92 into column C1 (with variable name Pounds), and values of viscosity into column C2 (with variable name Viscosity).
- Select Graph : Plot.
- In the Plot dialog box, select Viscosity for the Y (or vertical) axis and Pounds for the X (or horizontal) axis.
- Click OK in the Plot dialog box.
- The scatter plot appears in a graphics window and can be selected for printing or can be copied and pasted into a word processor document. (See Appendix 1.1.)

Appendix 2.2 ■ Descriptive Statistics Using Excel

The instructions in this section begin by describing the entry of data into an Excel spreadsheet. Please refer to Appendix 1.2 for further information about entering data and saving and printing results.

Frequency histogram for the gas mileages in Table 2.1 on page 40 (data file: GasMiles.xls):

- Enter the mileage data into column A with the variable name, MPG, in Line 1 of column A (that is, in cell A1) and with the 49 mileages in cells A2 to A50.
- Enter the variable name, MPG, in cell C1.
- Enter the upper class boundaries from Table 2.3 on page 45 (that is, 30.35, 30.95, 31.55, 32.15, 32.75, 33.35) into cells C2 to C7. (Skip this entry if you wish Excel to find automatic class limits.)

- Select Tools : Data Analysis : Histogram.
- Click OK in the Data Analysis dialog box.

- In the Histogram dialog box, enter the range A1:A50 into the "Input Range" box.

- In the "Bin Range" box enter the range, C1:C7. Leave this box blank if Excel is providing automatic class limits.

- Check the Labels check box.

- Click "Output Range" and enter a cell, say C10, in the "Output Range" box. The upper left portion of the output will be placed in the specified cell.

- Check the "Chart Output" check box.

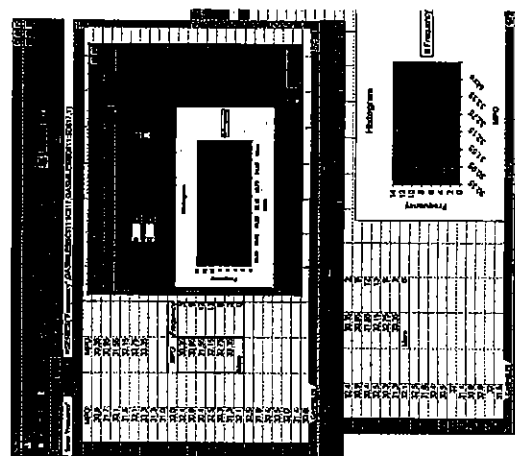
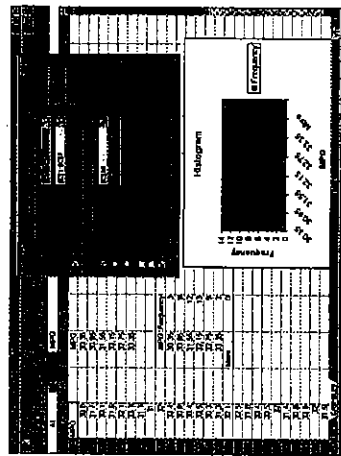
- Click OK in the Histogram dialog box.
- Click on the histogram that now appears, use the mouse to enlarge the chart to the desired size, and enlarge until the upper class limits are displayed along the horizontal axis.

To remove the gaps between the bars (optional):

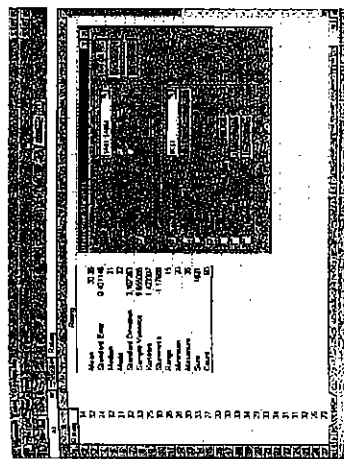
- Double-click on any of the bars.
- In the Format Data Series dialog box that appears, click on the Options tab.

- In the box for "Gap width," enter "0."
- Click OK in the Format Data Series dialog box.

- The histogram can be selected for printing or can be copied and pasted to a word processor document. (See Appendix 1.2.)



Appendix 2.2

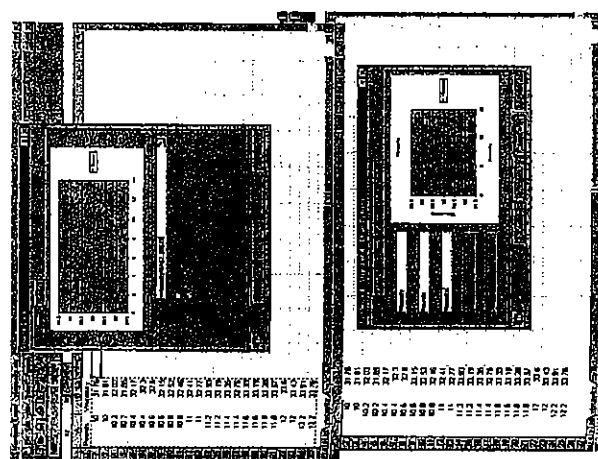


Numerical descriptive statistics in Figure 2.24(a) on page 60 (data file: Design.xls):

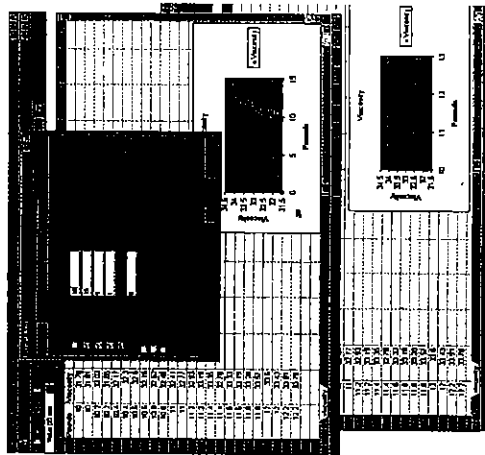
- Enter the package design ratings data into column A with the variable name Rating in cell A1 and with the 60 design ratings from Table 1.3 on page 7 in cells A2 to A61.
- Select Tools : Data Analysis : Descriptive Statistics.
- Click OK in the Data Analysis dialog box.
- Enter the range for the data, A1:A61, in the Input Range box.
- Check the "Labels in First Row" check box.
- Click "Output Range" and enter the cell for the upper left corner of the desired location of the output, say C1.
- Check the "Summary Statistics" check box.
- Click OK in the Descriptive Statistics dialog box.
- The descriptive statistics summary will appear in cells C1:D15. Drag the column C border to reveal complete labels for all statistics. The descriptive statistics can be printed directly from Excel or can be copied and pasted to a word processor document. (See Appendix 1.2.)

Scatter plot similar to Figure 2.47 on page 92 (data file: Viscosity.xls):

- Enter the viscosity data from Table 2.12 (page 92) into columns A and B. Place the variable name Pounds in cell A1 and enter the pounds of chemical figures in cells A2 to A25. Similarly, place the variable name Viscosity in cell B1 and enter the viscosity figures in cells B2 to B25. (Note: The order of the columns determines the axes in the plot. The variable for the vertical axis must be in the second column.)
- Select the Chart Wizard from the toolbar.
- In the Step 1 dialog box, select "XY (Scatter)" from the Chart Type menu. Click Next.
- In the Step 2 dialog box, enter the data range of both columns combined, A2:B25, into the "Data Range" box. Alternatively, click in the "Data Range" box and select cells A2 through B25 with the mouse. Click "Columns" in the Step 2 dialog box. Click Next.
- In the Step 3 dialog box, enter Pounds in the "Value(X) Axis" box and enter Viscosity in the "Value(Y) Axis" box. Click Next.
- In the Step 4 dialog box, click Finish.

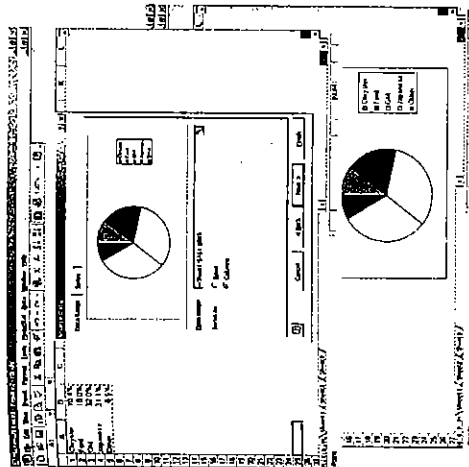
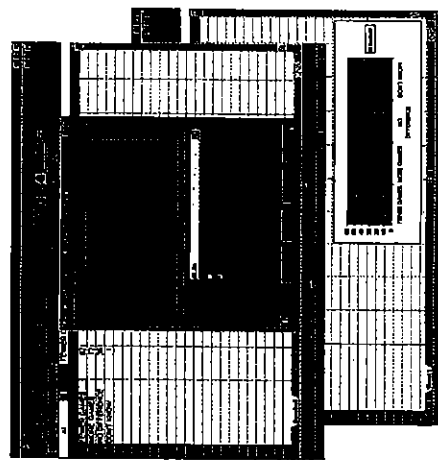


- Move the graph and make it the size you want by dragging edges with the mouse.
- Change the horizontal scale by double-clicking on any of the numerical labels on the horizontal scale (for instance, double-click on the 10). Then, in the Format Axis dialog box, click on the "Scale" tab and change the "minimum value" to 10. Click OK.
- The spreadsheet and scatter plot can be printed directly from Excel or can be copied and pasted to a word processor document. (See Appendix 1.2.)



Bar plots for categorical data in Figure 2.44(a) on page 89 (data file: Baseball.xls):

- Enter the data of Figure 2.44(a) into columns A and B as shown.
- Select the Chart Wizard from the toolbar.
- Select "Column" from the Chart Type menu and click Next.
- Enter A1:B4 into the "Data Range" box and click Finish.



- Pie chart of auto manufacturer percentages in Figure 2.40 on page 85:
- Enter the automanufacturer data into columns A and B as shown in the screenshot to the left.
 - Select the Chart Wizard from the toolbar.
 - Select "Pie" from the Chart Type menu and click Next.
 - Enter A1:B5 into the "Data Range" box and click Finish.

Appendix 2.3 Descriptive Statistics Using MegaStat

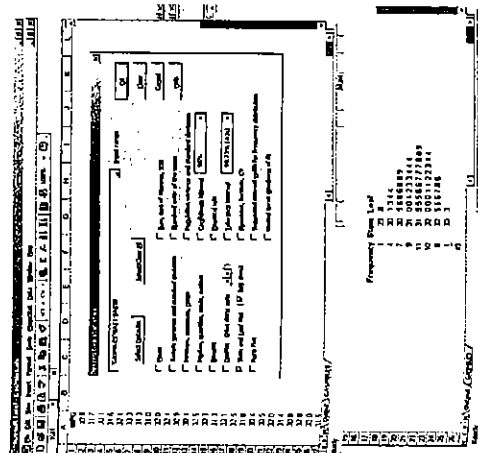
The instructions in this section begin by describing the entry of data into an Excel worksheet. Alternatively, the data may be loaded directly from the data disk included with the text. The appropriate data file name is given at

the top of each instruction block. Please refer to Appendix 1.2 for further information about entering data and saving and printing results in Excel. Please refer to Appendix 1.3 for more information about using MegaStat.

To analyze the gas mileage data in Table 2.1 on page 40 (data file: GasMiles.xls):

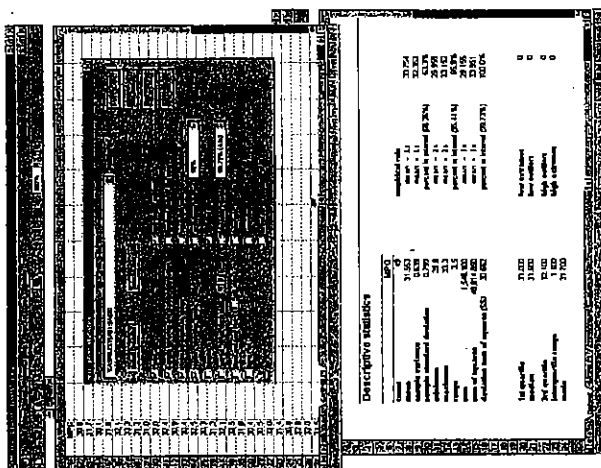
- Enter the mileage data from Table 2.1 into column A with the label MPG in cell A1 and with the 49 gas mileages in cells A2 to A50.
- To produce the stem-and-leaf display in Figure 2.1 on page 41:

- Select MegaStat: Descriptive Statistics.
- In the "Descriptive statistics" dialog box, use the AutoExpand feature to enter the range A1:A50 into the Input Range box.
- Check the "Stem and Leaf Plot" check box.
- (Optional) Check the "Split stem" check box—this depends on how you want the output to appear. You may wish to construct two plots—one with the split stem option and one without—and then choose the output you like best.
- Click OK in the "Descriptive statistics" dialog box. The output will be placed in an Output worksheet.



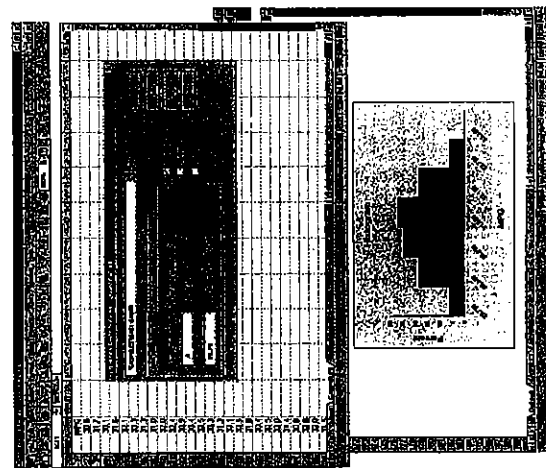
To compute the descriptive statistics given in Figure 2.25 on page 61:

- Check the check boxes that correspond to the desired statistics. If tolerance intervals based on the Empirical Rule are desired, check the "Empirical rule" check box.
- Click OK in the "Descriptive statistics" dialog box. The output will be placed in an Output worksheet.



To produce the histogram of gas mileages given in Figure 2.5 on page 46:

- Select MegaStat: Frequency Distribution: Quantitative.
- In the "Frequency Distribution—Quantitative" dialog box, use the AutoExpand feature to enter the input range A1:A50 into the Input Range box.
- Enter the class width (in this case equal to 6) into the Interval Width box.
- Enter the lower boundary of the first—leftmost—interval of the histogram (in this case equal to 29.75) into the appropriate box.
- Make sure that the Histogram check box is checked.
- Click OK in the "Frequency Distribution—Quantitative" dialog box.
- The histogram output will be placed in an Output worksheet. The histogram can then be edited using any of the usual Excel editing features (see Appendix 1.3).



We use the concept of *probability* to deal with uncertainty. Intuitively, the probability of an event is a number that measures the chance, or likelihood, that the event will occur. For instance, the probability that your favorite football team will win its next game measures the likelihood of a victory. The probability of an event is always a number between 0 and 1. The closer an event's probability is to 1, the higher is the likelihood that the event will occur; the closer the event's probability is to 0, the smaller is the likelihood that the event will occur. For example, if you believe that the probability that your favorite football team will win its next game is .95, then you are almost sure that your team will win. However, if you believe that the probability of victory is only .10, then you have very little confidence that your team will win.

Only 10, then you have very little confidence that you really want what you have. When performing statistical studies, we sometimes collect data by performing a controlled experiment. For instance, we might purposely vary the operating conditions of a manufacturing process in order to study the effects of these changes on the process output. Alternatively, we sometimes obtain data by observing uncontrolled events. For example, we might observe the closing price of a share of General Motors' stock every day for 30 trading days. In order to simplify our terminology, we will use the word *experiment* to refer to either method of data collection.

An experiment is any process of observation that has an uncertain outcome. The process must be defined so that on any single repetition of the experiment, *one and only one* of the possible outcomes will occur. The possible outcomes for an experiment are called **experimental outcomes**.

For example, if the experiment consists of tossing a coin, the experimental outcomes are "head" and "tail." If the experiment consists of rolling a die, the experimental outcomes are 1, 2, 3, 4, 5, and 6. If the experiment consists of subjecting an automobile to a tailpipe emissions test, the experimental outcomes are pass and fail.

Experimental outcomes are *ex post* and *tail*. We often wish to assign probabilities to experimental outcomes. This can be done by several methods. Regardless of the method used, probabilities must be assigned to the experimental outcomes so that two conditions are met:

- 1 The probability assigned to each experimental outcome must be between 0 and 1. That is, if E represents an experimental outcome and if $P(E)$ represents the probability of this outcome, then $0 \leq P(E) \leq 1$.
- 2 The probabilities of all of the experimental outcomes must sum to 1.

3.1 The Concept of Probability

3.2 Sample Spaces and Events

- ### 3.3 Some Elementary Probability Rules

Sometimes, when all of the experimental outcomes are equally likely, we can use logic to assign probabilities. This method, which is called the *classical method*, will be more fully discussed in the next section. As a simple example, consider the experiment of tossing a fair coin. Here, there are two equally likely experimental outcomes—head (H) and tail (T). Therefore, logic suggests that the probability of observing a head, denoted $P(H)$, is $1/2 = .5$, and that the probability of observing a tail, denoted $P(T)$, is also $1/2 = .5$. Notice that each probability is between 0 and 1. Furthermore, because H and T are all of the experimental outcomes, $P(H) + P(T) = 1$.

Probability is often interpreted to be a *long-run relative frequency*. As an example, consider repeatedly tossing a coin. If we get 6 heads in the first 10 tosses, then the relative frequency, or fraction, of heads is $6/10 = .6$. If we get 47 heads in the first 100 tosses, the relative frequency of heads is $47/100 = .47$. If we get 5,067 heads in the first 10,000 tosses, the relative frequency of heads is $5,067/10,000 = .5067$.¹ Since the relative frequency of heads is approaching (that is, getting closer to) .5, we might estimate that the probability of obtaining a head when tossing the coin is .5. When we say this, we mean that, if we tossed the coin an indefinitely large number of times (that is, a number of times *approaching infinity*), the relative frequency of heads obtained would approach .5. Of course, in actuality it is impossible to toss a coin (or perform any experiment) an indefinitely large number of times. Therefore, a relative frequency interpretation of probability is a mathematical idealization. To summarize, suppose that E is an experimental outcome that might occur when a particular experiment is performed. Then the probability that E will occur, $P(E)$, can be interpreted to be the number that would be approached by the relative frequency of E if we performed the experiment an indefinitely large number of times. It follows that we often think of a probability in terms of the percentage of the time the experimental outcome would occur in many repetitions of the experiment. For instance, when we say that the probability of obtaining a head when we toss a coin is .5, we are saying that, when we repeatedly toss the coin an indefinitely large number of times, we will obtain a head on 50 percent of the repetitions.

Sometimes it is either difficult or impossible to use the classical method to assign probabilities. Since we can often make a relative frequency interpretation of probability, we can estimate a probability by performing the experiment in which an outcome might occur many times. Then, we estimate the probability of the experimental outcome to be the proportion of the time that the outcome occurs during the many repetitions of the experiment. For example, to estimate the probability that a randomly selected consumer prefers Coca-Cola to all other soft drinks, we perform an experiment in which we ask a randomly selected consumer for his or her preference. There are two possible experimental outcomes: "prefers Coca-Cola" and "does not prefer Coca-Cola." However, we have no reason to believe that these experimental outcomes are equally likely, so we cannot use the classical method. We might perform the experiment, say, 1,000 times by surveying 1,000 randomly selected consumers. Then, if 140 of those surveyed said that they prefer Coca-Cola, we would estimate the probability that a randomly selected consumer prefers Coca-Cola to all other soft drinks to be $140/1,000 = .14$. This is called the *relative frequency method* for assigning probability.

If we cannot perform the experiment many times, we might estimate the probability by using our previous experience with similar situations, intuition, or special expertise that we may possess. For example, a company president might estimate the probability of success for a one-time business venture to be .7. Here, on the basis of knowledge of the success of previous similar ventures, the opinions of company personnel, and other pertinent information, the president believes that there is a 70 percent chance the venture will be successful.

When we use experience, intuitive judgement, or expertise to assess a probability, we call this a *subjective probability*. Such a probability may or may not have a relative frequency interpretation. For instance, when the company president estimates that the probability of a successful business venture is .7, this may mean that, if business conditions similar to those that are about to be encountered could be repeated many times, then the business venture would be successful in 70 percent of the repetitions. Or, the president may not be thinking in relative frequency terms but rather may consider the venture a "one-shot" proposition. We will discuss some other

¹The English mathematician John Kerrich actually obtained this result when he tossed a coin 10,000 times while imprisoned by the Germans during World War II.

subjective probabilities later. However, the interpretations of statistical inferences we will explain in later chapters are based on the relative frequency interpretation of probability. For this reason, we will concentrate on this interpretation.

3.2 Sample Spaces and Events

In order to calculate probabilities by using the classical method, it is important to understand and use the idea of a *sample space*.

The sample space of an experiment is the set of all possible experimental outcomes. The experimental outcomes in the sample space are often called *sample space outcomes*.



EXAMPLE 3.1

A company is choosing a new chief executive officer (CEO). It has narrowed the list of candidates to four finalists (identified by last name only)—Adams, Chung, Hill, and Rankin. If we consider our experiment to be making a final choice of the company's CEO, then the experiment's sample space consists of the four possible experimental outcomes:

A = Adams is chosen as CEO.

C = Chung is chosen as CEO.

H = Hill is chosen as CEO.

R = Rankin is chosen as CEO.

Each of these outcomes is a sample space outcome, and the set of these sample space outcomes is the sample space.

Next, suppose that industry analysts feel (subjectively) that the probabilities that Adams, Chung, Hill, and Rankin will be chosen as CEO are .1, .2, .5, and .2, respectively. That is, in probability notation

$$P(A) = .1 \quad P(C) = .2 \quad P(H) = .5 \quad \text{and} \quad P(R) = .2$$

Notice that each probability assigned to a sample space outcome is between 0 and 1 and that the sum of the probabilities equals 1.

EXAMPLE 3.2

A newly married couple plans to have two children. Naturally, they are curious about whether their children will be boys or girls. Therefore, we consider the experiment of having two children. In order to find the sample space of this experiment, we let B denote that a child is a boy and G denote that a child is a girl. Then, it is useful to construct the tree diagram shown in Figure 3.1. This diagram pictures the experiment as a two-step process—having the first child, which could be either a boy or a girl (B or G), and then having the second child, which could also be either a boy or a girl (B or G). Each branch of the tree leads to a sample space outcome. These outcomes are listed at the right ends of the branches. We see that there are four sample space outcomes. Therefore, the sample space (that is, the set of all the sample space outcomes) is

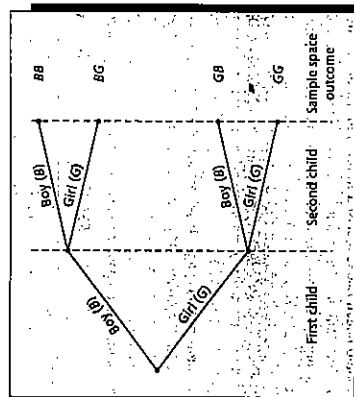
$BB \quad BG \quad GB \quad GG$

In order to consider the probabilities of these outcomes, suppose that boys and girls are equally likely each time a child is born. Intuitively, this says that each of the sample space outcomes is equally likely. That is, this implies that

$$P(BB) = P(BG) = P(GB) = P(GG) = \frac{1}{4}$$

This says that there is a 25 percent chance that each of these outcomes will occur. Again, notice that these probabilities sum to 1.

FIGURE 3.1 A Tree Diagram of the Genders of Two Children



A student takes a pop quiz that consists of three true-false questions. If we consider our experiment to be answering the three questions, each question can be answered correctly or incorrectly. We will let C denote answering a question correctly and I denote answering a question incorrectly. Then, Figure 3.2 depicts a tree diagram of the sample space outcomes for the experiment. The diagram portrays the experiment as a three-step process—answering the first question (correctly or incorrectly, that is, C or I), answering the second question, and answering the third question. The tree diagram has eight different branches, and the eight sample space outcomes are listed at the ends of the branches. We see that the sample space is

$$\begin{array}{l} CCC \quad CCI \quad CIC \quad CII \\ ICC \quad ICI \quad IIC \quad III \end{array}$$

Next, suppose that the student was totally unprepared for the quiz and had to blindly guess the answer to each question. That is, the student had a 50–50 chance (or .5 probability) of correctly answering each question. Intuitively, this would say that each of the eight sample space outcomes is equally likely to occur. That is,

$$P(CCC) = P(CCI) = \dots = P(III) = \frac{1}{8}$$

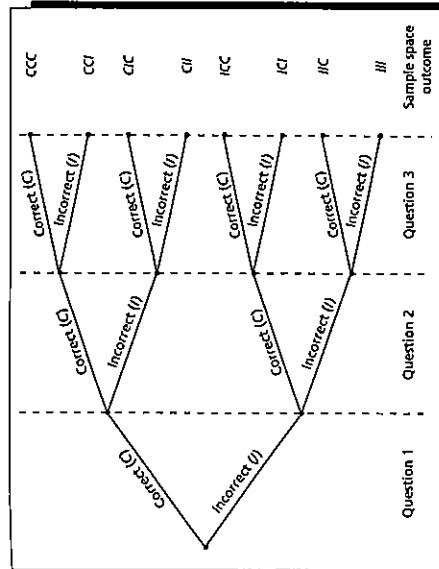
Here, as in Examples 3.1 and 3.2, the sum of the probabilities of the sample space outcomes is equal to 1.

Events and finding probabilities by using sample spaces At the beginning of this chapter, we informally talked about events. We now give the formal definition of an event.

An event is a set (or collection) of sample space outcomes.

For instance, if we consider the couple planning to have two children, the event “the couple will have at least one girl” consists of the sample space outcomes BG , GB , and GG . That is, the event “the couple will have at least one girl” will occur if and only if one of the sample

FIGURE 3.2 A Tree Diagram of Answering Three True-False Questions



space outcomes BG , GB , or GG occurs. As another example, in the pop quiz situation, the event “the student will answer at least two out of three questions correctly” consists of the sample space outcomes CCC , CCI , CIC , and ICC , while the event “the student will answer all three questions correctly” consists of the sample space outcome CCC . In general, we see that the word description of an event determines the sample space outcomes that correspond to the event.

Suppose that we wish to find the probability that an event will occur. We can find such a probability as follows:

The probability of an event is the sum of the probabilities of the sample space outcomes that correspond to the event.

As an example, in the CEO situation, suppose only Adams and Hill are internal candidates (they already work for the company). Letting INT denote the event that “an internal candidate is selected for the CEO position,” then INT consists of the sample space outcomes A and H (that is, INT will occur if and only if either of the sample space outcomes A or H occurs). It follows that $P(INT) = P(A) + P(H) = .1 + .5 = .6$. This says that the probability that an internal candidate will be chosen to be CEO is .6.

In general, we have seen that the probability of any sample space outcome (experimental outcome) is a number between 0 and 1, and we have also seen that the probabilities of all the sample space outcomes sum to 1. It follows that the probability of an event (that is, the probability of a set of sample space outcomes) is a number between 0 and 1. That is,

If A is an event, then $0 \leq P(A) \leq 1$.

Moreover:

- 1 If an event never occurs, then the probability of this event equals 0.
- 2 If an event is certain to occur, then the probability of this event equals 1.

Consider the couple that is planning to have two children, and suppose that each child is equally likely to be a boy or girl. Recalling that in this case each sample space outcome has a probability equal to $1/4$, we see that:

- 1 The probability that the couple will have two boys is

$$P(BB) = \frac{1}{4}$$

since two boys will be born if and only if the sample space outcome BB occurs.

- 2 The probability that the couple will have one boy and one girl is

$$P(BG) + P(GB) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

since one boy and one girl will be born if and only if one of the sample space outcomes BG or GB occurs.

- 3 The probability that the couple will have two girls is

$$P(GG) = \frac{1}{4}$$

since two girls will be born if and only if the sample space outcome GG occurs.

- 4 The probability that the couple will have at least one girl is

$$P(BG) + P(GB) + P(GG) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

since at least one girl will be born if and only if one of the sample space outcomes BG , GB , or GG occurs.

Again consider the pop quiz consisting of three true-false questions, and suppose that the student blindly guesses the answers. Remembering that in this case each sample space outcome has a probability equal to $1/8$, then:

- 1 The probability that the student will get all three questions correct is

$$P(CCC) = \frac{1}{8}$$

- 2 The probability that the student will get exactly two questions correct is

$$P(CCT) + P(CTC) + P(TCC) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

since two questions will be answered correctly if and only if one of the sample space outcomes CCT , CTC , or TCC occurs.

- 3 The probability that the student will get exactly one question correct is

$$P(CTT) + P(TCT) + P(TTC) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

since one question will be answered correctly if and only if one of the sample space outcomes CTT , TCT , or TTC occurs.

- 4 The probability that the student will get all three questions incorrect is

$$P(TTT) = \frac{1}{8}$$

- 5 The probability that the student will get at least two questions correct is

$$P(CCC) + P(CCT) + P(CTC) + P(TCC) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$$

since the student will get at least two questions correct if and only if one of the sample space outcomes CCC , CCT , CTC , or TCC occurs.

Notice that in the true-false question situation we find that, for instance, the probability that the student will get exactly two questions correct equals the ratio

$$\frac{\text{the number of sample space outcomes resulting in two correct answers}}{\text{the total number of sample space outcomes}} = \frac{3}{8}$$

In general, when a sample space is finite we can use the following method for computing the probability of an event.

If all of the sample space outcomes are equally likely, then the probability that an event will occur is equal to the ratio

$$\frac{\text{the number of sample space outcomes that correspond to the event}}{\text{the total number of sample space outcomes}}$$

When we use this rule, we are using the *classical method* for computing probabilities. Furthermore, it is important to emphasize that we can use this rule only when all of the sample space outcomes are equally likely (as they are in the true-false question situation). For example, if we were to use this rule in the CEO situation, we would find that the probability of choosing an internal candidate as CEO is

$$P(INT) = \frac{\text{the number of internal candidates}}{\text{the total number of candidates}} = \frac{2}{4} = .5$$

This result is not equal to the correct value of $P(INT)$, which we previously found to be equal to .6. Here, this rule does not give us the correct answer because the sample space outcomes A , C , H , and R are not equally likely—recall that $P(A) = .1$, $P(C) = .2$, $P(H) = .5$, and $P(R) = .2$.

EXAMPLE 3.6

Suppose that 650,000 of the 1,000,000 households in an eastern U.S. city subscribe to a newspaper called the *Atlantic Journal*, and consider randomly selecting one of the households in this city. That is, consider selecting one household by giving each and every household in the city the same chance of being selected. Let A be the event that the randomly selected household subscribes to the *Atlantic Journal*. Then, because the sample space of this experiment consists of 1,000,000 equally likely sample space outcomes (households), it follows that

$$\begin{aligned} P(A) &= \frac{\text{the number of households that subscribe to the } \textit{Atlantic Journal}}{\text{the total number of households in the city}} \\ &= \frac{650,000}{1,000,000} \\ &= .65 \end{aligned}$$

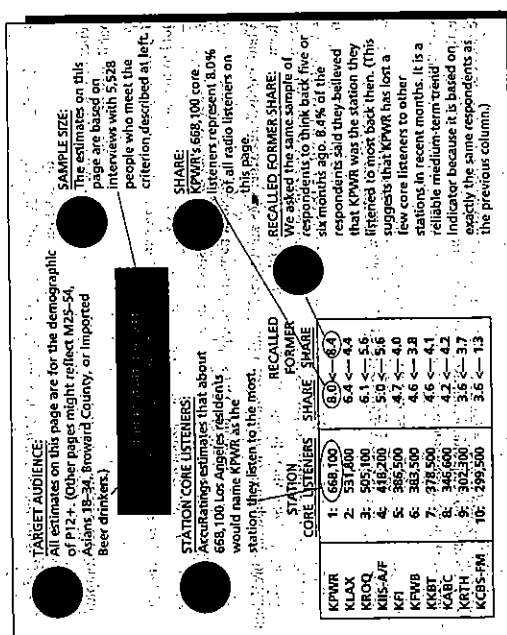
This says that the probability that the randomly selected household subscribes to the *Atlantic Journal* is .65.

EXAMPLE 3.7

As discussed in the introduction to this chapter, AccuRatings is a radio ratings service provided by Strategic Radio Research, a media research firm in Chicago, Illinois. Figure 3.3 gives portions of an AccuRatings report on radio ratings in the Los Angeles market. This report, based on interviews with 5,528 randomly selected persons 12 years of age or older, gives estimates of the number and the percentage of Los Angeles residents who would name each of the top 10 radio stations in Los Angeles as the station they listen to most.

To better understand the estimates in Figure 3.3, we will consider how they were obtained. AccuRatings asked each of the 5,528 sampled residents to name which station (if any) he or

FIGURE 3.3 Portions of an AccuRatings Report on Radio Ratings in the Los Angeles Market



Source: Strategic Radio Research, AccuRatings Introduction for Broadcasters.



she listens to most. AccuRatings then used the responses of the sampled residents to calculate the proportion of these residents who favored each station. The sample proportion of the residents who favored a particular station is an estimate of the population proportion of all Los Angeles residents (12 years of age or older) who favor the station, or, equivalently, of the probability that a randomly selected Los Angeles resident would favor the station. For example, if 445 of the 5,528 sampled residents favored station KPWR, then $445/5,528 = .080499276$ is an estimate of $P(\text{KPWR})$, the probability that a randomly selected Los Angeles resident would favor station KPWR. Furthermore, assuming that there are 8,300,000 Los Angeles residents 12 years of age or older, an estimate of the number of these residents who favor station KPWR is

$$(8,300,000) \times (.080499276) = 668,143.99$$

Now, if we

- 1 Round the estimated number of residents favoring station KPWR to 668,100, and
 - 2 Express the estimated probability $P(\text{KPWR})$ as the rounded percentage 8.0%,
- we obtain what the AccuRatings report in Figure 3.3 states are (1) the estimated number of core listeners for station KPWR and (2) the estimated share of all listeners for station KPWR. These measures of listenership would be determined for other stations in a similar manner (see Figure 3.3).

To conclude this section, we note that in Appendix B of this book we discuss several *counting rules* that can be used to count the number of sample space outcomes in an experiment. These

rules are particularly useful when there are many sample space outcomes and thus these outcomes are difficult to list.

Exercises for Sections 3.1 and 3.2

CONCEPTS

- 3.1 Define the following terms: *experiment*, *event*, *probability*, *sample space*.
- 3.2 Explain the properties that must be satisfied by a probability.

METHODS AND APPLICATIONS

- 3.3 Two randomly selected grocery store patrons are each asked to take a blind taste test and to then state which of three diet colas (marked as A, B, or C) he or she prefers.

a Draw a tree diagram depicting the sample space outcomes for the test results.

b List the sample space outcomes that correspond to each of the following events:

- Both patrons prefer diet cola A.
 - The two patrons prefer the same diet cola.
 - The two patrons prefer different diet colas.
 - Diet cola A is preferred by at least one of the two patrons.
 - Neither of the patrons prefers diet cola C.
- c Assuming that all sample space outcomes are equally likely, find the probability of each of the events given in part b.

3.3, 3.6, 3.7

b1 AA

b2 AA, BB, CC

b3 AB, AC, BA, BC

CA, CB

b4 AA, AB, AC, BA,

CA

b5 AA, AB, BA, BB

c 1/9, 1/3, 2/3, 5/9,

4/9

- 3.4 Suppose that a couple will have three children. Letting B denote a boy and G denote a girl:

a Draw a tree diagram depicting the sample space outcomes for this experiment.

b List the sample space outcomes that correspond to each of the following events:

- All three children will have the same gender.
 - Exactly two of the three children will be girls.
 - Exactly one of the three children will be a girl.
 - None of the three children will be a girl.
- c Assuming that all sample space outcomes are equally likely, find the probability of each of the events given in part b.

BBB, GGG

BGG, GBG, GGB

GBB, BGG, BGG

BBB

1/4, 3/8, 3/8, 1/8

- 3.5 Four people will enter an automobile showroom, and each will either purchase a car (P) or not purchase a car (N).

a Draw a tree diagram depicting the sample space of all possible purchase decisions that could potentially be made by the four people.

b List the sample space outcomes that correspond to each of the following events:

- Exactly three people will purchase a car.
 - Two or fewer people will purchase a car.
 - One or more people will purchase a car.
 - All four people will make the same purchase decision.
- c Assuming that all sample space outcomes are equally likely, find the probability of each of the events given in part b.

b1 PPPP, PPNP,

PNPP, NPPP

b2 Outcomes

b3 Outcomes

with ≥ 2 P's

b4 Outcomes

with ≥ 1 P

b5 Outcomes

c 1/4, 11/16,

15/16, 1/8

3.6 THE ACCURATING CASE

Using the information given in the AccuRatings report of Figure 3.3 (page 128), find estimates of each of the following:

- The probability that a randomly selected Los Angeles resident (12 years or older) would name station KLAX as the station that he or she listens to most.
- The probability that a randomly selected Los Angeles resident (12 years or older) would name station KABC as the station that he or she listens to most.
- The percentage of all Los Angeles residents (12 years or older) who would name KCBS-FM as the station that he or she listens to most.
- The number of the 5,528 sampled residents who named station KFI as the station he or she listens to most.
- The number of the 5,528 sampled residents who named station KROQ as the station he or she listens to most.

.054

.042

3.6%

260

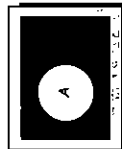
337

- 3.7 Let A, B, C, D, and E be sample space outcomes forming a sample space. Suppose that $P(A) = .2$, $P(B) = .15$, $P(C) = .3$, and $P(D) = .2$. What is $P(E)$? Explain how you got your answer.

3.3 Some Elementary Probability Rules

We can often calculate probabilities by using formulas called **probability rules**. We will begin by presenting the simplest probability rule: the **rule of complements**. To start, we define the complement of an event:

FIGURE 3.4
The Complement of an Event (the Shaded Region is $\bar{A}$)



Given an event A , the complement of A is the event consisting of all sample space outcomes that do not correspond to the occurrence of A . The complement of A is denoted $\bar{A}$. Furthermore, $P(\bar{A})$ denotes the probability that A will not occur.

Figure 3.4 is a Venn diagram depicting the complement $\bar{A}$ of an event A . In any probability situation, either an event A or its complement $\bar{A}$ must occur. Therefore, we have

$$P(A) + P(\bar{A}) = 1$$

This implies the following result:

The Rule of Complements

Consider an event A . Then, the probability that A will not occur is

$$P(\bar{A}) = 1 - P(A)$$

Recall from Example 3.6 that the probability that a randomly selected household in an eastern U.S. city subscribes to the *Atlantic Journal* is .65. It follows that the probability of the complement of this event (that is, the probability that a randomly selected household in the eastern U.S. city does not subscribe to the *Atlantic Journal*) is $1 - .65 = .35$.

Consider Example 3.6, and recall that 650,000 of the 1,000,000 households in an eastern U.S. city subscribe to the *Atlantic Journal*. Also, suppose that 500,000 households in the city subscribe to a competing newspaper, the *Beacon News*, and further suppose that 250,000 households subscribe to both the *Atlantic Journal* and the *Beacon News*. As in Example 3.6, we consider randomly selecting one household in the city, and we define the following events.

A = the randomly selected household subscribes to the *Atlantic Journal*.

$\bar{A}$ = the randomly selected household does not subscribe to the *Atlantic Journal*.

B = the randomly selected household subscribes to the *Beacon News*.

$\bar{B}$ = the randomly selected household does not subscribe to the *Beacon News*.

Using the notation $A \cap B$ to denote both A and B , we also define

$A \cap B$ = the randomly selected household subscribes to both the *Atlantic Journal* and the *Beacon News*.

Since 650,000 of the 1,000,000 households subscribe to the *Atlantic Journal* (that is, correspond to the event A occurring), then 350,000 households do not subscribe to the *Atlantic Journal* (that is, correspond to the event $\bar{A}$ occurring). Similarly, 500,000 households subscribe to the *Beacon News* (B), so 500,000 households do not subscribe to the *Beacon News* ($\bar{B}$). We summarize this information, as well as the 250,000 households that correspond to the event $A \cap B$ occurring, in Table 3.1.

TABLE 3.1 A Summary of the Number of Households Corresponding to the Events A , $\bar{A}$, B , $\bar{B}$, and $A \cap B$

Events	Subscribes to <i>Atlantic Journal</i> , A	Does Not Subscribe to <i>Atlantic Journal</i> , $\bar{A}$	Total
Subscribes to <i>Beacon News</i> , B	250,000	500,000	750,000
Does Not Subscribe to <i>Beacon News</i> , $\bar{B}$	400,000	500,000	900,000
Total	650,000	1,000,000	1,650,000

Next, consider the events

$A \cap B$ = the randomly selected household subscribes to the *Atlantic Journal* and does not subscribe to the *Beacon News*.

$\bar{A} \cap B$ = the randomly selected household does not subscribe to the *Atlantic Journal* and does subscribe to the *Beacon News*.

$\bar{A} \cap \bar{B}$ = the randomly selected household does not subscribe to the *Atlantic Journal* and does not subscribe to the *Beacon News*.

Since 650,000 households subscribe to the *Atlantic Journal* (A) and 250,000 households subscribe to both the *Atlantic Journal* and the *Beacon News* ($A \cap B$), it follows that $650,000 - 250,000 = 400,000$ households subscribe to the *Atlantic Journal* but do not subscribe to the *Beacon News* ($\bar{A} \cap B$). This subtraction is illustrated in Table 3.2(a). By similar logic, it also follows that:

1 As illustrated in Table 3.2(b), $500,000 - 250,000 = 250,000$ households do not subscribe to the *Atlantic Journal* but do subscribe to the *Beacon News* ($A \cap \bar{B}$).

2 As illustrated in Table 3.2(c), $350,000 - 250,000 = 100,000$ households do not subscribe to the *Atlantic Journal* and do not subscribe to the *Beacon News* ($\bar{A} \cap \bar{B}$).

TABLE 3.2 Subtracting to Find the Number of Households Corresponding to the Events $A \cap B$, $A \cap \bar{B}$, and $\bar{A} \cap \bar{B}$

(a) The Number of Households Corresponding to A and B

Events	Subscribes to <i>Beacon News</i> , B	Does Not Subscribe to <i>Beacon News</i> , $\bar{B}$	Total
Subscribes to <i>Atlantic Journal</i> , A	250,000	650,000 - 250,000 = 400,000	900,000
Does Not Subscribe to <i>Atlantic Journal</i> , $\bar{A}$	500,000	500,000	1,000,000

(b) The Number of Households Corresponding to $\bar{A}$ and B

Events	Subscribes to <i>Beacon News</i> , B	Does Not Subscribe to <i>Beacon News</i> , $\bar{B}$	Total
Subscribes to <i>Atlantic Journal</i> , A	250,000	650,000 - 250,000 = 400,000	650,000
Does Not Subscribe to <i>Atlantic Journal</i> , $\bar{A}$	500,000 - 250,000 = 250,000	500,000	750,000

TABLE 3.2 (continued)

(c) The Number of Households Corresponding to $(\bar{A} \text{ and } \bar{B})$

Events	Subscribes to Beacon News, $\bar{B}$	Does Not Subscribe to Beacon News, $\bar{B}$	Total
Subscribes to Atlantic Journal, $\bar{A}$	250,000	650,000 - 250,000 = 400,000	650,000
Does Not Subscribe to Atlantic Journal, $\bar{A}$	500,000 - 250,000 = 250,000	350,000 - 250,000 = 100,000	350,000
Total	500,000	500,000	1,000,000

TABLE 3.3 A Contingency Table Summarizing Subscription Data for the Atlantic Journal and the Beacon News

Events	Subscribes to Beacon News, B	Does Not Subscribe to Beacon News, $\bar{B}$	Total
Subscribes to Atlantic Journal, A	250,000	400,000	650,000
Does Not Subscribe to Atlantic Journal, $\bar{A}$	250,000	100,000	350,000
Total	500,000	500,000	1,000,000

We summarize all of these results in Table 3.3, which is called a contingency table. Because we will randomly select one household (making all of the households equally likely to be chosen), the probability of any of the previously defined events is the ratio of the number of households corresponding to the event's occurrence to the total number of households in the city. Therefore, for example,

$$P(A) = \frac{650,000}{1,000,000} = .65 \quad P(B) = \frac{500,000}{1,000,000} = .5$$

$$P(A \cap B) = \frac{250,000}{1,000,000} = .25$$

This last probability says that the probability that the randomly selected household subscribes to both the *Atlantic Journal* and the *Beacon News* is .25.

Next, letting $A \cup B$ denote A or B (or both), we consider finding the probability of the event

$A \cup B \equiv$ the randomly selected household subscribes to the *Atlantic Journal* or the *Beacon News* (or both)—that is, subscribes to at least one of the two newspapers.

Looking at Table 3.3, we see that the households subscribing to the *Atlantic Journal* or the *Beacon News* are (1) the 400,000 households that subscribe to only the *Atlantic Journal*, $A \cap \bar{B}$, (2) the 250,000 households that subscribe to only the *Beacon News*, $\bar{A} \cap B$, and (3) the 250,000 households that subscribe to both the *Atlantic Journal* and the *Beacon News*, $A \cap B$. Therefore, since a total of 900,000 households subscribe to the *Atlantic Journal* or the *Beacon News*, it follows that

$$P(A \cup B) = \frac{900,000}{1,000,000} = .9$$

This says that the probability that the randomly selected household subscribes to the *Atlantic Journal* or the *Beacon News* is .90. That is, 90 percent of the households in the city subscribe to the *Atlantic Journal* or the *Beacon News*. Notice that $P(A \cup B) = .90$ does not equal

$$P(A) + P(B) = .65 + .5 = 1.15$$

Logically, the reason for this is that both $P(A) = .65$ and $P(B) = .5$ count the 25 percent of the households that subscribe to both newspapers. Therefore, the sum of $P(A)$ and $P(B)$ counts this

25 percent of the households once too often. It follows that if we subtract $P(A \cap B) = .25$ from the sum of $P(A)$ and $P(B)$, then we will obtain $P(A \cup B)$. That is,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= .65 + .5 - .25 = .90$$

In order to generalize the ideas in the previous example, we make the following definitions:

The Intersection and Union of Two Events

Given two events A and B ,

- 1 The intersection of A and B is the event consisting of the sample space outcomes belonging to both A and B . The intersection is denoted by $A \cap B$. Furthermore, $P(A \cap B)$ denotes the probability that both A and B will simultaneously occur.
- 2 The union of A and B is the event consisting of the sample space outcomes belonging to A or B (or both). The union is denoted $A \cup B$. Furthermore, $P(A \cup B)$ denotes the probability that A or B (or both) will occur.

Noting that Figure 3.5 shows Venn diagrams depicting the events A , B , $A \cap B$, and $A \cup B$, we have the following general result:

The Addition Rule

Let A and B be events. Then, the probability that A or B (or both) will occur is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The reasoning behind this result has been illustrated at the end of Example 3.9. Similarly, the Venn diagrams in Figure 3.5 show that when we compute $P(A) + P(B)$, we are counting each of the sample space outcomes in $A \cap B$ twice. We correct for this by subtracting $P(A \cap B)$.

We next define the idea of *mutually exclusive events*:

Mutually Exclusive Events

Two events A and B are *mutually exclusive* if they have no sample space outcomes in common. In this case, the events A and B cannot occur simultaneously, and thus

$$P(A \cap B) = 0$$

Noting that Figure 3.6 is a Venn diagram depicting two mutually exclusive events, we consider the following example.

EXAMPLE

Consider randomly selecting a card from a standard deck of 52 playing cards. We define the following events:

J = the randomly selected card is a jack.

Q = the randomly selected card is a queen.

R = the randomly selected card is a red card (that is, a diamond or a heart).

Because there is no card that is both a jack and a queen, the events J and Q are mutually exclusive. On the other hand, there are two cards that are both jacks and red cards—the jack of diamonds and the jack of hearts—so the events J and R are not mutually exclusive.

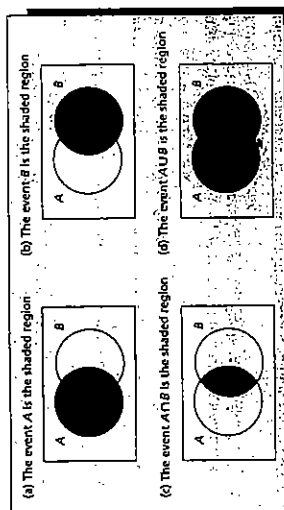
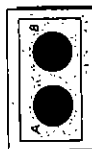
FIGURE 3.5 Venn Diagrams Depicting the Events A , B , $A \cap B$, and $A \cup B$ 

FIGURE 3.6 Two Mutually Exclusive Events



We have seen that for any two events A and B , the probability that A or B (or both) will occur is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Therefore, when calculating $P(A \cup B)$, we should always subtract $P(A \cap B)$ from the sum of $P(A)$ and $P(B)$. However, when A and B are mutually exclusive, $P(A \cap B)$ equals 0. Therefore, in this case—and only in this case—we have the following:

The Addition Rule for Two Mutually Exclusive Events

Let A and B be mutually exclusive events. Then, the probability that A or B will occur is

$$P(A \cup B) = P(A) + P(B)$$

Again consider randomly selecting a card from a standard deck of 52 playing cards, and define the events

J = the randomly selected card is a jack.

Q = the randomly selected card is a queen.

R = the randomly selected card is a red card (a diamond or a heart).

Since there are four jacks, four queens, and 26 red cards, we have $P(J) = \frac{4}{52}$, $P(Q) = \frac{4}{52}$, and $P(R) = \frac{26}{52}$. Furthermore, since there is no card that is both a jack and a queen, the events J and Q are mutually exclusive and thus $P(J \cap Q) = 0$. It follows that the probability that the randomly selected card is a jack or a queen is

$$\begin{aligned} P(J \cup Q) &= P(J) + P(Q) \\ &= \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13} \end{aligned}$$

Since there are two cards that are both jacks and red cards—the jack of diamonds and the jack of hearts—the events J and R are not mutually exclusive. Therefore, the probability that the randomly selected card is a jack or a red card is

$$\begin{aligned} P(J \cup R) &= P(J) + P(R) - P(J \cap R) \\ &= \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13} \end{aligned}$$

We now consider an arbitrary group of events— $A_1, A_2, \dots, A_n$. We will denote the probability that A_1 or A_2 or $\dots$ or A_n occurs (that is, the probability that at least one of the events occurs) as $P(A_1 \cup A_2 \cup \dots \cup A_n)$. Although there is a formula for this probability, it is quite complicated and we will not present it in this book. However, sometimes we can use sample spaces to reason out such a probability. For instance, in the playing card situation of Example 3.11, there are four jacks, four queens, and 22 red cards that are not jacks or queens (the 26 red cards minus the two red jacks and the two red queens). Therefore, because there are a total of 30 cards corresponding to the event $J \cup Q \cup R$, it follows that

$$P(J \cup Q \cup R) = \frac{30}{52} = \frac{15}{26}$$

Because some cards are both jacks and red cards, and because some cards are both queens and red cards, we say that the events J , Q , and R are not mutually exclusive. When, however, a group of events is mutually exclusive, there is a simple formula for the probability that at least one of the events will occur:

The Addition Rule for N Mutually Exclusive Events

The events $A_1, A_2, \dots, A_n$ are mutually exclusive if no two of the events have any sample space outcomes in common. In this case, no two of the events can occur simultaneously, and

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

As an example of using this formula, again consider the playing card situation and the events J and Q . If we define the event

K = the randomly selected card is a king

then the events J , Q , and K are mutually exclusive. Therefore,

$$\begin{aligned} P(J \cup Q \cup K) &= P(J) + P(Q) + P(K) \\ &= \frac{4}{52} + \frac{4}{52} + \frac{4}{52} = \frac{12}{52} = \frac{3}{13} \end{aligned}$$

EXAMPLE 3.12 THE ACCURATING

Recall that Figure 3.3 (page 128) gives the AccuRatings estimates of the number and the percentage of Los Angeles residents who favor each of the 10 top radio stations in Los Angeles. We will let the call letters of each station denote the event that a randomly selected Los Angeles resident would favor the station. Since the AccuRatings survey asked each resident to name the single station (if any) that he or she listens to most, the 10 events

KPWR KLAX KROQ KIIS-A/F KFI
KFWB KKBT KABC KRTH and KCBS-FM

are mutually exclusive. Therefore, for example, the probability that a randomly selected Los Angeles resident would favor a station that is rated among the top 10

$$P(KPWR \cup KLAX \cup \dots \cup KCBS-FM)$$

is the sum of the 10 individual station probabilities

$$P(KPWR) + P(KLAX) + \dots + P(KCBS-FM)$$

Since we can estimate each individual station probability by dividing the share for the station in Figure 3.3 by 100, we estimate that the probability that a randomly selected Los Angeles resident would favor a station that is rated among the top 10 is

$$.08 + .064 + .061 + .050 + .047 + .046 + .042 + .036 + .036 = .508$$

Note that these probabilities sum to less than 1 because there are far more than 10 stations in Los Angeles.



CONCEPTS

3.8 Explain what it means for two events to be mutually exclusive; for N events.

3.9 If A and B are events, define $\bar{A}$, $A \cup B$, $A \cap B$, and $\bar{A} \cap \bar{B}$.

METHODS AND APPLICATIONS

3.10, 3.11, 3.13, 3.15 Consider a standard deck of 52 playing cards, a randomly selected card from the deck, and the following events:

R = red B = black A = ace N = nine D = diamond C = club

- Describe the sample space outcomes that correspond to each of these events.
- For each of the following pairs of events, indicate whether the events are mutually exclusive. In each case, if you think the events are mutually exclusive, explain why the events have no common sample space outcomes. If you think the events are not mutually exclusive, list the sample space outcomes that are common to both events.

- R and A
- R and C
- A and N
- A and D
- N and C

3.11 Of 10,000 students at a college, 2,500 have a Mastercard (M), 4,000 have a VISA (V), and 1,000 have both.

- Find the probability that a randomly selected student

- Has a Mastercard.
- Has a VISA.
- Has both credit cards.

3.12 Construct and fill in a contingency table summarizing the credit card data. Employ the following pairs of events: M and $\bar{M}$, V and $\bar{V}$.

- Use the contingency table to find the probability that a randomly selected student

- Has a Mastercard or a VISA.
- Has neither credit card.
- Has exactly one of the two credit cards.

3.13 The card game of Euchre employs a deck that consists of all four of the aces, kings, queens, jacks, tens, and nines (one of each suit—clubs, diamonds, spades, and hearts). Find the probability that a randomly selected card from a Euchre deck is

- A jack (J).
- A spade (S).
- A jack or an ace (A).
- A jack or a spade.

3.14 Are the events J and A mutually exclusive? J and S ? Why or why not?

3.15 Each month a brokerage house studies various companies and rates each company's stock as being either "low risk" or "moderate to high risk." In a recent report, the brokerage house summarized its findings about 15 aerospace companies and 25 food retailers in the following table:

Company Type	Low Risk	Moderate to High Risk
Aerospace company	6	9
Food retailer	15	10

If we randomly select one of the total of 40 companies, find

- The probability that the company is a food retailer.
- The probability that the company's stock is "low risk."
- The probability that the company's stock is "moderate to high risk."
- The probability that the company is a food retailer and has a stock that is "low risk."
- The probability that the company is a food retailer or has a stock that is "low risk."

3.16 In the book *Essentials of Marketing Research*, William R. Dillon, Thomas J. Madden, and Neil H. Firtle present the results of a concept study for a new wine cooler. Three hundred consumers between 21 and 49 years old were randomly selected. After sampling the new beverage, each was asked to rate the appeal of the phrase

Not sweet like wine coolers; not filling like beer; and more refreshing than wine or mixed drinks

as it relates to the new wine cooler. The rating was made on a scale from 1 to 5, with 5 representing "extremely appealing" and with 1 representing "not at all appealing." The results obtained are given in Table 3.4.

TABLE 3.4 Results of a Concept Study for a New Wine Cooler

Rating	Extremely appealing (5)	(4)	(3)	(2)	Not at all appealing (1)	Gender		Age Group	
						Male	Female	21-24	25-49
Total	151	68	83	37	37				
Extremely appealing (5)	(4)	91	51	40	36	36	19		
(3)	(3)	36	21	15	9	12	15		
(2)	(2)	13	7	6	4	6	3		
Not at all appealing (1)	(1)	9	3	6	4	3	2		

Sources: W. R. Dillon, T. J. Madden, and N. H. Firtle, *Essentials of Marketing Research* (Burr Ridge, IL: Richard D. Irwin, Inc., 1993), p. 390.

Based on these results, estimate the probability that a randomly selected 21- to 49-year-old consumer

- Would give the phrase a rating of 5.
- Would give the phrase a rating of 3 or higher.
- Is in the 21-24 age group; the 25-49 age group; the 35-49 age group.
- Is a male who gives the phrase a rating of 4.
- Is a 35- to 49-year-old who gives the phrase a rating of 1.

3.15 THE ACCURATINGS CASE

Using the information in Figure 3.3 (page 128), find an estimate of the probability that a randomly selected Los Angeles resident (12 years or older) would

- Name one of the top three rated stations (KPWR, KLA-X, or KROQ) as the station that he or she listens to most.
- Not name one of the top five rated stations as the station that he or she listens to most.
- Name a station that is not rated among the top seven stations as the station that he or she listens to most.
- Name a station that is not rated among the top three stations nor is rated lower than 10th as the station that he or she listens to most.

3.4 Conditional Probability and Independence

Conditional probability In Table 3.5 we repeat the contingency table summarizing the subscription data for the *Atlantic Journal* and the *Bacon News*. Suppose that we randomly select a household, and that the chosen household reports that it subscribes to the *Bacon News*. Given this new information, we wish to find the probability that the household subscribes to the *Atlantic Journal*. This new probability is called a **conditional probability**.

The probability of the event A , given the condition that the event B has occurred, is written as $P(A|B)$ —pronounced "the probability of A given B ." We often refer to such a probability as the **conditional probability of A given B** .

In order to find the conditional probability that a household subscribes to the *Atlantic Journal*, given that it subscribes to the *Bacon News*, notice that if we know that the randomly selected household subscribes to the *Bacon News*, we know that we are considering one of 500,000 households (see Table 3.5). That is, we are now considering what we might call a reduced sample space of 500,000 households. Since 250,000 of these 500,000 *Bacon News* subscribers also subscribe to the *Atlantic Journal*, we have

$$P(A|B) = \frac{250,000}{500,000} = .5$$

This says that the probability that the randomly selected household subscribes to the *Atlantic Journal*, given that the household subscribes to the *Bacon News*, is .5. That is, 50 percent of the *Bacon News* subscribers also subscribe to the *Atlantic Journal*.

Next, suppose that we randomly select another household from the community of 1,000,000 households, and suppose that this newly chosen household reports that it subscribes to the *Atlantic Journal*. We now wish to find the probability that this household subscribes to the

TABLE 3.5 A Contingency Table Summarizing Subscription Data for the *Atlantic Journal* and the *Beacon News*

Events	Subscribers to <i>Beacon News</i> , B	Does Not Subscribe to <i>Beacon News</i> , $\bar{B}$	Total
Subscribers to <i>Atlantic Journal</i> , A	250,000	400,000	650,000
Does Not Subscribe to <i>Atlantic Journal</i> , $\bar{A}$	250,000	100,000	350,000
Total	500,000	500,000	1,000,000

Beacon News. We write this new probability as $P(B|A)$. If we know that the randomly selected household subscribes to the *Atlantic Journal*, we know that we are considering a reduced sample space of 650,000 households (see Table 3.5). Since 250,000 of these 650,000 *Atlantic Journal* subscribers also subscribe to the *Beacon News*, we have

$$P(B|A) = \frac{250,000}{650,000} = .3846$$

This says that the probability that the randomly selected household subscribes to the *Beacon News*, given that the household subscribes to the *Atlantic Journal*, is .3846. That is, 38.46 percent of the *Atlantic Journal* subscribers also subscribe to the *Beacon News*.

If we divide both the numerator and denominator of each of the conditional probabilities $P(A|B)$ and $P(B|A)$ by 1,000,000, we obtain

$$P(A|B) = \frac{250,000}{500,000} = \frac{250,000/1,000,000}{500,000/1,000,000} = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{250,000}{650,000} = \frac{250,000/1,000,000}{650,000/1,000,000} = \frac{P(A \cap B)}{P(A)}$$

We express these conditional probabilities in terms of $P(A)$, $P(B)$, and $P(A \cap B)$ in order to obtain a more general formula for a conditional probability. We need a more general formula because, although we can use the reduced sample space approach we have demonstrated to find conditional probabilities when all of the sample space outcomes are equally likely, this approach may not give correct results when the sample space outcomes are not equally likely. We now give expressions for conditional probability that are valid for any sample space.

Conditional Probability

1. The conditional probability that A will occur given that B will occur is written $P(A|B)$ and is defined to be

Here we assume that $P(B)$ is greater than 0.

2. The conditional probability that B will occur given that A will occur is written $P(B|A)$ and is defined to be

Here we assume that $P(A)$ is greater than 0.

If we multiply both sides of the equation

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

by $P(B)$, we obtain the equation

$$P(A \cap B) = P(B)P(A|B)$$

Similarly, if we multiply both sides of the equation

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

by $P(A)$, we obtain the equation

$$P(A \cap B) = P(A)P(B|A)$$

In summary, we now have two equations that can be used to calculate $P(A \cap B)$. These equations are often referred to as the general multiplication rule for probabilities.

The General Multiplication Rule—Two Ways to Calculate $P(A \cap B)$

Given any two events A and B ,

$$\begin{aligned} P(A \cap B) &= P(A)P(B|A) \\ &= P(B)P(A|B) \end{aligned}$$

In a soft drink taste test, each of 1,000 consumers chose between two colas—Cola 1 and Cola 2—and stated whether they preferred their cola drinks *sweet* or *very sweet*. Unfortunately, some of the survey information was lost. The following information remains:

- 1 68.3 percent of the consumers (that is, 683 consumers) preferred Cola 1 to Cola 2.
- 2 62 percent of the consumers (that is, 620 consumers) preferred their cola *sweet* (rather than *very sweet*).
- 3 85 percent of the consumers who said that they liked their cola *sweet* preferred Cola 1 to Cola 2.

To recover all of the lost survey information, consider randomly selecting one of the 1,000 survey participants, and define the following events:

$C1$ = the randomly selected consumer prefers Cola 1.

$C2$ = the randomly selected consumer prefers Cola 2.

S = the randomly selected consumer prefers *sweet* cola drinks.

V = the randomly selected consumer prefers *very sweet* cola drinks.

From the survey information that remains, (1) says that $P(C1) = .683$, (2) says that $P(S) = .62$, and (3) says that $P(C1|S) = .85$.

We will see that we can recover all of the lost survey information if we can find $P(C1 \cap S)$. The general multiplication rule says that

$$P(C1 \cap S) = P(C1)P(S|C1) = P(S)P(C1|S)$$

Although we know that $P(C1) = .683$, we do not know $P(S|C1)$. Therefore, we cannot calculate $P(C1 \cap S)$ as $P(C1)P(S|C1)$. However, because we know that $P(S) = .62$ and that $P(C1|S) = .85$, we can calculate

$$P(C1 \cap S) = P(S)P(C1|S) = (.62)(.85) = .527$$

This implies that 527 consumers preferred Cola 1 and preferred their cola *sweet*. Since 683 consumers preferred Cola 1, and 620 consumers preferred *sweet* cola drinks, we can summarize the numbers of consumers corresponding to the events $C1$, $C2$, S , V , and $C1 \cap S$ as shown in Table 3.6. Furthermore, by performing subtractions as shown in Table 3.7, the numbers of consumers corresponding to the events $C1 \cap V$, $C2 \cap S$, and $C2 \cap V$ can be obtained. We summarize all of our results in Table 3.8. We will use these results in the next subsection to investigate the relationship between cola preference and sweetness preference.

TABLE 3.6 A Summary of the Number of Consumers Corresponding to the Events $C1$, $C2$, S , V , and $C1 \cap S$

Events	S (Sweet)	V (Very Sweet)	Total
$C1$ (Cola 1)	527	683	830
$C2$ (Cola 2)	317	317	634
Total	620	380	1,000

TABLE 3.7 Subtractions to Obtain the Number of Consumers Corresponding to the Events $C1 \cap V$, $C2 \cap S$, and $C2 \cap V$

Events	S (Sweet)	V (Very Sweet)	Total
$C1$ (Cola 1)	527	683	830
$C2$ (Cola 2)	620 - 527 = 93	380 - 156 = 224	317
Total	620	380	1,000

TABLE 3.8 A Contingency Table Summarizing the Cola Brand and Sweetness Preferences

Events	S (Sweet)	V (Very Sweet)	Total
$C1$ (Cola 1)	527	156	683
$C2$ (Cola 2)	93	224	317
Total	620	380	1,000

Independence We have seen in Example 3.13 that $P(C1) = .683$, while $P(C1|S) = .85$. Because $P(C1|S)$ is greater than $P(C1)$, the probability that a randomly selected consumer will prefer Cola 1 is higher if we know that the person prefers sweet cola than it is if we have no knowledge of the person's sweetness preference. Another way to see this is to use Table 3.8 to calculate

$$P(C1|V) = \frac{P(C1 \cap V)}{P(V)} = \frac{156/1,000}{380/1,000} = .4105$$

Since $P(C1|S) = .85$ is greater than $P(C1|V) = .4105$, the probability that a randomly selected consumer will prefer Cola 1 is higher if the consumer prefers sweet colas than it is if the consumer prefers very sweet colas. Since the probability of the event $C1$ is influenced by whether the event S occurs, we say that the events $C1$ and S are dependent. If $P(C1|S)$ were equal to $P(C1)$, then the probability of the event $C1$ would not be influenced by whether S occurs. In this case we would say that the events $C1$ and S are independent. This leads to the following definition of independence:

Independent Events

Two events A and B are independent if and only if

- $P(A|B) = P(A)$ or, equivalently,
- $P(B|A) = P(B)$

Here we assume that $P(A)$ and $P(B)$ are greater than 0.

When we say that conditions (1) and (2) are equivalent, we mean that condition (1) holds if and only if condition (2) holds. Although we will not prove this, we will demonstrate it in the next example.

In the soft drink taste test of Example 3.13, we have seen that $P(C1|S) = .85$ does not equal $P(C1) = .683$. This implies that $P(S|C1)$ does not equal $P(S)$. To demonstrate this, note from Table 3.8 that

$$P(S|C1) = \frac{P(C1 \cap S)}{P(C1)} = \frac{527/1,000}{683/1,000} = .7716$$

This probability is larger than $P(S) = 620/1,000 = .62$. In summary:

- A comparison of $P(C1|S) = .85$ and $P(C1) = .683$ says that a consumer is more likely to prefer Cola 1 if the consumer prefers sweet colas.
- A comparison of $P(S|C1) = .7716$ and $P(S) = .62$ says that a consumer is more likely to prefer sweet colas if the consumer prefers Cola 1.

This suggests, but does not prove, that one reason Cola 1 is preferred to Cola 2 is that Cola 1 is sweeter (as opposed to very sweet).

If the occurrences of the events A and B have nothing to do with each other, then we know that A and B are independent events. This implies that $P(A|B)$ equals $P(A)$ and that $P(B|A)$ equals $P(B)$. Recall that the general multiplication rule tells us that, for any two events A and B , we can say that

$$P(A \cap B) = P(A)P(B|A)$$

Therefore, if $P(B|A)$ equals $P(B)$, it follows that

$$P(A \cap B) = P(A)P(B)$$

which is called the **multiplication rule for independent events**. To summarize:

The Multiplication Rule for Two Independent Events

If A and B are independent events, then

$$P(A \cap B) = P(A)P(B)$$

As a simple example, define the events C and P as follows:

C = your favorite college football team wins its first game next season.

P = your favorite professional football team wins its first game next season.

Suppose you believe that for next season $P(C) = .6$ and $P(P) = .6$. Then, because the outcomes of a college football game and a professional football game would probably have nothing to do with each other, it is reasonable to assume that C and P are independent events. It follows that

$$P(C \cap P) = P(C)P(P) = (.6)(.6) = .36$$

This probability might be surprisingly low. That is, since you believe that each of your teams has a 60 percent chance of winning, you might feel reasonably confident that both your college and professional teams will win their first game. Yet, the chance of this happening is really only 36!

Next, consider a group of events $A_1, A_2, \dots, A_N$. Intuitively, the events $A_1, A_2, \dots, A_N$ are independent if the occurrences of these events have nothing to do with each other. Denoting the probability that A_1 and A_2 and $\dots$ and A_N will simultaneously occur as $P(A_1 \cap A_2 \cap \dots \cap A_N)$, we have the following:

The Multiplication Rule for N Independent Events

If $A_1, A_2, \dots, A_N$ are independent events, then

$$P(A_1 \cap A_2 \cap \dots \cap A_N) = P(A_1)P(A_2) \dots P(A_N)$$

This says that the multiplication rule for two independent events can be extended to any number of independent events.

This example is based on a real situation encountered by a major producer and marketer of consumer products. The company assessed the service it provides by surveying the attitudes of its customers regarding 10 different aspects of customer service—order filled correctly, billing amount on invoice correct, delivery made on time, and so forth. When the survey

results were analyzed, the company was dismayed to learn that only 59 percent of the survey participants indicated that they were satisfied with all 10 aspects of the company's service. Upon investigation, each of the 10 departments responsible for the aspects of service considered in the study insisted that it satisfied its customers 95 percent of the time. That is, each department claimed that its error rate was only 5 percent. Company executives were confused and felt that there was a substantial discrepancy between the survey results and the claims of the departments providing the services. However, a company statistician pointed out that there was no discrepancy. To understand this, consider randomly selecting a customer from among the survey participants, and define 10 events (corresponding to the 10 aspects of service studied):

A_1 = the customer is satisfied that the order is filled correctly (aspect 1).

A_2 = the customer is satisfied that the billing amount on the invoice is correct (aspect 2).

...

A_{10} = the customer is satisfied that the delivery is made on time (aspect 10).

Also, define the event

S = the customer is satisfied with all 10 aspects of customer service.

Since 10 different departments are responsible for the 10 aspects of service being studied, it is reasonable to assume that all 10 aspects of service are independent of each other. For instance, billing amounts would be independent of delivery times. Therefore, $A_1, A_2, \dots, A_{10}$ are independent events, and

$$P(S) = P(A_1 \cap A_2 \cap \dots \cap A_{10}) \\ = P(A_1)P(A_2) \dots P(A_{10})$$

If, as the departments claim, each department satisfies its customers 95 percent of the time, then the probability that the customer is satisfied with all 10 aspects is

$$P(S) = (.95)(.95) \dots (.95) = (.95)^{10} = .5987$$

This result is almost identical to the 59 percent satisfaction rate reported by the survey participants.

If the company wants to increase the percentage of its customers who are satisfied with all 10 aspects of service, it must improve the quality of service provided by the 10 departments. For example, to satisfy 95 percent of its customers with all 10 aspects of service, the company must require each department to raise the fraction of the time it satisfies its customers to x , where

$$(x)^{10} = .95$$

It follows that

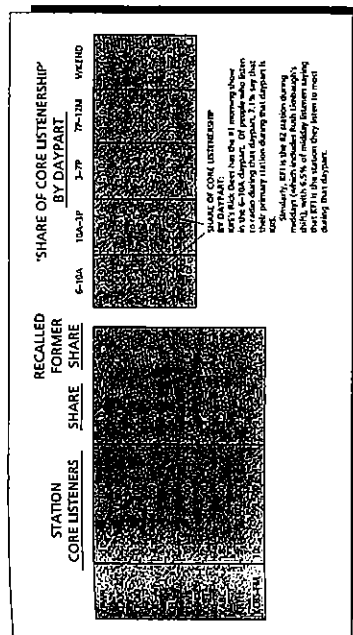
$$x = (.95)^{1/10} = .9949$$

and that each department must satisfy its customers 99.49 percent of the time (rather than the current 95 percent of the time).

A real-world application of conditional probability, independence, and dependence

In addition to asking each of the 5,528 sampled Los Angeles residents to name which station (if any) he or she listens to most on an overall basis, AccuRatings asked each resident to name which station (if any) he or she listens to most during various parts of the day. The various parts of the day considered by AccuRatings and the results of the survey are given in Figure 3.7. To explain these results, suppose that 2,827 of the 5,528 sampled residents said that they listen to the radio during

FIGURE 3.7 Further Portions of an AccuRatings Report on Radio Ratings in the Los Angeles Market



Source: Strategic Radio Research, AccuRatings Introduction for Broadcasters.

some portion of the 6–10 A.M. daypart. Furthermore, suppose that 201 of these 2,827 residents named station KIIS as the station that they listen to most during that daypart. It follows that

$$\frac{201}{2,827} = .07100106$$

is an estimate of $P(KIIS | 6-10 \text{ A.M.})$, the probability that a randomly selected Los Angeles resident who listens to the radio during the 6–10 A.M. daypart would name KIIS as his or her primary station during that daypart. Said equivalently, station KIIS has an estimated share of 7.1 percent of the 6–10 A.M. radio listeners. In general, Figure 3.7 gives the estimated shares during the various dayparts for the five stations that are rated best overall (KPWR, KLAX, KROQ, KIIS, and KFI). Examination of this figure seems to reveal that a station's share depends somewhat on the daypart being considered. For example, note that Figure 3.7 tells us that the estimate of $P(KIIS | 6-10 \text{ A.M.})$ is .071, whereas the estimate of $P(KIIS | 3-7 \text{ P.M.})$ is .049. This says that station KIIS's estimated share of the 6–10 A.M. radio listeners is higher than its estimated share of the 3–7 P.M. radio listeners.

Estimating Probabilities of Radio Station Listenership

AccuRatings provides the sort of estimates given in Figures 3.3 and 3.7 not only for the Los Angeles market but for other markets as well. In addition, AccuRatings provides (for a given market) hour-by-hour estimates of the probabilities of different stations being listened to in the market. How this is done is an excellent real-world application of the general multiplication rule. As an example, consider how AccuRatings might find an estimate of "the probability that a randomly selected Los Angeles resident will be listening to station KIIS at an average moment from 7 to 8 A.M." To estimate this probability, AccuRatings estimates

- 1 The probability that a randomly selected Los Angeles resident will be listening to the radio at an average moment from 7 to 8 A.M.
- 2 The probability that a randomly selected Los Angeles resident who is listening to the radio at an average moment from 7 to 8 A.M. will be listening to station KIIS at that average moment.

Because the hour of 7 to 8 A.M. is in the 6–10 A.M. daypart, it is reasonable to estimate the probability in (2) by using an estimate of $P(KIIS | 6-10 \text{ A.M.})$, which Figure 3.7 tells us is .071. To find an estimate of the probability in (1), AccuRatings uses a 2,000-person national study. Here, each person is interviewed to obtain a detailed, minute-by-minute reconstruction of the times that the person listened to the radio on the previous day (with no attempt to identify the specific stations listened to). Then, for each minute of the day the proportion of the 2,000 people who listened to the radio during that minute is determined. The average of the 60 such proportions for a particular hour is the estimate of the probability that a randomly selected person will listen to the radio at an average moment during that hour. Using a national study is reasonable because the detailed reconstruction made by AccuRatings would be extremely time-consuming to construct for individual markets and because AccuRatings' studies show very consistent hour-by-hour patterns of radio usage across markets, across seasons, and across demographics. This implies that the national study applies to individual markets (such as the Los Angeles market). Suppose, then, that the national study estimate of the 7 to 8 A.M. radio listening probability in (1) is .242. Since (as previously discussed) an estimate of the station KIIS conditional listening probability in (2) is .071, it follows that an estimate of the desired probability is $.242 \times .071 = .017182 \approx .017$. This says that we estimate that 1.7 percent of all Los Angeles residents will be listening to station KIIS at an average moment from 7 to 8 A.M. Assuming that there are 8,300,000 Los Angeles residents, we estimate that

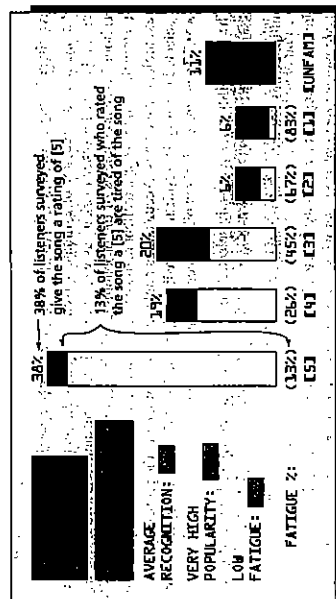
$$(8,300,000) \times (.017) = 141,000$$

of these residents will be listening to station KIIS at an average moment from 7 to 8 A.M. Finally, note that in making its hour-by-hour radio station listening estimates, AccuRatings makes a separate set of estimates for the hours on a weekday, for the hours on Saturday, and for the hours on Sunday. The above 7 to 8 A.M. estimate is for the 7 to 8 A.M. hour on a weekday.

Estimating Song Ratings

In addition to providing AccuRatings reports to radio stations, Strategic Radio Research does music research for clients such as MTV. Figure 3.8 gives a portion of a *title-by-title analysis* for the song "Gangsta's Paradise" by Coolio. Listeners are surveyed and are asked to rate the song on a 1 to 5 rating scale with 1 being the lowest possible rating and 5 being the highest. Figure 3.8 gives a histogram of these ratings; notice that UNFAM indicates that the listener was not familiar with this particular song. The percentages above the bars of the histogram give the percentages

FIGURE 3.8 A Portion of a Title-by-Title Analysis for the Song "Gangsta's Paradise" by Coolio



Source: Strategic Radio Research, Chicago, Illinois.

of listeners rating the song 5, 4, 3, 2, 1, and UNFAM, respectively. If we let the symbol denoting a particular rating also denote the event that a randomly selected listener would give the song the rating, it follows that we estimate that

$$P(5) = .38 \quad P(4) = .19 \quad P(3) = .20 \\ P(2) = .06 \quad P(1) = .06 \quad P(\text{UNFAM}) = .11$$

The three boxes on the left of Figure 3.8 give recognition, popularity, and fatigue indexes for the song being analyzed. Although we will have to wait until Chapter 4 to learn the meaning of the popularity index, we now explain the meaning of the recognition and fatigue indexes. The recognition index estimates the probability that a randomly selected listener is familiar with the song. We have seen that the estimate of $P(\text{UNFAM})$ is .11, so the recognition index is $1 - .11 = .89$, which is expressed as the 89 percent in Figure 3.8. This index says we estimate that 89 percent of all listeners are familiar with the song. The fatigue index, 28 percent, estimates the percentage of listeners who are tired of the song. That is, if 7 denotes the event that a randomly selected listener is tired of the song, we estimate that $P(7) = .28$. Finally, note that at the bottom of each histogram bar in Figure 3.8, and shaded in blue, is the fatigue percentage corresponding to the rating described by the bar. This percentage is an estimate of the conditional probability that a randomly selected listener giving the song that rating is tired of the song. Therefore, we estimate that $P(7 | 1) = .83$, $P(7 | 2) = .67$, $P(7 | 3) = .45$, $P(7 | 4) = .26$, and $P(7 | 5) = .13$. From these conditional probabilities we might conclude that the higher the song is rated, the lower is its fatigue percentage.

EXERCISES SECTION 3.4

CONCEPTS

- 3.16 Explain the concept of a conditional probability. Give an example of a conditional probability that would be of interest to a college student; to a business.
- 3.17 Explain what it means for two events to be independent.

METHODS AND APPLICATIONS

- 3.18 Recall from Exercise 3.11 (page 136) that of 10,000 students at a college, 2,500 have a Mastercard (M), 4,000 have a VISA (V), and 1,000 have both. Find
- The proportion of Mastercard holders who have VISA cards. Interpret and write this proportion as a conditional probability.
 - The proportion of VISA cardholders who have Mastercards. Interpret and write this proportion as a conditional probability.
 - Are the events *having a Mastercard* and *having a VISA* independent? Justify your answer.

$$\text{Yes; } P(M|V) = P(M)$$

- 3.19 Recall from Exercise 3.13 (page 136) that each month a brokerage house studies various companies and rates each company's stock as being either "low risk" or "moderate to high risk." In a recent report, the brokerage house summarized its findings about 15 aerospace companies and 25 food retailers in the following table:

Company Type	Low Risk	Moderate to High Risk
Aerospace company	6	9
Food retailer	15	10

If we randomly select one of the total of 40 companies, find

- The probability that the company's stock is moderate to high risk given that the firm is an aerospace company.
- The probability that the company's stock is moderate to high risk given that the firm is a food retailer.

- Determine if the company type is independent of the level of risk of the firm's stock.

3.20 John and Jane are married. The probability that John watches a certain television show is .4. The probability that Jane watches the show is .5. The probability that John watches the show, given that Jane does, is .7.

- Find the probability that both John and Jane watch the show.
- Find the probability that Jane watches the show, given that John does.
- Do John and Jane watch the show independently of each other? Justify your answer.

3.21 In Exercise 3.20, find the probability that either John or Jane watches the show.



$$.3, .17, .3, .19, .3, .20, \\ .3, .21, .3, .29, .3, .30$$

$$.4, P(V|M)$$

$$.25, P(M|V)$$

$$\text{Yes; } P(M|V) = P(M)$$

$$.6$$

$$.4$$

Dependent

$$.35$$

$$.875$$

$$\text{No; } P(\text{John} | \text{Jane}) \neq P(\text{John})$$

$$.55$$

3.22 In the July 29, 2001, issue of *The Journal News* (Hamilton, Ohio), Lynn Elber of the Associated Press reported that "while 40 percent of American families own a television set with a V-chip installed to block designated programs with sex and violence, only 17 percent of those parents use the device."

.058 a Use the report's results to find an estimate of the probability that a randomly selected American family has used a V-chip to block programs containing sex and violence.

Much higher

b According to the report, more than 50 percent of parents have used the TV rating system (TV-14, etc.) to control their children's TV viewing. How does this compare to the percentage using the V-chip?

.1692

3.23 According to the Associated Press report (in Exercise 3.22), 47 percent of parents who have purchased TV sets after V-chips became standard equipment in January 2000 are aware that their sets have V-chips, and of those who are aware of the option, 36 percent have programmed their V-chips. Using these results, find an estimate of the probability that a randomly selected parent who has bought a TV set since January 2000 has programmed the V-chip.

3.24 Fifteen percent of the employees in a company have managerial positions, and 25 percent of the employees in the company have MBA degrees. Also, 60 percent of the managers have MBA degrees. Using the probability formulas,

.09 a Find the proportion of employees who are managers and have MBA degrees.

.36 No; $P(MBA | MBA) \neq P(MBA)$

.31 b Find the proportion of MBAs who are managers.

c Are the events *being a manager* and *having an MBA* independent? Justify your answer.

3.25 In Exercise 3.24, find the proportion of employees who either have MBAs or are managers.

3.26 Consider Exercise 3.14 (page 136). Using the results in Table 3.4 (page 137), estimate the probability that a randomly selected 21- to 49-year-old consumer would

.793, 820 a Give the phrase a rating of 4 or 5 given that the consumer is male; give the phrase a rating of 4 or 5 given that the consumer is female. Based on these results, is the appeal of the phrase among males much different from the appeal of the phrase among females? Explain.

.832, b Give the phrase a rating of 4 or 5, given that the consumer is in the 21-24 age group; given that the consumer is in the 25-34 age group; given that the consumer is in the 35-49 age group.

.737

21-24, 35-49 Based on these results, which age group finds the phrase most appealing? Least appealing?

3.27 In a survey of 100 insurance claims, 40 are fire claims (FIRE), 16 of which are fraudulent (FRAUD). Also, there are a total of 40 fraudulent claims.

.40 a Construct a contingency table summarizing the claims data. Use the pairs of events FIRE and FRAUD.

Yes, $P(FRAUD | FIRE) = P(FRAUD)$

b What proportion of the fire claims are fraudulent?

c Are the events *a claim is fraudulent* and *a claim is a fire claim* independent? Use your probability of part b to prove your answer.

3.28 Recall from Exercise 3.3 (page 129) that two randomly selected customers are each asked to take a blind taste test and then to state which of three diet colas (labeled as A, B, or C) he or she prefers. Suppose that cola A's distributor claims that 80 percent of all people prefer cola A and that only 10 percent prefer each of colas B and C.

.04 a Assuming that the distributor's claim is true and that the two taste test participants make independent cola preference decisions, find the probability of each sample space outcome.

b Find the probability that neither taste test participant will prefer cola A.

c If, when the taste test is carried out, neither participant prefers cola A, use the probability you computed in part b to decide whether the distributor's claim seems valid. Explain.

3.29 A sprinkler system inside an office building has two types of activation devices, D1 and D2, which operate independently. When there is a fire, if either device operates correctly, the sprinkler system is turned on. In case of fire, the probability that D1 operates correctly is .92, and the probability that D2 operates correctly is .92. Find the probability that

.874 a Both D1 and D2 will operate correctly.

.996 b The sprinkler system will come on.

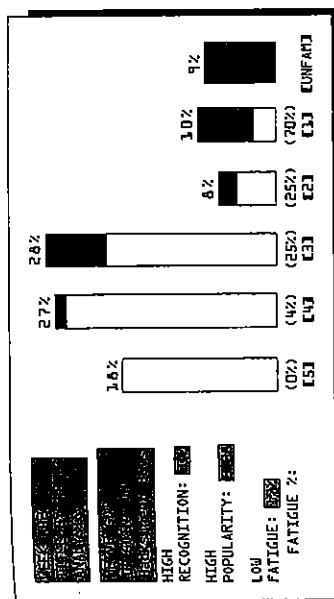
.004 c The sprinkler system will fail.

3.30 A product is assembled using 10 different components, each of which must meet specifications for five different quality characteristics. Suppose that there is a .9973 probability that each individual specification will be met.

.874 a Assuming that all 50 specifications are met independently, find the probability that the product meets all 50 specifications.

[Source: *The Journal News* (Hamilton, Ohio), July 29, 2001, p. C5.]

FIGURE 3.9 A Portion of a Title-by-Title Analysis for the Song "We've Got It Goin' On" by the Backstreet Boys



Source: Strategic Radio Research, Chicago, Illinois.

b Suppose that we wish to have a 99.73 percent chance that all 50 specifications will be met. If each specification will have the same chance of being met, how large must we make the probability of meeting each individual specification?

3.31 THE ACCURATINGS CASE

Consider the share of core listenership by daypart information given in Figure 3.7 (page 143).

a Find an estimate of $P(KPWR | 3-7 \text{ P.M.})$, the probability that a randomly selected Los Angeles resident who listens to the radio during the 3-7 P.M. daypart would name KPWR as his or her primary station during that daypart.

b Find $P(KLAX | 3-7 \text{ P.M.})$, $P(KROQ | 3-7 \text{ P.M.})$, $P(KIIS | 3-7 \text{ P.M.})$, and $P(KFI | 3-7 \text{ P.M.})$.

c Suppose that the Accuratings national survey estimates that the probability that a randomly selected Los Angeles resident will be listening to the radio at an average moment between 5 and 6 P.M. is .256. Use this survey result and the estimate in part a to estimate the probability that a randomly selected Los Angeles resident will be listening to station KPWR at an average moment between 5 and 6 P.M.

d Repeat part c for each of KLAX, KROQ, KIIS, and KFI.

e Find an estimate of the probability that a randomly selected Los Angeles resident will be listening to one of the five most highly rated stations in the Los Angeles market (KPWR, KLAX, KROQ, KIIS, or KFI) at an average moment between 5 and 6 P.M.

3.32 THE ACCURATINGS CASE

Figure 3.9 gives a portion of a title-by-title analysis for the song "We've Got It Goin' On" by the Backstreet Boys. The ratings information given in this figure is the same type given in Figure 3.8 (page 144) and explained in Example 3.16 (pages 144-145). Using the ratings information:

a Find an estimate of the probability that a randomly selected listener would give the song each of the ratings 5, 4, 3, 2, and 1.

b Find an estimate of the probability that a randomly selected listener is (1) familiar with the song; (2) tired of the song.

c Find estimates of each of $P(T | 5)$, $P(T | 4)$, $P(T | 3)$, $P(T | 2)$, and $P(T | 1)$, where T denotes the event that a listener is tired of the song.

3.33 In a murder trial in Los Angeles, the prosecution claims that the defendant was cut on the left middle finger at the murder scene, but the defendant claims the cut occurred in Chicago, the day after the murders had been committed. Because the defendant is a sports celebrity, many people noticed him before he reached Chicago. Twenty-two people saw him casually, one person on the plane to Chicago carefully studied his hands looking for a championship ring, and another person stood with him as he signed autographs and drove him from the airport to the hotel. None of these 24 people saw a cut on the defendant's finger. If in fact he was not cut at all, it would be extremely unlikely that he left blood at the murder scene.

- .0295 a Since a person casually meeting the defendant would not be looking for a cut, assume that the probability is 0 that such a person would not have seen the cut, even if it was there. Furthermore, assume that the person who carefully looked at the defendant's hands had a .5 probability of not seeing the cut even if it was there and that the person who drove the defendant from the airport to the hotel had a .6 probability of not seeing the cut even if it was there. Given these assumptions, and also assuming that all 24 people looked at the defendant independently of each other, what is the probability that all 24 people would not have seen the cut, even if it was there?
- .9705 b What is the probability that at least one of the 24 people would have seen the cut if it was there?
- Probably not c Given the result of part b and given the fact that none of the 24 people saw a cut, do you think the defendant had a cut on his hand before he reached Chicago?
- d How might we estimate what the assumed probabilities in a would actually be? (Note: This would not be easy.)

Chapter Summary

In this chapter we studied probability. We began by defining an event to be an experimental outcome that may or may not occur and by defining the probability of an event to be a number that measures the likelihood that the event will occur. We learned that a probability is often interpreted as a long-run relative frequency, and we saw that probabilities can be found by examining

sample spaces and by using probability rules. We learned several important probability rules—addition rules, multiplication rules, and the rule of complements. We also studied a special kind of probability called a conditional probability, which is the probability that one event will occur given that another event occurs, and we used probabilities to define independent events.

Glossary of Terms

complement (of an event): If A is an event, the complement of A is the event that A will not occur. (page 130)

conditional probability: The probability that one event will occur given that we know that another event occurs. (page 137)

dependent events: When the probability of one event is influenced by whether another event occurs, the events are said to be dependent. (page 140)

event: A set of sample space outcomes. (page 124)

experiment: A process of observation that has an uncertain outcome. (page 121)

independent events: When the probability of one event is not influenced by whether another event occurs, the events are said to be independent. (page 140)

mutually exclusive events: Events that have no sample space outcomes in common, and, therefore, cannot occur simultaneously. (page 133)

probability (of an event): A number that measures the chance, or likelihood, that an event will occur when an experiment is carried out. (page 125)

sample space: The set of all possible experimental outcomes (sample space outcomes). (page 123)

sample space outcome: A distinct outcome of an experiment (that is, an element in the sample space). (page 123)

subjective probability: A probability assessment that is based on experience, intuitive judgment, or expertise. (page 122)

Important Formulas

Probabilities when all sample space outcomes are equally likely: page 127

The rule of complements: page 130

The addition rule for two events: page 133

The addition rule for two mutually exclusive events: page 134

The addition rule for N mutually exclusive events: page 135

Conditional probability: page 138

The general multiplication rule: page 139

Independence: page 140

The multiplication rule for two independent events: page 141

The multiplication rule for N independent events: page 141

Supplementary Exercises

Exercises 3.24 through 3.37 are based on the following situation: An investor holds two stocks, each of which can rise (R), remain unchanged (U), or decline (D) on any particular day.

3.34 Construct a tree diagram showing all possible combined movements for both stocks on a particular day (for instance, RR, RD, and so on, where the first letter denotes the movement of the first stock, and the second letter denotes the movement of the second stock).

- 3.35 If all outcomes are equally likely, find the probability that both stocks rise; that both stocks decline; that exactly one stock declines.
- 3.36 Find the probabilities you found in Exercise 3.33 by assuming that for each stock $P(R) = .6$, $P(U) = .1$, and $P(D) = .3$, and assuming that the two stocks move independently.
- 3.37 Assume that for the first stock (on a particular day)
- $$P(R) = .4, \quad P(U) = .2, \quad P(D) = .4$$
- and that for the second stock (on a particular day)
- $$P(R) = .8, \quad P(U) = .1, \quad P(D) = .1$$
- Assuming that these stocks move independently, find the probability that both stocks decline; the probability that exactly one stock rises; the probability that exactly one stock is unchanged; the probability that both stocks rise.

Exercises 3.38 through 3.48 are based on the following table, which has been constructed using information on consumer ratings of refrigerators in various price ranges given in the February 1994 issue of *Consumer Reports* (vol. 59, no. 2).

	Rating	\$600–799	\$800–999	\$1,000–1,199	\$1,200–1,399
Excellent	1	3	0	0	0
Very good	4	5	4	5	5
Good	1	1	1	0	0

Suppose that we randomly select one of these refrigerators.

- 3.38 Find the probability that the refrigerator's rating is excellent. $1/6$
- 3.39 Find the probability that the refrigerator's rating is very good. $3/4$
- 3.40 Find the probability that the refrigerator's rating is good. $1/12$
- 3.41 Find the probability that the refrigerator's rating is very good or excellent. $22/24$
- 3.42 Find the probability that the refrigerator's price is in the \$800–999 range. $9/24$
- 3.43 Find the probability that the refrigerator's price is in the \$1,200–1,399 range. $5/24$
- 3.44 Find the probability that the refrigerator has an excellent rating and a price in the \$800–999 range. $1/8$
- 3.45 Find the probability that the refrigerator's rating is excellent, if its price is in the \$800–999 range. $1/3$
- 3.46 Find the probability that the refrigerator's rating is excellent, if its price is in the \$1,200–1,399 range. 0
- 3.47 Find the probability that the refrigerator's rating is very good or excellent, if its price is in the \$800–999 range. $8/9$
- 3.48 Find the probability that the refrigerator's rating is very good or excellent, if its price is in the \$1,200–1,399 range. 1
- 3.49 Suppose that in a survey of 1,000 U.S. residents, 721 residents believed that the amount of violent television programming had increased over the past 10 years, 454 residents believed that the overall quality of television programming had decreased over the past 10 years, and 362 residents believed both.
- a What proportion of the 1,000 U.S. residents believed that the amount of violent programming had increased over the past 10 years? $.721$
- b What proportion of the 1,000 U.S. residents believed that the overall quality of programming had decreased over the past 10 years? $.454$
- c What proportion of the 1,000 U.S. residents believed that both the amount of violent programming had increased and the overall quality of programming had decreased over the past 10 years? $.362$
- d What proportion of the 1,000 U.S. residents believed that either the amount of violent programming had increased or the overall quality of programming had decreased over the past 10 years? $.813$
- e What proportion of the 1,000 U.S. residents believed that neither the amount of violent programming had increased nor the overall quality of programming had decreased over the past 10 years? $.187$
- f What proportion of the U.S. residents who believed that the amount of violent programming had increased believed that the overall quality of programming had decreased? $.502$
- g What proportion of the U.S. residents who believed that the overall quality of programming had decreased believed that the amount of violent programming had increased? $.797$

- Slight dependence**
- h What sort of dependence seems to exist between whether U.S. residents believed that the amount of violent programming had increased and whether U.S. residents believed that the overall quality of programming had decreased? Explain your answer.

3.50 Enterprise Industries has been running a television advertisement for Fresh liquid laundry detergent. When a survey was conducted, 21 of the individuals surveyed had purchased Fresh, 41 of the individuals surveyed had recalled seeing the advertisement, and 13 of the individuals surveyed had purchased Fresh and recalled seeing the advertisement.

- 3.51** a What proportion of the individuals surveyed who recalled seeing the advertisement had purchased Fresh?
b Based on your answer to part a, does the advertisement seem to have been effective? Explain.
c A company employs 400 salespeople. Of these, 83 received a bonus last year, 100 attended a special sales training program at the beginning of last year, and 42 both attended the special sales training program and received a bonus. (Note: the bonus was based totally on sales performance.)
d What proportion of the 400 salespeople received a bonus last year?
e What proportion of the 400 salespeople attended the special sales training program at the beginning of last year?
f What proportion of the 400 salespeople both attended the special sales training program and received a bonus?
g What proportion of the salespeople who attended the special sales training program received a bonus?
h Based on your answers to parts a and d, does the special sales training program seem to have been effective? Explain your answer.

Exercises 3.52, 3.53, and 3.54 extend Exercise 3.32 (page 147). Recall that Figure 3.9 (page 147) gives an Accutrackings analysis for the song "We've Got It Going On" by the Backstreet Boys. Also recall that

- 1 Estimates of the probabilities that a randomly selected listener would give the song the ratings 5, 4, 3, 2, and 1 are $P(5) = .18$, $P(4) = .27$, $P(3) = .28$, $P(2) = .08$, and $P(1) = .10$.
2 An estimate of the probability that a randomly selected listener is tired of the song is $P(T) = .17$.
3 We estimate that $P(T|5) = 0$, $P(T|4) = .04$, $P(T|3) = .25$, $P(T|2) = .25$, and $P(T|1) = .70$.
4 We estimate that the probability that a randomly selected listener is familiar with the song is $P(FAM) = .91$.

3.52 Find estimates of $P(5|T)$, $P(4|T)$, $P(3|T)$, $P(2|T)$, and $P(1|T)$.

Hint:

$$P(1|T) = \frac{P(1 \cap T)}{P(T)} = \frac{P(1)P(T|1)}{P(T)}$$

and the other probabilities are calculated similarly.

- 3.53** Let NT denote the event that a randomly selected listener is not tired of the song. Because we estimate that $P(T) = .17$ and $P(T|1) = .70$, we estimate that

$$P(NT) = 1 - P(T) = .83 \quad \text{and} \quad P(NT|1) = 1 - P(T|1) = .30$$

$$P(NT|5) = P(NT|4) = P(NT|3) = P(NT|2) = .70$$

$$P(NT|5) = P(NT|4) = P(NT|3) = P(NT|2) = .70$$

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Note here that $P(5 \cap FAM)$ equals $P(5)$ because the event $5 \cap FAM$ and the event 5 are equivalent. That is, a randomly selected listener would give the song a rating of 5 if and only if the listener is familiar with the song and would give the song a rating of 5. By using similar reasoning, find $P(4 \cap FAM)$, $P(3 \cap FAM)$, $P(2 \cap FAM)$, and $P(1 \cap FAM)$.

3.55 Suppose that A and B are events and that $P(A)$ and $P(B)$ are both positive.

a If A and B are mutually exclusive, what is $P(A \cap B)$?

b If A and B are independent events, explain why $P(A \cap B)$ is positive.

c Can two mutually exclusive events, each having a positive probability of occurrence, also be independent? Prove your answer using your answers to parts a and b.

3.56 Below we give two contingency tables of data from reports submitted by airlines to the U.S. Department of Transportation. The data concern the numbers of on-time and delayed flights for Alaska Airlines and America West Airlines at five major airports.

Alaska Airlines					America West				
	On Time	Delayed	Total		On Time	Delayed	Total		
Los Angeles	497	62	559	Los Angeles	694	117	811		
Phoenix	221	12	233	Phoenix	4,840	415	5,255		
San Diego	212	20	232	San Diego	383	65	448		
San Francisco	503	102	605	San Francisco	320	129	449		
Seattle	1,841	305	2,146	Seattle	201	61	262		
Total	3,274	501	3,775	Total	6,438	787	7,225		

Source: A. Barnett, "How Numbers Can Trick You," *Technology Review*, October 1994, pp. 38–45. Copyright © 1994 MIT Technology Review. Reprinted by permission of the publisher via Copyright Clearance Center.

- a What percentage of all Alaska Airlines flights were delayed? That is, use the data to estimate the probability that an Alaska Airlines flight will be delayed. Do the same for America West Airlines. Which airline does best overall?
b For Alaska Airlines find the percentage of delayed flights at each airport. That is, use the data to estimate each of the probabilities $P(\text{delayed} | \text{Los Angeles})$, $P(\text{delayed} | \text{Phoenix})$, and so on. Then do the same for America West Airlines. Which airline does best at each individual airport?
c We find that America West Airlines does worse at every airport, yet America West does best overall. This seems impossible, but it is true! By looking carefully at the data, explain how this can happen. Hint: Consider the weather in Phoenix and Seattle. (This exercise is an example of what is called *Simpson's paradox*.)

3.57 Internet Exercise

What is the age, gender, and ethnic composition of U.S. college students? As background for its 1995 study of college students and their risk behaviors, the Centers for Disease Control and Prevention collected demographic data—age, gender, and ethnicity—about college students. A report on the 1995 National Health Risk Behavior Survey can be found at the CDC website [http://www.cdc.gov/Data/Statistics/YouthRiskBehaviorSurveillanceSystem/DataProducts/1995NationalCollegeHealthRiskBehaviorSurveyofMHWRF050606.htm]. This report includes a large number of tables, the first of which summarizes the demographic information for the sample of $n = 4609$ college students. An excerpt of Table 1 is given on the right.

Using conditional probabilities, discuss (a) the dependence between age and gender and (b) the dependence between age and ethnicity for U.S. college students.

TABLE 1. Demographic Characteristics of Undergraduate College Students Aged 18–24 Years, by Age Group — United States: National College Health Risk Behavior Survey, 1995

Category	Total (2)	18–24 Years	25 Years
Total	4609	3636	973
Sex			
Male	55.5	52.0	61.8
Female	44.5	48.0	38.2
Race/ethnicity			
White	72.8	70.9	76.1
Black	10.3	10.5	9.1
Hispanic	7.1	6.9	7.4
Other	9.9	11.7	6.4